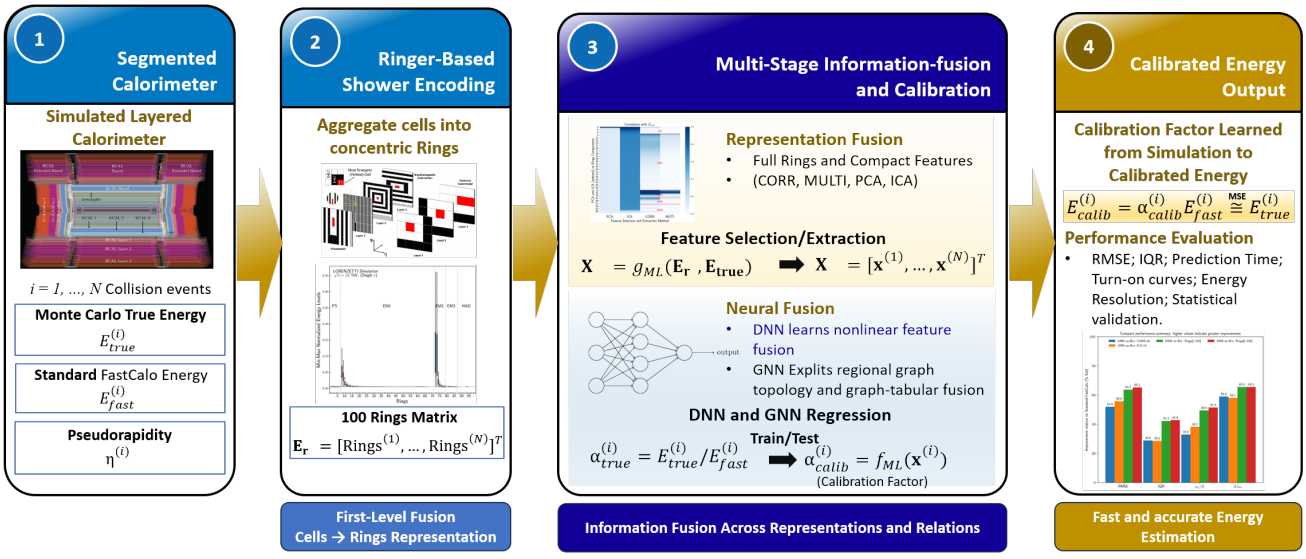
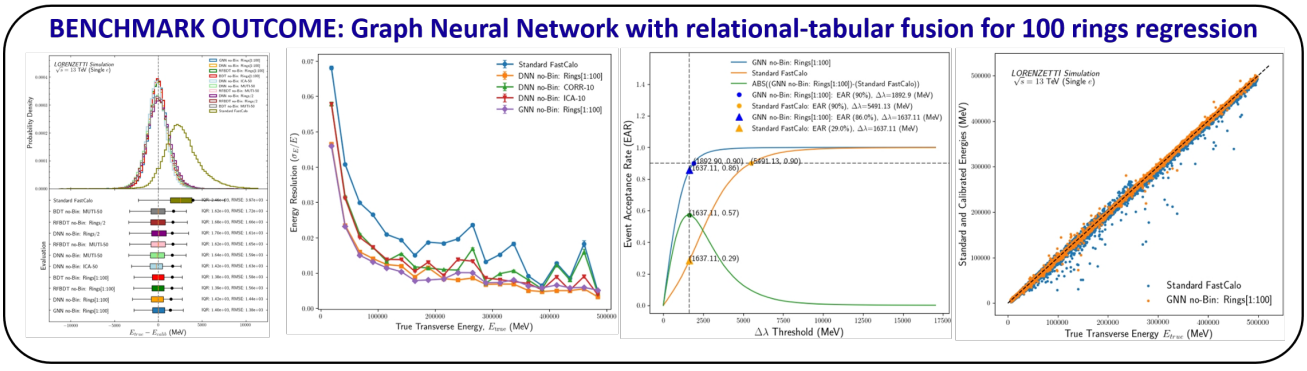
Graphical Abstract

Multi-Stage Neural Information Fusion for Fast Energy Calibration in Segmented Particle-Detector Calorimeters

Paulo Roberto Araujo da Silva, Eduardo Furtado de Simas Filho





Highlights

Multi-Stage Neural Information Fusion for Fast Energy Calibration in Segmented Particle-Detector Calorimeters

Paulo Roberto Araujo da Silva, Eduardo Furtado de Simas Filho

- A neural fusion pipeline is proposed for fast calorimeter energy calibration.
- Ring-based shower profiles are evaluated in complete and compact forms.
- Graph-based calibration combines calorimeter topology with shower summaries.
- Neural calibration reduces error, relative energy resolution, and truth-referenced acceptance-window width.
- Compact neural models preserve strong accuracy–efficiency trade-offs.

Multi-Stage Neural Information Fusion for Fast Energy Calibration in Segmented Particle-Detector Calorimeters

Paulo Roberto Araujo da Silva [a,*], Eduardo Furtado de Simas Filho [a]

[a] PPGEEC – Federal University of Bahia, R Aristídes Novis 2, Federação, Salvador, 40210-630, BA, Brazil

ARTICLE INFO

Keywords:
Multi-stage neural fusion
Segmented calorimeters
Graph neural networks
Ring-based feature extraction
Energy calibration

ABSTRACT

Accurate and low-latency energy calibration is essential for fast calorimetric reconstruction in high-energy physics experiments operating under strict trigger-throughput constraints. This work formulates fast transverse-energy calibration in segmented particle-detector calorimeters as a multi-stage information-fusion problem. Calorimeter-cell measurements are fused into ring-based shower descriptors, the representation is preserved in full or transformed into compact feature spaces, and neural regressors estimate event-level multiplicative calibration factors. Deep Neural Network calibrators are evaluated using full-ring and compact representations, while a representative topology-aware Graph Neural Network exploits the complete 100-ring shower profile through region-specific calorimeter graphs and relational–tabular fusion. Tree-based and hybrid tree-based regressors are retained as reference baselines. The methodology is assessed on an open Lorenzetti-generated benchmark configured as an ATLAS-inspired electromagnetic calorimeter. The full-ring Graph Neural Network achieves the best absolute performance, although its gain over the full-ring Deep Neural Network is modest and is interpreted as evidence of a promising topology-aware alternative rather than proof of intrinsic architectural superiority. Relative to Standard FastCalo, it reduces RMSE by 65.25%, IQR by 42.82%, the global relative energy-resolution summary by 51.46%, and the residual-window half-width required to contain 90% of the independent test events by 65.53%. Compact ten-component Deep Neural Network models based on independent component analysis and correlation ranking retain competitive accuracy with reduced input dimensionality. Statistical validation supports significant paired residual shifts and substantial residual-variance reductions. Overall, the results support ring-based deep and graph neural calibration as a promising information-fusion strategy for improving candidate-level fast calorimetric energy estimates, while detector-specific, resonance-level, controlled pile-up, and hardware-level validation remain necessary before deployment conclusions can be drawn.

1. Introduction

Fast and accurate energy calibration is a central requirement in modern particle-detector calorimetry. In high-energy physics experiments, calorimeters provide highly segmented measurements of the energy deposited by particles as they interact with detector material. These measurements support the reconstruction and selection of electrons and photons in final states involving electroweak and Higgs-boson processes, where the reconstructed transverse energy affects trigger decisions, downstream reconstruction quality, and physics sensitivity [1]. In trigger-oriented environments, the calibration problem is further constrained by the need to improve the energy estimate without introducing computational costs incompatible with high-throughput event selection.

The challenge is intrinsically an information-fusion problem. A segmented calorimeter can be viewed as a multi-sensor measurement system in which many localized readout cells provide partial and noisy observations of the same particle shower. These observations are affected by electronic noise, pile-up, shower fluctuations, detector nonlinearities, and geometry-dependent response effects [2, 3]. Reliable fast calibration therefore requires combining spatially distributed detector information into a compact and informative representation from which an event-level energy correction can be estimated.

Ring-based calorimetric representations provide a natural intermediate level of fusion for this task. In the Ringer approach, cell-level transverse-energy deposits inside a region of interest are aggregated into concentric rings centered on the shower core [4, 5]. The resulting ring profile is not a set of independent physical sensors, but a structured descriptor obtained by fusing many segmented calorimeter-cell measurements. It preserves relevant radial shower information while reducing the dimensionality of the original cell-level readout. This makes ring-based representations suitable for fast calibration studies in which both predictive performance and computational compactness are important.

In this work, fast calorimetric energy calibration is formulated as a multi-stage information-fusion pipeline. The first stage fuses segmented cell-level measurements into ring-based shower descriptors. The second stage either preserves the complete ring representation or transforms it into compact representations through feature-selection and feature-extraction methods. The third stage uses neural regression models to fuse the resulting representation into an

This work was supported by FAPESB, FAPERJ, CNPq, RENAFAE, CAPES-COFECUB program and Coordination for the Improvement of Higher Education Personnel- Brazil (CAPES)- Granting Code 001 in Brazil.

*Corresponding author.

araujop@ufba.br (Paulo Roberto Araujo da Silva);
eduardo.simas@ufba.br (Eduardo Furtado de Simas Filho)

event-level multiplicative calibration factor. This formulation is consistent with the broader information-fusion principle of combining multiple sources, features, or representations to obtain more reliable estimates than those available from isolated measurements alone [6, 7].

Learning-based fusion has become increasingly relevant in multi-sensor and multi-source systems. Previous studies in *Information Fusion* have addressed machine-learning approaches for data fusion [7], graph-based feature-fusion strategies for structured learning problems [8], and information fusion for multi-scale data representations [9]. The present study applies this perspective to fast calorimetric calibration, where the information to be fused is not multimodal in the conventional image–text sense, but is instead distributed across detector cells, ring descriptors, compact latent or selected features, graph-structured regional representations, and event-level calibration outputs.

Within the proposed framework, deep neural models provide nonlinear feature fusion over either the complete ring profile or compact representations obtained from correlation ranking, mutual-information ranking, principal component analysis, and independent component analysis. Graph neural models provide an additional relational-fusion mechanism by incorporating the ordered and region-dependent structure of the calorimeter-ring representation into the model architecture. In the graph-based formulation evaluated here, the complete 100-ring profile is partitioned into calorimeter-inspired regions, represented as regional graphs, and processed through local message passing before being fused with tabular shower-shape information for calibration-factor estimation. This graph architecture is used as a representative topology-aware neural fusion model, motivated by the structured organization of the calorimeter rings, rather than as an exhaustive comparison of all possible neural architectures for calorimetric calibration.

Published deep-learning studies have demonstrated the applicability of neural methods to calorimetric reconstruction and energy regression across different detector geometries and calorimeter representations [10, 11, 12]. However, no external calorimetric energy-regression result is numerically reproduced in the present Lorenzetti benchmark. Differences in detector geometry, simulation fidelity, input observables, calibration targets, event spectra, and evaluation metrics prevent literature results from being interpreted as controlled numerical baselines for the present dataset. Accordingly, these external neural-calorimetry studies are used as methodological context, while all numerical performance comparisons in this work are restricted to methods evaluated under the common Lorenzetti dataset and experimental protocol.

Previous Lorenzetti-based studies established the feasibility of ring-based machine-learning calibration using conventional regressors and hybrid tree-based models [13, 14]. The present work deliberately uses the same open benchmark as a controlled reference environment, but addresses a different methodological question: whether deep and graph neural information-fusion models can exploit full-ring and compact ring-based representations for fast calorimetric calibration. In this sense, Lorenzetti is used as the data-generation and benchmark layer, whereas the methodological novelty of the present article lies in the neural calibration formulation, the topology-aware graph representation, and the expanded performance analysis. Reusing the same benchmark makes it possible to isolate the effect of the newly evaluated neural fusion models from changes in the simulation environment. Tree-based and hybrid tree-based methods are therefore retained only as reference baselines, while the main methodological focus is the evaluation of neural and graph-based fusion strategies.

The proposed methodology is general at the level of its mathematical formulation and data interface. It requires segmented calorimetric readout information that can be transformed into ring-based or Ringer-like observables, together with cluster-level quantities and supervised calibration targets. In this study, the data interface is instantiated using the Lorenzetti Showers framework [15, 16], configured as an ATLAS-inspired electromagnetic calorimeter [17]. Lorenzetti is used as an open and reproducible simulation environment for benchmark evaluation, not as evidence of detector-independent deployment readiness or simulation-to-data closure. The reported results are therefore benchmark-level evidence under the simulated detector-response and pile-up conditions considered here; the specific validation steps required for transfer to experimental data are discussed in Section 3.8.

The main contribution of this work is not the proposal of a new standalone learning algorithm or the causal isolation of one neural branch. The contribution is the formulation and benchmark evaluation of a multi-stage information-fusion pipeline for fast calorimetric calibration: segmented calorimeter measurements are aggregated into ring descriptors; the representation is preserved or compacted; neural models fuse the resulting shower information into an event-level calibration factor; and the graph configuration further combines relational regional embeddings with deterministic tabular shower summaries. The experiments compare these complete pipelines under a common benchmark. They do not isolate the marginal causal contribution of graph, tabular, descriptor, skip, or fusion components; that attribution requires the controlled ablations summarized in Section 3.8.

The specific contributions of this work are as follows: (i) A multi-stage information-fusion formulation is introduced for fast energy calibration in segmented particle-detector calorimeters, connecting cell-level measurements, ring-based shower descriptors, compact feature representations, and event-level calibration factors; (ii) Deep neural calibration models are evaluated across both the complete 100-ring representation and compact representations obtained through correlation ranking, mutual-information ranking, principal component analysis, and independent component analysis; (iii) A representative topology-aware graph neural calibration model is evaluated for the complete 100-ring shower profile, using calorimeter-inspired regional

graphs and relational message passing over the ordered ring representation; (iv) A hybrid relational–tabular graph neural architecture is evaluated, combining graph embeddings, derived shower-shape descriptors, tabular neural processing, and final fusion for calibration-factor estimation; (v) Full-ring and compact neural calibration strategies are compared with tree-based and hybrid tree-based baselines under an integrated protocol covering global calibration error, energy resolution, acceptance-window behavior, bin-wise stability, statistical validation, input compactness, and CPU-based trained-model prediction time.

The remainder of this paper is organized as follows. Section 2 describes the methodology and simulation setup, including the event-generation tool, ring-based feature construction, dimensionality-reduction strategies, neural and graph-based calibration models, baseline regressors, and performance-evaluation protocol. Section 3 presents the experimental results, including global accuracy metrics, energy-resolution studies, threshold-response analyses, bin-wise stability, statistical validation, pile-up-aware interpretation, computational-efficiency comparisons, and implications for physics-process reconstruction. Section 4 summarizes the main findings and discusses possible directions for future developments.

2. Methodology

This section describes the simulation benchmark, calorimetric observables, ring-based encoding, neural calibration formulation, feature-selection and feature-extraction procedures, regression models, training protocol, and performance-evaluation criteria used in this work.

The proposed methodology follows a multi-stage information-fusion pipeline. First, segmented calorimeter-cell measurements are aggregated into ring-based shower descriptors. Second, the resulting 100-ring representation is either preserved in full or transformed into compact representations through feature-selection and feature-extraction methods. Third, regression models estimate an event-level multiplicative calibration factor that corrects the Standard FastCalo transverse-energy estimate.

The neural contribution of this work consists of introducing Deep Neural Network and Graph Neural Network calibration models into this ring-based pipeline. Tree-based and hybrid tree-based methods are retained as reference baselines because they were previously shown to provide strong performance in the same Lorenzetti-generated benchmark [14]. All data-dependent transformations and hyperparameter-selection steps are fitted within the training data only, and final performance is assessed on an independent test partition.

2.1. Simulation Setup and Calorimetric Observables

The dataset used in this study was generated with the Lorenzetti Showers framework [15, 16], using the ATLAS-inspired electromagnetic calorimeter configuration previously adopted in Lorenzetti-based calibration studies [13, 14]. Lorenzetti provides an open simulation chain including event generation, shower propagation in a segmented calorimeter geometry, electronic response emulation, and reconstruction of calorimetric observables. This environment gives access to low-level cell information and ring-based energy representations, which are required for the calibration methodology evaluated here.

Approximately 3×10^5 reconstructed single-electron particle-gun events were used. The particle-gun setup injects electrons with predefined energies and incident directions into the detector volume, enabling broad coverage of the kinematic region considered in this benchmark. This choice is suitable for event-wise calibration studies because it avoids restricting the training sample to a specific decay topology, such as Z-boson decays, whose energy and angular distributions would cover a narrower region of phase space [3, 18].

Pile-up contributions were represented by merging minimum-bias proton–proton events into the primary single-electron event. The simulation assumed a luminosity of $10^{34}\ \mathrm{cm}^{-2}\ \mathrm{s}^{-1}$, consistent with the benchmark configuration described in [17]. Electronic noise is included during the pulse-shaping stage of the Lorenzetti simulation, which converts simulated hits into readout signals and emulates the response of calorimeter electronics. The noise contribution follows a Gaussian distribution centered at zero, with layer-dependent standard deviation. The complete simulated dataset is publicly available at [19].

In this work, Lorenzetti is used as an open and reproducible detector-inspired benchmark, not as a replacement for an official experimental reconstruction chain. The configuration preserves several features relevant to fast calorimetric calibration, including segmentation, electronic noise, pile-up treatment, and access to cell-level and ring-level observables. Nevertheless, the results should be interpreted as simulation-based relative-performance evidence under the adopted benchmark conditions. Validation with collaboration-grade simulation, experimental data, and deployment oriented trigger software or hardware would be required before operational conclusions could be drawn.

In the adopted simulation, pile-up refers to additional detector activity produced by multiple proton–proton interactions occurring within the same, or nearby, readout time window. Minimum-bias events correspond to generic proton–proton collision events generated with loose selection requirements and are used to emulate the diffuse background activity associated with additional collisions. Nearby calorimetric activity denotes energy deposits from particles close to the electron candidate within, or close to, the region of interest; such activity may originate from pile-up particles or from other particles produced in the same hard interaction.

Figure 1 shows the distribution of the average pile-up parameter $\langle\mu\rangle$ in the simulated configuration. The distribution contains 10,010 pile-up profile entries, with mean $\langle\mu\rangle = 38.94$ and standard deviation 9.999. Therefore, the calibration benchmark corresponds to a pile-up-inclusive scenario

with a substantial level of additional collision activity. However, the dataset was not generated as a systematic scan over independently controlled pile-up levels. Consequently, the reported RMSE, IQR, energy-resolution, and acceptance-window results quantify the relative calibration performance under the simulated pile-up distribution considered here, but do not establish invariance across separately defined low-, medium-, and high-pile-up regimes.

The 10,010 entries in Fig. 1 are records of the pile-up profile used to define the simulated pile-up distribution; they are not reconstructed single-electron events and therefore are not expected to match the 302,619-event calibration dataset one-to-one. The reconstructed electron events are generated under this pile-up-inclusive configuration. The available benchmark documentation does not require a one-to-one correspondence between profile records and reconstructed electron events, and no such correspondence is assumed in the analysis.

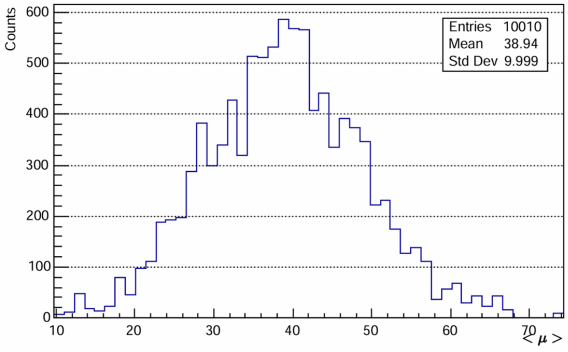


Figure 1: Distribution of the average pile-up parameter $\langle\mu\rangle$ in the Lorenzetti simulated dataset used in this work. The distribution contains 10,010 pile-up profile entries, with mean $\langle\mu\rangle = 38.94$ and standard deviation 9.999, indicating that the calibration benchmark includes a substantial level of additional collision activity.

A cross-sectional view of the simulated calorimeter is shown in Fig. 2, and the corresponding segmentation parameters are summarized in Table 1. The electromagnetic and hadronic calorimeters are segmented into longitudinal layers and preceded by a presampler layer, allowing the simulation to reproduce the main features of a sampling calorimeter relevant to electron energy reconstruction.

For each simulated event, a fixed region of interest (RoI) of 0.4×0.4 in (η, ϕ) is considered. When all seven Lorenzetti calorimeter layers are included, this RoI contains approximately 1,000 calorimeter cells per event. To obtain a compact representation of the electromagnetic shower topology, the *Ringer* algorithm [20, 5] is applied within the RoI. The algorithm aggregates transverse-energy deposits into 100 concentric rings centered on the shower barycenter, as illustrated in Fig. 3. The resulting 100-dimensional vector preserves radial shower information while reducing the original cell-level dimensionality, making it suitable for low-latency machine-learning calibration studies [17].

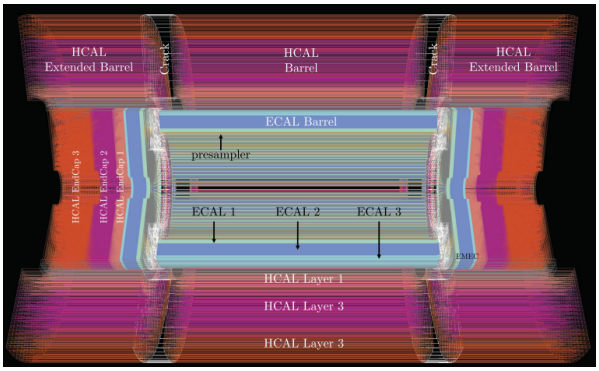


Figure 2: Cross-sectional view of the general-purpose sampling calorimeter used for simulations with the Lorenzetti Showers framework. The single-electron benchmark analyzed in this work is restricted to the range $|\eta| \leq 1.46$, extending to the barrel-side transition neighborhood of the ATLAS-inspired geometry. Source: [15].

Table 1
Segmentation, coverage in $|\eta|$, and cell granularity used in the simulated calorimeter. Electromagnetic and hadronic calorimeters are segmented into three longitudinal layers and preceded by a presampler layer. Source: [15].

Layer	Sampling	Coverage	Granularity ($\Delta\eta \times \Delta\phi$)
Presampler	Barrel	$0.00 < \|\eta\| < 1.58$	0.025×0.1
	End-Cap	$1.50 < \|\eta\| < 1.80$	0.025×0.1
Electromagnetic Calorimeter			
Layer 1	Barrel	$0.00 < \|\eta\| < 1.55$	0.003×0.1
		$1.37 < \|\eta\| < 1.80$	0.003×0.1
	End-Cap	$1.80 < \|\eta\| < 2.00$	0.025×0.1
		$2.00 < \|\eta\| < 2.37$	0.006×0.1
		$2.37 < \|\eta\| < 3.20$	0.1×0.1
Layer 2	Barrel	$0.00 < \|\eta\| < 1.50$	0.025×0.025
	End-Cap	$1.35 < \|\eta\| < 2.50$	0.025×0.025
		$2.50 < \|\eta\| < 3.20$	0.1×0.1
Layer 3	Barrel	$0.00 < \|\eta\| < 1.58$	0.05×0.1
	End-Cap	$1.35 < \|\eta\| < 2.50$	0.05×0.025
		$2.50 < \|\eta\| < 3.20$	0.1×0.1
Hadronic Calorimeter			
Layer 1	Barrel	$0.00 < \|\eta\| < 1.09$	0.1×0.1
	Extended Barrel	$0.94 < \|\eta\| < 1.77$	0.1×0.1
	End-Cap	$1.50 < \|\eta\| < 2.50$	0.1×0.1
		$2.50 < \|\eta\| < 3.20$	0.2×0.2
Layer 2	Barrel	$0.00 < \|\eta\| < 1.09$	0.1×0.1
	Extended Barrel	$0.85 < \|\eta\| < 1.41$	0.1×0.1
	End-Cap	$1.50 < \|\eta\| < 2.50$	0.1×0.1
		$2.50 < \|\eta\| < 3.20$	0.2×0.2
Layer 3	Barrel	$0.85 < \|\eta\| < 0.72$	0.2×0.1
	Extended Barrel	$0.85 < \|\eta\| < 1.41$	0.2×0.1
	End-Cap	$1.50 < \|\eta\| < 2.50$	0.1×0.1
		$2.50 < \|\eta\| < 3.20$	0.2×0.2

In addition to the normalized ring-energy vector, three event-level quantities are used throughout the calibration pipeline: E_{true} is the true transverse energy of the generated particle, used to define the reference calibration target; E_{fast}

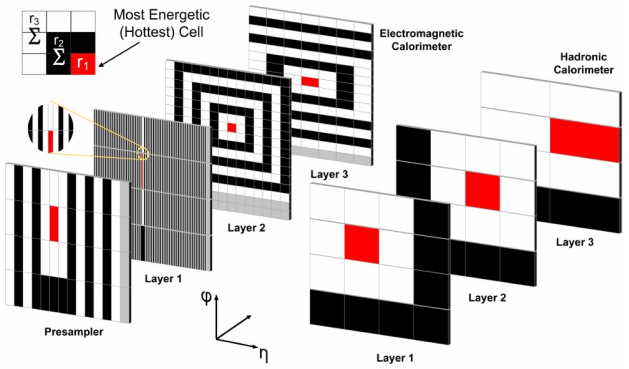


Figure 3: Graphical representation of the ring-based reconstruction concept. The Ringer algorithm centers the reconstruction on the highest-energy cell and groups surrounding calorimeter cells into concentric rings across different calorimeter layers. Each ring aggregates the transverse energy deposited by the cells it encompasses, preserving the spatial topology of the electromagnetic shower in a one-dimensional profile. Source: [4, 21].

is the Standard FastCalo baseline transverse-energy estimate, corresponding to the uncalibrated fast reconstruction output; η is the pseudorapidity of the reconstructed electromagnetic object, used for regional performance evaluation across the detector acceptance.

The global workflow is summarized in Fig. 4. Starting from the Lorenzetti simulation output, normalized ring variables are used either directly or after feature selection or extraction. The resulting representation is provided to regression models that estimate event-level calibration factors. The calibrated energy is then compared with E_{true} to quantify the improvement over the Standard FastCalo baseline.

From an information-fusion perspective, the pipeline contains four distinct fusion levels. First, cell-level calorimetric measurements are aggregated into ring descriptors, reducing the original segmented readout while preserving radial shower topology. Second, representation-level fusion is performed by either retaining the complete 100-ring profile or transforming it into compact selected or extracted feature spaces. Third, neural feature fusion is performed by the DNN models, while relational fusion is performed by the GNN through regional message passing over the ordered calorimeter-ring topology. Finally, output-level fusion occurs when the predicted calibration factor is combined multiplicatively with the Standard FastCalo estimate to obtain the calibrated transverse energy. This layered structure is the main reason why the calibration task is treated here as an information-fusion problem rather than as a conventional standalone regression exercise.

2.2. Standard FastCalo Energy and Ring Encoding

This section defines the Standard FastCalo baseline energy estimate and the normalized ring representation used as input to the calibration models. The same Lorenzetti-based reconstruction interface was used in previous ring-based calibration studies [13, 14]; it is summarized here only to make the present calibration formulation self-contained.

Starting from the raw calorimeter signals, the energy is reconstructed at the cell level, aggregated into clusters, corrected for detector effects and pile-up, and finally transformed into structured ring features suitable for ML-based energy calibration [17]. The reconstructed energy in the k-th calorimeter cell is estimated using optimal filtering of the digitized signal:

$$E_k^{\text{cell}} = \sum_{j=1}^{N_s} w_j S_{kj}, \tag{1}$$

where S_{kj} is the signal sampled at time j, w_j are the optimal-filter weights, and N_s is the number of signal samples [17].

The reconstructed cell energies are aggregated within an adaptively defined electromagnetic shower cluster to obtain the raw cluster energy:

$$E_{\text{raw}} = \sum_{k\in\text{cluster}} E_k^{\text{cell}}. \tag{2}$$

The transverse component of the fast energy estimate, hereafter referred to as Standard FastCalo energy, is then computed as

$$E_{\text{fast}} = \frac{E_{\text{raw}}}{\cosh(\eta)}, \tag{3}$$

where η is the pseudorapidity of the cluster centroid.

To construct a compact topological representation of the electromagnetic shower, the *Ringer* algorithm aggregates the transverse energy deposits into concentric rings centered on the shower core, as illustrated in Fig. 3. The transverse energy associated with the j-th ring is defined as

$$E_{r,j} = \sum_{k\in\text{ring}\,j} \frac{E_k^{\text{cell}}}{\cosh(\eta)}, \tag{4}$$

where E_k^{cell} is the reconstructed energy in the k-th calorimeter cell belonging to ring j.

The ring energies are normalized to form a dimensionless shower-shape profile:

$$\tilde{E}_{r,j} = \frac{E_{r,j}}{\sum_{\ell=1}^{100} E_{r,\ell}}, \tag{5}$$

where $\tilde{E}_{r,j}$ is the normalized transverse energy of the j-th ring. This normalization emphasizes the spatial distribution of the shower rather than its absolute magnitude, allowing the calibration model to exploit topological information independently of the original energy scale.

Although normalization removes the overall magnitude of the ring vector, it does not remove all correlations between shower morphology and the correction required by the fast energy estimate. The regression models used in this work do

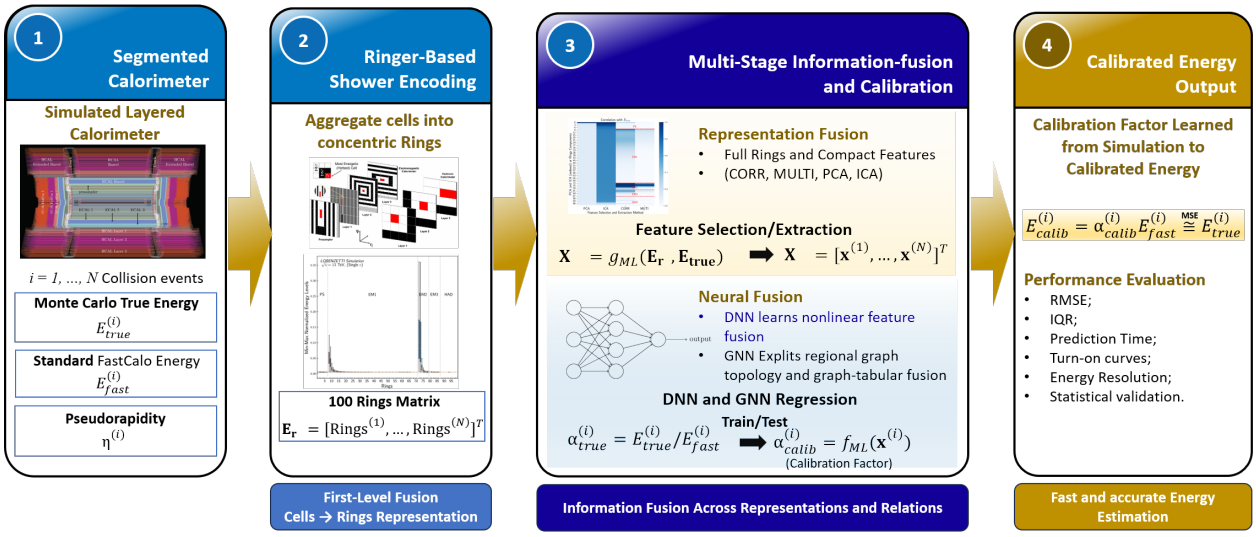


Figure 4: Schematic workflow of the proposed energy-calibration system integrated with Lorenzetti Showers. The feature-selection/extraction and machine-learning regression blocks constitute the main processing stages used to estimate the event-level calibration factors.

not receive E_{fast}, E_{raw}, or another unnormalized energy variable as an input. Consequently, they cannot deterministically recover the absolute event energy from the normalized ring vector. Instead, they estimate α_{calib} from statistical correlations between the normalized spatial shower profile and the event-wise correction target. Energy-dependent shower development, longitudinal sharing, leakage, detector-response effects, and nearby calorimetric activity can modify the relative distribution of energy among rings even after normalization. Absolute scale re-enters only after inference through Eq. 8, where the predicted calibration factor is multiplied by the measured E_{fast}.

This distinction limits the physical interpretation of the model. A purely scale-dependent nonlinearity that produced no correlated change in the normalized ring pattern would not be identifiable from the present inputs alone. The observed correction of energy-dependent response structures therefore indicates that the normalized shower morphology contains predictive information correlated with the required calibration factor; it does not imply that the network has direct access to the absolute energy regime. Evaluating E_{fast} or unnormalized cluster energy as an additional model input would constitute a separate controlled experiment and is left for future work. The normalized ring representation should likewise be interpreted as a compact shower-topology descriptor, not as an intrinsically pile-up-invariant observable.

For a dataset with N events, the normalized ring-energy matrix is written as

$$\tilde{\mathbf{E}}_r = \begin{bmatrix} \tilde{E}^{(1)}_{r,1} & \cdots & \tilde{E}^{(1)}_{r,100} \\ \tilde{E}^{(2)}_{r,1} & \cdots & \tilde{E}^{(2)}_{r,100} \\ \vdots & \ddots & \vdots \\ \tilde{E}^{(N)}_{r,1} & \cdots & \tilde{E}^{(N)}_{r,100} \end{bmatrix}, \tag{6}$$

where each row corresponds to one event and encodes the normalized shower profile across the 100 rings. For notational convenience, the full ring vector of the i-th event is denoted by

$$\text{Rings}[1:100]^{(i)} = \left[\tilde{E}^{(i)}_{r,1}, \ldots, \tilde{E}^{(i)}_{r,100}\right]. \tag{7}$$

The Standard FastCalo estimate E_{fast} provides the initial trigger-level transverse-energy measurement. Because this estimate is affected by detector response, noise, pile-up, and resolution effects, a data-driven calibration factor is introduced to obtain a corrected energy estimate closer to E_{true}.

2.3. Proposed Calibration Method

The calibration procedure follows an event-wise multiplicative correction strategy. For the i-th event, the calibrated transverse energy is defined as

$$E^{(i)}_{\text{calib}} = \alpha^{(i)}_{\text{calib}} E^{(i)}_{\text{fast}}, \tag{8}$$

where $E^{(i)}_{\text{fast}}$ is the Standard FastCalo transverse-energy estimate and $\alpha^{(i)}_{\text{calib}}$ is the event-level calibration factor estimated by a supervised regression model.

The target calibration factor is obtained from the reference and fast transverse-energy estimates:

$$\alpha^{(i)}_{\text{true}} = \frac{E^{(i)}_{\text{true}}}{E^{(i)}_{\text{fast}}}. \tag{9}$$

Thus, the calibration problem is formulated as a supervised regression task in which a machine-learning model estimates the multiplicative correction factor $\alpha^{(i)}_{\text{calib}}$ from a ring-based calorimetric representation.

Let

$$\tilde{\mathbf{E}}_r \in \mathbb{R}^{N\times 100} \tag{10}$$

denote the normalized ring-energy matrix, where each row corresponds to one event and each column corresponds to one ring descriptor. The representation-construction stage is denoted by $g_{\mathrm{ML}}(\cdot)$. This operator is defined using only the training partition, as detailed in Section 2.4, and is then applied unchanged to the corresponding ring matrices:

$$\mathbf{X}_{\mathrm{train}} = g_{\mathrm{ML}}\left(\tilde{\mathbf{E}}_{r,\mathrm{train}}\right), \qquad \mathbf{X}_{\mathrm{test}} = g_{\mathrm{ML}}\left(\tilde{\mathbf{E}}_{r,\mathrm{test}}\right). \quad (11)$$

For the full-ring configuration, $g_{\mathrm{ML}}(\cdot)$ is the identity mapping and the input representation contains all 100 rings. For compact configurations, $g_{\mathrm{ML}}(\cdot)$ denotes the corresponding feature-selection or feature-extraction operator defined from the training data. Therefore, E_{true} may be used only inside training-partition representation-definition procedures, such as supervised ranking or component ordering, but it is never used as an input variable during validation or test inference.

The resulting feature vector for the i-th event is

$$\mathbf{x}^{(i)} = \left[x_1^{(i)}, \ldots, x_F^{(i)}\right] \in \mathbb{R}^F. \quad (12)$$

For full-ring models, $F = 100$. For compact models, F is the number of selected or extracted features. In the graph-based configuration, the complete full-ring vector is used as the input representation from which the graph and auxiliary tabular descriptors are internally constructed by the implemented regression architecture, as detailed in Section 2.5.2.

A regression model $f_{\mathrm{ML}}(\cdot)$ is then trained to estimate the event-level calibration factor:

$$\alpha_{\mathrm{calib}}^{(i)} = f_{\mathrm{ML}}\left(\mathbf{x}^{(i)}\right). \quad (13)$$

Here, $f_{\mathrm{ML}}(\cdot)$ denotes the trained regression function. Depending on the evaluated configuration, this function may correspond to a Deep Neural Network, a Graph Neural Network, or one of the reference baseline regressors. The model architecture, trainable parameters, optimization procedure, and selected hyperparameters are part of the specific regression model and are described separately in Section 2.5 and Appendix C.

The final calibrated transverse energy is therefore given by

$$E_{\mathrm{calib}}^{(i)} = f_{\mathrm{ML}}\left(\mathbf{x}^{(i)}\right) E_{\mathrm{fast}}^{(i)} \approx E_{\mathrm{true}}^{(i)}. \quad (14)$$

During training, the model is supervised using the target calibration factor $\alpha_{\mathrm{true}}^{(i)}$. A generic factor-level objective can be written as

$$\mathrm{MSE}(\alpha) = \frac{1}{N}\sum_{i=1}^{N}\left(\alpha_{\mathrm{true}}^{(i)} - f_{\mathrm{ML}}\left(\mathbf{x}^{(i)}\right)\right)^2. \quad (15)$$

After inference, the predicted factor is applied to the Standard FastCalo estimate according to Eq. 14. Therefore, the learning problem is defined at the calibration-factor level, whereas the final performance assessment is performed at the calibrated-energy level through the residuals, energy resolution, acceptance-window behavior, and statistical procedures described in Section 2.7.

This formulation defines the calibration pipeline used throughout the paper. Calorimeter-cell measurements are first fused into ring-based descriptors, the ring representation is either preserved in full or transformed into a compact feature space by $g_{\mathrm{ML}}(\cdot)$, and the trained regression model $f_{\mathrm{ML}}(\cdot)$ estimates the event-level correction factor applied to the Standard FastCalo energy. The role of the neural architectures evaluated in this work is therefore to instantiate $f_{\mathrm{ML}}(\cdot)$: the DNN performs nonlinear feature fusion over ring-based inputs, whereas the GNN implements a topology-aware relational–tabular regression model from the complete ring representation.

2.4. Feature Selection and Extraction

Feature selection and extraction are used to reduce the original 100-dimensional ring representation while preserving information related to the physical shower-energy content. The representation-ranking protocol follows the preceding Lorenzetti calibration studies: CORR and MUTI rank original rings by their relation to E_{true}, while PCA and ICA components are ordered by their relation to E_{true} after the unsupervised transformation has been fitted. The regression target itself remains the event-level calibration factor α_{true}. This E_{true}-based ordering was retained to preserve controlled comparability with the established representation study; it should not be interpreted as evidence that E_{true}-based ranking is the target-optimal feature-selection rule for α_{true}. A target-specific re-ranking by α_{true} would define a different representation experiment and would require re-running model selection and evaluation. For each representation, candidate variables or components are retained at fixed cardinalities $m \in \{50, 10, 5, 2\}$, enabling a controlled comparison of CORR, MUTI, PCA, and ICA under identical external input dimensionalities.

In addition to these data-driven compact representations, an expert-guided 50-ring subset, denoted by `Rings/2`, is included as a physically motivated reference. This subset was selected from the original ring profile using prior inspection of the calorimeter-region energy-deposition pattern and is not obtained by correlation ranking, mutual information, PCA, or ICA.

Let $\tilde{\mathbf{E}}_r \in \mathbb{R}^{N\times 100}$ denote the normalized ring-energy matrix. The role of the representation stage is to construct an input matrix $\mathbf{X} \in \mathbb{R}^{N\times F}$, where each row $\mathbf{x}^{(i)} \in \mathbb{R}^F$ represents the i-th event in the selected or extracted feature space, and F is the number of input features used by the regression model. For reduced configurations, $F = m$, with $m \in \{50, 10, 5, 2\}$. Thus, dimensionality reduction is performed with respect to the original 100-ring representation, so that compact configurations satisfy $F < 100$, reaching as few as $F = 2$.

To make explicit the separation between representation fitting and representation application, the feature-construction operator is denoted by $g_{\mathrm{ML}}(\cdot)$. This operator maps a ring matrix with 100 columns into a feature matrix with F columns. During training, $g_{\mathrm{ML}}(\cdot)$ is fitted or defined using

only the training partition:

$$g_{\mathrm{ML}} = \mathcal{T}_g\left(\tilde{\mathbf{E}}_{r,\mathrm{train}}, \mathbf{E}_{\mathrm{true,train}}\right), \tag{16}$$

where $\mathcal{T}_g(\cdot)$ denotes the procedure used to define the representation operator. Depending on the representation, this procedure may correspond to the identity mapping, a fixed expert-guided selector, a supervised feature-ranking rule, or a fitted feature-extraction transformation followed by target-based component ordering. The argument $\mathbf{E}_{\mathrm{true,train}}$ is used only when the representation-definition procedure requires target-based ranking or ordering; it is ignored by the identity mapping and by the fixed expert-guided selector.

Once g_{ML} has been defined using the training data, it is applied unchanged to the training, validation, or test ring matrices:

$$\mathbf{X}_{\mathrm{train}} = g_{\mathrm{ML}}\left(\tilde{\mathbf{E}}_{r,\mathrm{train}}\right), \qquad \mathbf{X}_{\mathrm{test}} = g_{\mathrm{ML}}\left(\tilde{\mathbf{E}}_{r,\mathrm{test}}\right). \tag{17}$$

This notation emphasizes that E_{true} may be used to define or rank the representation within the training partition, but it is not used as an input variable during validation or test inference.

For the complete `Rings[1:100]` representation, g_{ML} is the identity mapping and $F = 100$. For the expert-guided `Rings/2` representation, g_{ML} is a fixed selector defined from prior inspection of the calorimeter-region energy-deposition pattern; it does not use E_{true} and retains $F = 50$ original ring variables.

For the data-driven compact representations, g_{ML} is fitted using the training partition. In CORR and MUTI, g_{ML} selects the top m original ring variables ranked according to their relevance to E_{true}. In PCA and ICA, the transformation is fitted on the training ring matrix, and the resulting components are subsequently ordered according to their relevance to E_{true} within the same training partition. The fitted selector, transformation, and component ordering are then applied unchanged to validation or test data.

Four data-driven strategies are evaluated: two feature-selection methods, based on Pearson correlation and mutual information, and two feature-extraction methods, based on principal component analysis and independent component analysis [22, 23]. In addition, the expert-guided `Rings/2` subset is retained as a physically motivated compact reference.

In the definitions below, the quantities used to compute relevance scores or component orderings refer to the training partition when defining g_{ML}, consistently with Eq. 16.

For correlation-based selection, each original ring variable is scored according to its Pearson correlation coefficient with the true transverse energy:

$$\rho_j = \rho(\tilde{\mathbf{E}}_{r,j}, \mathbf{E}_{\mathrm{true}}), \qquad j = 1, \ldots, 100, \tag{18}$$

where $\tilde{\mathbf{E}}_{r,j}$ denotes the j-th column of $\tilde{\mathbf{E}}_r$. The ring variables are ranked by $|\rho_j|$, and the top m variables are retained. The resulting configurations are denoted by `CORR-`m.

For mutual-information-based selection, each ring variable is scored using its mutual information with the true transverse energy:

$$I_j = \mathrm{MI}(\tilde{\mathbf{E}}_{r,j}, \mathbf{E}_{\mathrm{true}}), \qquad j = 1, \ldots, 100. \tag{19}$$

The variables are ranked by I_j, and the top m variables are retained. These configurations are denoted by `MUTI-`m. Unlike the correlation score, mutual information can capture nonlinear dependencies between the ring response and E_{true} [24].

For principal component analysis, the normalized ring matrix is transformed into an orthogonal basis of uncorrelated components. The resulting components are ordered by their relevance to $\mathbf{E}_{\mathrm{true}}$, measured through the absolute Pearson correlation:

$$|\rho(\mathbf{PCA}^{(1)}, \mathbf{E}_{\mathrm{true}})| \geq \cdots \geq |\rho(\mathbf{PCA}^{(m)}, \mathbf{E}_{\mathrm{true}})|. \tag{20}$$

The top m components define the `PCA-`m configurations.

For independent component analysis, the normalized ring matrix is decomposed into statistically independent latent components. As with PCA, the components are ranked according to their absolute correlation with the true transverse energy:

$$|\rho(\mathbf{ICA}^{(1)}, \mathbf{E}_{\mathrm{true}})| \geq \cdots \geq |\rho(\mathbf{ICA}^{(m)}, \mathbf{E}_{\mathrm{true}})|. \tag{21}$$

The top m components are retained to form the `ICA-`m configurations. This representation is useful for investigating whether non-Gaussian latent structures in the shower profile provide compact information for calibration [23].

The notation used throughout the paper is therefore as follows: `CORR-`m and `MUTI-`m denote the top m original ring variables selected by correlation and mutual information, respectively, whereas `PCA-`m and `ICA-`m denote the top m extracted components retained after E_{true}-relevance ordering. The notation `Rings[1:100]` denotes the complete ring representation, and `Rings/2` denotes the expert-guided 50-ring subset. This fixed-cardinality design makes it possible to compare feature-selection and feature-extraction methods under identical dimensionality constraints.

This training-only construction prevents information leakage. CORR and MUTI rankings, PCA and ICA transformations, and PCA/ICA component orderings are defined using only the training partition. The resulting selected variables, fitted transformations, and component orderings are then applied unchanged to validation or held-out test data.

For PCA and ICA, the variance associated with each retained component is also analyzed to support interpretability. In PCA, the explained variance is obtained from the eigenvalues of the covariance or correlation matrix. In ICA, although the components are not inherently ordered by variance, a post-hoc variance analysis is used to assess the relative contribution of the retained components. The resulting matrix $\mathbf{X} \in \mathbb{R}^{N \times F}$ is then used as input to the Deep Neural Network models and to the corresponding baseline regressors described in the next section. The Graph Neural Network model uses the complete `Rings[1:100]` representation, with $F = 100$, in order to preserve the topology required by the graph-based formulation; additional graph-node and tabular descriptors used by the GNN are deterministic internal quantities derived from the same full-ring input.

2.5. Neural and Baseline Regression Models

The calibration task is formulated as a supervised regression problem in which a trained model maps the ring-derived feature vector $\mathbf{x}^{(i)}$ to the event-level calibration factor, as defined in Eq. 13. In this section, different choices for the regression function $f_{\mathrm{ML}}(\cdot)$ are described. These include Deep Neural Network and Graph Neural Network calibrators, as well as tree-based and hybrid tree-based reference baselines because they were previously shown to provide strong performance in the same Lorenzetti-generated benchmark [14]. Architecture-specific quantities, such as graph node features, regional edge sets, tabular shower descriptors, hidden embeddings, or fusion layers, are treated as internal components of the corresponding regressor and do not alter the generic calibration formulation defined in Section 2.3.

All models are trained globally, without prior $(|\eta|, E_T)$ binning. For the neural models, an internal validation subset drawn from the training partition is used for early stopping and hyperparameter selection, while the independent test partition remains untouched until final evaluation. The label `no-Bin` is therefore used to indicate that a single regressor is trained over the complete feature space, rather than fitting separate models in disjoint kinematic regions. After training, the models are evaluated both globally and within distinct $(|\eta|, E_T)$ regions. This strategy allows the stability of the global calibration function to be assessed across different kinematic regimes while avoiding the statistical fragmentation and boundary effects associated with bin-specific training.

2.5.1. Deep Neural Network Calibration

Deep Neural Network models are used to perform nonlinear feature fusion over the ring-based calorimetric representation. Multilayer neural networks are well established as universal nonlinear function approximators for supervised learning tasks and have become a standard component of modern deep learning [25, 22]. In high-energy physics, deep learning has been widely investigated for detector reconstruction, classification, simulation, and calorimetric applications [26, 27]. In the present work, the DNN is used as a nonlinear regression model that maps full or compact ring-based shower descriptors to an event-level calibration factor.

$$\alpha_{\mathrm{calib}}^{(i)} = f_{\mathrm{DNN}}\left(\mathbf{x}^{(i)}\right), \tag{22}$$

where $f_{\mathrm{DNN}}(\cdot)$ denotes the trained Deep Neural Network regressor associated with the selected full or compact ring-based representation. The trainable weights, optimization settings, and hyperparameters define how this regression function is obtained during training, but after training they are absorbed into the fixed mapping $f_{\mathrm{DNN}}(\cdot)$.

The Deep Neural Network models are evaluated under two complementary input regimes. In the first regime, the complete 100-ring representation is used as input, allowing the network to learn directly from the full shower profile. In the second regime, compact representations obtained through correlation ranking, mutual-information ranking, principal component analysis, or independent component analysis are used as inputs. This design makes it possible to assess whether nonlinear neural calibration can preserve competitive performance under strong dimensionality reduction.

The hidden layers apply nonlinear transformations to the input representation, progressively combining calibration-relevant information distributed across the ring descriptors or compact components. In this sense, the Deep Neural Network acts as a nonlinear feature-fusion model: it does not rely on a single calorimetric observable, but learns a multivariate correction factor from the combined shower-shape information. The DNN regressors were implemented with the scikit-learn `MLPRegressor` interface. The selected configurations use a training batch size of 512, as detailed in Appendix C. The output layer produces a scalar estimate of α_{calib}, which is then multiplied by the Standard FastCalo energy according to Eq. 8.

2.5.2. Graph Neural Network Calibration

In the graph-based configuration, the generic regression function $f_{\mathrm{ML}}(\cdot)$ in Eq. 13 is instantiated as a topology-aware relational–tabular Graph Neural Network operating on the complete Rings[1:100] representation. The motivation is that the ring profile is not an unordered feature vector: it is an ordered, region-dependent descriptor derived from a segmented calorimeter geometry. Graph neural networks are appropriate for this type of problem because they introduce a relational inductive bias, allowing information associated with entities and their connections to be processed explicitly [28]. In high-energy physics, graph-based learning has also been increasingly used when detector information is naturally represented as structured sets of hits, clusters, particles, or calorimeter cells [29, 30].

The architecture selected in this work is not intended to be a minimal graph-only baseline. A minimal graph-only model could be obtained by assigning one ring value to each node, connecting adjacent rings, pooling the resulting node embeddings, and regressing the calibration factor directly. Such a model would isolate the effect of local message passing, but it would be a restrictive model for fast calorimetric calibration.

The calibration factor depends not only on local structure in the ordered Ringer profile, but also on global shower-shape information, regional energy balance, and physically meaningful ratios between calorimeter regions. These aspects are important in sampling calorimeters, where the measured response depends on longitudinal shower development, energy leakage, detector segmentation, and region-dependent response effects [2, 3].

For this reason, the implemented model is designed as a representative relational–tabular fusion architecture rather than as a graph-only ablation. The graph branch evaluates whether the ordered and region-dependent topology of the ring representation provides useful relational information. The tabular branch preserves the full standardized ring profile and derived shower-shape summaries, ensuring that

global and regional information remains available to the calibration model. The final fusion stage combines both sources of information to estimate the event-level calibration factor. This design follows the general information-fusion principle of combining complementary representations to obtain a more informative estimate than would be available from either representation alone [6, 7, 8].

The regional graph construction is motivated by the ordered organization of the calorimeter response in the Ringer representation. The presampler, electromagnetic, and hadronic regions are represented as separate graphs instead of being treated as a single homogeneous chain. Within each region, bidirectional edges are introduced between consecutive ring indices in the ordered Ringer descriptor. These edges encode sequential adjacency in the adopted ring representation rather than direct Euclidean distance between calorimeter cells. This construction is consistent with the neighborhood-aggregation principle of GraphSAGE, in which node embeddings are learned from local graph neighborhoods [31]. No edges are introduced between PS, EM, and HAD in the selected architecture; the global interaction among regions is subsequently handled by the tabular and fusion paths.

The node descriptors combine event-dependent and structural information. The standardized and original normalized ring values encode the local shower response, while the local position, global position, and region indicator encode where that response occurs within the calorimeter representation. This is important because similar numerical ring values may have different calibration relevance depending on whether they occur in the presampler, electromagnetic, or hadronic part of the shower descriptor.

The use of two GraphSAGE layers provides a compact local message-passing mechanism over the ordered regional chains. Because the graph edges connect one-hop neighboring rings, two successive GraphSAGE layers allow an internal node to incorporate information from nodes located up to two graph hops away, subject to the corresponding regional boundaries. The PS, EM, and HAD graph blocks use the same architecture but independent trainable parameters. The graph model was implemented with PyTorch and PyTorch Geometric, using `SAGEConv` layers with mean neighborhood aggregation. The selected GNN uses a training batch size of 256, as detailed in Appendix C. Mean and max pooling are concatenated to form regional summaries: mean pooling summarizes the distributed regional response, whereas max pooling preserves dominant latent activations. The resulting regional embeddings are subsequently concatenated and fused with the tabular representation to estimate the calibration factor.

The linear skip connection, nonlinear tabular output, and graph–tabular fusion output are included to stabilize calibration-factor regression. The skip and tabular paths provide a direct global calibration route, while the graph branch acts as a topology-aware relational correction. This design is related to the broader use of shortcut paths in deep architectures to improve optimization stability [32]. The trainable graph-scaling factor initialized at a small value prevents the randomly initialized graph branch from dominating the early training stage and allows its contribution to be learned from data.

The selected GNN should therefore be interpreted as a complete relational–tabular calibration pipeline rather than as an isolated test of graph topology. Although its external measurement input remains the same 100 normalized rings, the implemented pipeline additionally constructs 20 deterministic shower-shape descriptors and combines graph, tabular, nonlinear-output, and linear-skip paths. The full-ring DNN, in contrast, is a flat nonlinear regressor over the 100-ring representation. Consequently, their numerical difference mixes relational inductive bias, deterministic feature engineering, model capacity, and fusion design, and cannot be causally attributed to graph message passing alone. A component-level attribution study would require, at minimum, a DNN receiving the same 120-dimensional internal representation, graph-only and tabular-only variants, removal of the derived descriptors, and parameter-matched controls. These models require separate training and hyperparameter selection and are therefore outside the experimental protocol evaluated here. Accordingly, the present paper compares complete calibration pipelines and does not claim that graph topology alone causes the observed numerical advantage.

In the notation of Eq. 13, the GNN configuration receives the complete full-ring input vector,

$$\mathbf{x}^{(i)} = \mathbf{x}^{(i)}_{\text{full}} \in \mathbb{R}^{100}.$$

The graph and tabular quantities described below are deterministic internal representations constructed from this same full-ring input vector. They are not additional external measurements and do not modify the generic calibration equation. Thus, the GNN prediction can be written at the calibration-method level as

$$\alpha^{(i)}_{\text{calib}} = f_{\text{GNN}}\left(\mathbf{x}^{(i)}_{\text{full}}\right), \tag{23}$$

where the internal graph construction, regional message passing, tabular embedding, and final fusion stage define the functional form of $f_{\text{GNN}}(\cdot)$.

For each event i, the complete full-ring vector is partitioned into three calorimeter-inspired regions:

$$\mathbf{x}^{(i)}_{\text{full}} = \left[\mathbf{x}^{(i)}_{\text{PS}}, \mathbf{x}^{(i)}_{\text{EM}}, \mathbf{x}^{(i)}_{\text{HAD}}\right], \tag{24}$$

where $\mathbf{x}^{(i)}_{\text{PS}} \in \mathbb{R}^8$ corresponds to Rings 1–8, $\mathbf{x}^{(i)}_{\text{EM}} \in \mathbb{R}^{80}$ corresponds to Rings 9–88, and $\mathbf{x}^{(i)}_{\text{HAD}} \in \mathbb{R}^{12}$ corresponds to Rings 89–100. In the underlying 100-ring Ringer organization, the finer longitudinal mapping is PS: Rings 1–8, EM1: Rings 9–72, EM2: Rings 73–80, EM3: Rings 81–88, and hadronic rings: Rings 89–100. The graph architecture deliberately coarsens EM1–EM3 into one electromagnetic subgraph and the hadronic rings into one hadronic subgraph. Thus, PS/EM/HAD is an architecture-level grouping of the

ordered ring descriptor, whereas the finer layer organization remains the physical segmentation used to construct the 100-ring representation. The graph edges connect neighboring positions in this ordered descriptor after cell-to-ring aggregation and should not be interpreted as direct Euclidean adjacency between individual calorimeter cells.

Three independent regional graphs are constructed:

$$\mathcal{G}_r^{(i)} = \left(\mathcal{V}_r, \mathcal{E}_r, \mathbf{Z}_r^{(i)}\right), \qquad r \in \{\mathrm{PS}, \mathrm{EM}, \mathrm{HAD}\}. \quad (25)$$

The graph topology is fixed across events, and only the node features depend on the event. Within each region, adjacent rings are connected through a bidirectional sequential topology. In the selected configuration, the neighborhood range is one ring index, such that

$$(j, j+1) \in \mathcal{E}_r \quad \text{and} \quad (j+1, j) \in \mathcal{E}_r, \quad (26)$$

for all valid neighboring ring indices j inside region r. No edges are introduced between different calorimeter regions. Thus, the graph branch models local sequential relations independently within the presampler, electromagnetic, and hadronic ring groups.

Each node is represented by a seven-dimensional descriptor:

$$\mathbf{z}_v^{(i)} = \left[x_{v,z}^{(i)}, x_{v,\mathrm{raw}}^{(i)}, p_v^{\mathrm{local}}, p_v^{\mathrm{global}}, \mathbf{e}_r\right], \quad (27)$$

where $x_{v,z}^{(i)}$ is the standardized ring value, $x_{v,\mathrm{raw}}^{(i)}$ is the original normalized ring value, $p_v^{\mathrm{local}} \in [0,1]$ is the normalized node position inside its region, $p_v^{\mathrm{global}} \in [0,1]$ is the normalized position of the corresponding ring in the complete 100-ring sequence, and $\mathbf{e}_r \in \mathbb{R}^3$ is a one-hot encoding of the calorimeter region. These node features combine event-dependent shower information with fixed positional and regional descriptors, allowing the Graph Neural Network to distinguish both local ring values and their structural location in the calorimetric representation.

The initial node representation is obtained with a linear encoder followed by GELU activation and layer normalization:

$$\mathbf{h}_v^{(0)} = \mathrm{LayerNorm}\left[\mathrm{GELU}\left(\mathbf{W}_{\mathrm{enc}}\mathbf{z}_v^{(i)} + \mathbf{b}_{\mathrm{enc}}\right)\right]. \quad (28)$$

The encoded node representations are then processed independently in each region by two GraphSAGE layers with mean neighborhood aggregation:

$$\mathbf{h}_v^{(\ell+1)} = \mathrm{SAGEConv}_\ell\left(\mathbf{h}_v^{(\ell)}, \left\{\mathbf{h}_u^{(\ell)} : u \in \mathcal{N}_r(v)\right\}\right), \quad (29)$$

where $\mathcal{N}_r(v)$ denotes the neighboring rings of node v inside region r. In the selected configuration, the node encoder has hidden dimension 64 and the GraphSAGE layers have hidden dimension 128. A GELU activation and dropout are applied after the first GraphSAGE layer, whereas the second GraphSAGE layer provides the final regional node embeddings.

The node embeddings in each region are converted into a fixed-size regional representation by concatenating mean and max pooling:

$$\mathbf{g}_r^{(i)} = \left[\frac{1}{|\mathcal{V}_r|}\sum_{v\in\mathcal{V}_r}\mathbf{h}_v^{(L)}, \max_{v\in\mathcal{V}_r}\mathbf{h}_v^{(L)}\right], \qquad L = 2. \quad (30)$$

The complete graph embedding is formed by concatenating the three regional embeddings:

$$\mathbf{g}^{(i)} = \left[\mathbf{g}_{\mathrm{PS}}^{(i)}, \mathbf{g}_{\mathrm{EM}}^{(i)}, \mathbf{g}_{\mathrm{HAD}}^{(i)}\right]. \quad (31)$$

Because each regional embedding concatenates mean- and max-pooled vectors of dimension 128, the final graph embedding has dimension $3 \times 2 \times 128 = 768$.

In parallel to the graph branch, the model constructs a tabular representation from the complete 100-ring profile and 20 deterministic shower-shape descriptors derived from the same calorimetric information. The descriptors summarize four complementary classes of information: regional PS/EM/HAD energy content; relative energy concentration among those regions; the energy content of the physical EM1, EM2, and EM3 samplings and their contributions relative to the complete EM region; and weighted positional moments inside PS, EM, and HAD. The positional moments use the min–max weights defined in Appendix B. The exact definitions of all 20 descriptors, including EM1 = Rings 9–72, EM2 = Rings 73–80, and EM3 = Rings 81–88, are given in that appendix.

The resulting internal tabular representation is

$$\mathbf{x}_{\mathrm{tab}}^{(i)} = \left[\mathbf{x}_{\mathrm{full}}^{(i)} \| \mathbf{d}^{(i)}\right] \in \mathbb{R}^{120}, \quad (32)$$

where $\mathbf{d}^{(i)} \in \mathbb{R}^{20}$ is the deterministic descriptor vector. These descriptors are internal functions of the same full-ring input and do not introduce additional external detector measurements. No target-derived quantity, including E_{true} or α_{true}, is used as an input during validation or test inference.

The tabular branch maps $\mathbf{x}_{\mathrm{tab}}^{(i)}$ into a 256-dimensional embedding:

$$\mathbf{t}^{(i)} = f_{\mathrm{tab}}\left(\mathbf{x}_{\mathrm{tab}}^{(i)}\right), \qquad \mathbf{t}^{(i)} \in \mathbb{R}^{256}. \quad (33)$$

The implemented tabular branch consists of three hidden transformations of width 256. The first two transformations use a linear layer, GELU activation, layer normalization, and dropout, whereas the last transformation uses a linear layer followed by GELU activation. The nonlinear tabular output head maps the 256-dimensional tabular embedding through a hidden layer of width 128 with GELU activation, followed by a scalar output layer.

Internally, the trained GNN mapping in Eq. 23 combines a linear skip connection from the tabular representation, a nonlinear tabular prediction, and a graph-based correction obtained from the fusion of graph and tabular embeddings:

$$f_{\mathrm{GNN}}\left(\mathbf{x}_{\mathrm{full}}^{(i)}\right) = f_{\mathrm{lin}}\left(\mathbf{x}_{\mathrm{tab}}^{(i)}\right) + f_{\mathrm{tab,out}}\left(\mathbf{t}^{(i)}\right) + f_{\mathrm{fus}}\left(\left[s_{\mathrm{g}}\mathbf{g}^{(i)}, \mathbf{t}^{(i)}\right]\right),$$

(34)

where s_g is a trainable scalar initialized to 0.05. The fusion network receives the concatenation of the scaled graph embedding and the tabular embedding, resulting in an input dimension of $768 + 256 = 1024$. It then applies a hidden layer of width 256, GELU activation, layer normalization, dropout, a second hidden layer of width 128, and a final scalar output layer. The selected GNN is trained directly against α_{true} using the combined objective $\mathcal{L}_{\text{GNN}} = \lambda_{\text{MSE}}\mathcal{L}_{\text{MSE}} + (1-\lambda_{\text{MSE}})\mathcal{L}_{\text{SmoothL1}}(\beta)$, with $\lambda_{\text{MSE}} = 0.70$ and $\beta = 1.10$ for the selected configuration. The optimizer and remaining training settings are reported in Appendix C.

This architecture implements a hybrid relational–tabular information-fusion model. The graph branch extracts local and regional relational patterns from the complete ring topology, while the tabular branch preserves global shower-shape information and derived regional summaries. The final fusion stage combines both representations to estimate the event-level calibration factor applied to the Standard FastCalo energy.

The comparison with the flat DNN is therefore interpreted as a benchmark between complete neural calibration pipelines, not as a causal ablation of graph topology. The controls required for component-level architectural attribution are summarized in Section 3.8.

2.5.3. Tree-Based and Hybrid Baseline Models

Tree-based and hybrid tree-based models are included as reference baselines to contextualize the gains obtained with the neural calibration architectures. The baseline models include Decision Trees, Gradient Boosted Decision Trees, Random Forests, and hybrid combinations based on residual correction or stacked generalization.

For consistency with Eq. 13, each tree-based or hybrid baseline is treated as a trained regression function that receives the representation vector $\mathbf{x}^{(i)}$ and predicts the calibration factor $\alpha_{\text{calib}}^{(i)}$. The hybrid models modify only the internal construction of this regression function, for example through residual correction or stacked generalization, but the calibration interface remains unchanged.

Decision Trees recursively partition the feature space using hierarchical decision rules and can capture nonlinear interactions, although they are prone to overfitting if not properly regularized [22]. Gradient Boosted Decision Trees build an additive ensemble of trees trained sequentially to reduce prediction errors [33, 34, 22]. Random Forests train an ensemble of decorrelated trees on bootstrap samples with random feature subsampling and combine their predictions by averaging [35, 22]:

$$f_{\text{RF},\alpha}(\mathbf{x}^{(i)}) = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}^{(i)}), \tag{35}$$

where T_b denotes the b-th regression tree.

The hybrid tree-based baselines follow the residual-correction and stacked-generalization strategies previously evaluated in [14]. In residual-correction models, a first regressor estimates the calibration factor and a second regressor learns the remaining factor-level residual structure. In stacked models, the predictions of multiple base regressors are combined by a regularized meta-learner. These baselines are not the main methodological contribution of the present work; they are retained to provide a controlled comparison against previously competitive non-neural calibration strategies under the same feature representations and evaluation protocol.

All Gradient Boosted Decision Tree models evaluated in this work were implemented using the LightGBM library. Comparative studies indicate that the relative performance of modern boosted-tree implementations, including XGBoost, LightGBM, and CatBoost, depends on the dataset characteristics, hyperparameter-optimization procedure, evaluation metric, and computational setting [36, 37]. Accordingly, LightGBM was selected as the common boosted-tree implementation for the controlled comparison of the calibration architectures.

Overall, the selected regressors provide a controlled comparison between nonlinear neural feature fusion, graph-based relational fusion, and established tree-based calibration baselines. This design enables the identification of calibration strategies that are accurate, compact, statistically stable, and computationally feasible for fast energy-reconstruction pipelines.

2.6. Binning, Training, and Hyperparameter Selection

The detector response in calorimetric systems is not uniform over the full phase space, as it may depend on both the pseudorapidity $|\eta|$ and the transverse energy E_T. For this reason, segmentation in $(|\eta|, E_T)$ regions is commonly used in calorimetric performance and calibration studies to inspect spatial and energy-dependent variations in reconstruction performance [3, 18].

In this work, binning is used for performance evaluation rather than for model training. All regression models are trained globally, using the full training sample without prior $(|\eta|, E_T)$ segmentation. The bin-wise analysis is then performed a posteriori to assess whether the globally trained calibration models remain stable across different detector and energy regions. This strategy avoids the statistical fragmentation and boundary effects that may arise when separate models are trained independently in each bin, while still allowing localized performance differences to be identified.

The $(|\eta|, E_T)$ boundaries adopted for the regional analysis are the same segmentation previously obtained with the genetic-algorithm optimization reported in our IEEE TNS study [13]. That optimization selected five transverse-energy intervals and three absolute-pseudorapidity intervals to avoid sparsely populated regions and improve the statistical balance among the resulting 15 segments. The same boundaries are retained here to preserve direct comparability with the previous calibration study. Table 2 reports the population of the complete simulated dataset inside those

regions; the binning is used only for a posteriori regional evaluation and does not partition the data for model training.

Table 2
Number of events from the complete simulated dataset in each $(|\eta|, E_T)$ region defined by the GA-optimized segmentation. The 15 bins contain 302,618 of the 302,619 reconstructed events; one event lies marginally above the adopted upper boundary $|\eta| = 1.46$.

E_T interval (GeV)	$\|\eta\| \in [0, 0.43]$	$\|\eta\| \in [0.43, 0.83]$	$\|\eta\| \in [0.83, 1.46]$
$[0, 44.7]$	11043	13643	30715
$[44.7, 71.5]$	12642	14247	25023
$[71.5, 103.5]$	16354	17853	29538
$[103.5, 164.3]$	31019	31064	15582
$[164.3, 499.9]$	26866	15233	11796

The experimental design compares full-ring and compact ring-based representations across neural and baseline regression models. The full-ring configuration uses the complete `Rings[1:100]` representation. Compact configurations are obtained from correlation ranking, mutual-information ranking, principal component analysis, and independent component analysis, using fixed cardinalities selected from $m \in \{50, 10, 5, 2\}$.

Deep Neural Network models are evaluated across the full-ring representation and the compact feature-selection and feature-extraction configurations. The Graph Neural Network model is evaluated using the complete 100-ring representation, because its purpose is to exploit the constructive topology of the full ring system. Tree-based and hybrid tree-based methods are retained as reference baselines for comparison with previously established calibration strategies.

The reconstructed dataset contains 302,619 events. A fixed split assigns 211,833 events (70%) to the development partition and 90,786 events (30%) to the independent test partition. No permanent third validation set is defined. The 90,786-event test partition remains outside representation fitting, hyperparameter selection, early stopping, and model training until the final evaluation.

For the compact representations, the representation operator $g_{\mathrm{ML}}(\cdot)$ is fitted or defined using only the development partition and is then held fixed. Conditional on that predefined representation, regressor hyperparameters are selected inside the 211,833-event development partition using five outer folds and five inner folds. The inner folds select numerical model hyperparameters, while the outer folds characterize the stability of that regressor-selection procedure within development data. Thus, the nested cross-validation described here applies to the regressor conditional on the representation already defined from the development partition; it is not an end-to-end nested refitting of the complete $f_{\mathrm{ML}} \circ g_{\mathrm{ML}}$ pipeline.

For neural training calls, the early-stopping validation subset is drawn only from the data available to the corresponding development/training call. After hyperparameter selection, the final model is refitted using the development partition and evaluated on the independent 90,786-event test partition. The GNN setting `max_train_samples`=300000 is only an upper cap; because it is larger than the number of development events available to any training call, it does not induce subsampling in the experiments reported here. The final selected hyperparameters for the representative Deep Neural Network and Graph Neural Network configurations are reported in Appendix C.

2.7. Performance Evaluation

The performance of each calibration strategy is evaluated from complementary perspectives: global energy-reconstruction accuracy, localized stability across $(|\eta|, E_T)$ regions, energy resolution, trigger-level response, statistical significance, input compactness, and CPU-based trained-model prediction time. This multi-criteria evaluation is intended to identify models that are accurate, stable across the evaluated benchmark regions, compact, and computationally efficient in the relative CPU-based comparison considered here.

2.7.1. Global Energy-Error Metrics

The global calibration performance is quantified by comparing the calibrated energy E_{calib} with the true energy E_{true}. For each event, the calibration error is defined as

$$\varepsilon_{\mathrm{calib}} = E_{\mathrm{true}} - E_{\mathrm{calib}}, \tag{36}$$

where calib denotes the machine-learning model used to estimate the calibration factor. For the Standard FastCalo baseline, the corresponding error is

$$\varepsilon_{\mathrm{fast}} = E_{\mathrm{true}} - E_{\mathrm{fast}}. \tag{37}$$

An ideal calibration method would yield errors centered around zero and with minimal dispersion. Therefore, two complementary metrics are used: Root Mean Squared Error (RMSE), which measures the average magnitude of the error and penalizes large deviations; Interquartile Range (IQR), which quantifies the central dispersion of the error distribution and is robust to outliers and non-Gaussian tails.

To summarize the joint behavior of these two error-based metrics, we define

$$d_0 = \sqrt{\mathrm{RMSE}^2 + \mathrm{IQR}^2}. \tag{38}$$

Lower values of d_0 indicate better simultaneous performance in terms of global accuracy and error dispersion. This score is restricted to predictive performance and does not include computational cost or model compactness.

2.7.2. Computational Efficiency and Global Ranking

Since the intended application involves high-throughput trigger-level environments, predictive performance must be assessed together with computational efficiency. Two additional quantities are therefore considered: (i) Number of external ring-based input features, F, which reflects the dimensionality of the independent representation provided to the trained model during inference. For the GNN, $F = 100$ denotes the complete full-ring input; graph-node descriptors

and tabular shower-shape summaries are deterministic internal quantities derived from the same 100-ring vector; (ii) Amortized prediction time per event, T, computed as the total wall-clock prediction time for the test batch divided by the number of events in that batch. This quantity measures the average throughput-oriented cost of the trained model after the input features are available, and should not be interpreted as single-event trigger latency.

Training time is not included in this evaluation because training is performed offline and does not affect real-time operation. The Standard FastCalo baseline is treated as a reference with $F_{\text{fast}} = 1$ and zero additional trained-model prediction time, since no machine-learning inference step is applied to produce this baseline estimate in the present comparison.

To combine predictive accuracy, compactness, and trained-model prediction time into a single global score, we define:

$$D_0 = \sqrt{\text{IQR}_{(N)}^2 + \text{RMSE}_{(N)}^2 + F_{(N)}^2 + T_{(N)}^2}, \qquad (39)$$

where $\text{IQR}_{(N)}$, $\text{RMSE}_{(N)}$, $F_{(N)}$, and $T_{(N)}$ are min–max normalized values computed across all evaluated configurations. Normalization is required because the four quantities have different scales and physical units.

Equal weights are assigned to the four min–max-normalized terms in D_0 deliberately as a neutral baseline aggregation rule; the weights are not derived from detector physics, trigger policy, or a specific deployment cost function. The purpose of D_0 is therefore to expose how the ranking changes when predictive error, residual dispersion, input dimensionality, and measured model-prediction time are given equal numerical importance. It is not proposed as a universal deployment metric. RMSE, IQR, feature count, and prediction time are consequently reported separately, so that another application can impose different weights or hard latency/memory constraints without relying on D_0. The measured timing terms remain relative CPU throughput benchmarks rather than production-trigger latency guarantees.

Prediction times were measured using the following hardware configuration: Intel® Core™ i7-7500U CPU, 20 GB DDR4 RAM, 1 TB SSD, and NVIDIA® GeForce® 940MX GPU with 2 GB of memory. All reported prediction times correspond to CPU-based inference and were computed as amortized per-event times by dividing the total wall-clock prediction time of the full test batch by the number of events in that batch. The CPU thread count was not explicitly pinned in the timing script and therefore followed the default threading behavior of the underlying numerical libraries.

For the selected full-ring configurations, the DNN contains 161,701 trainable parameters and the relational–tabular GNN contains approximately 642,812 trainable parameters. With 32-bit floating-point parameters, these correspond to approximately 0.62 MiB and 2.45 MiB of parameter storage, respectively; these figures do not include intermediate activations, framework workspaces, or data buffers. The measured trained-model prediction times are 0.00405 ms/event for the full-ring DNN and 0.153 ms/event for the GNN, making the latter approximately 37.8 times slower in this CPU throughput benchmark. The corresponding RMSE values are 1439.35 and 1378.77 MeV, respectively, so the GNN provides a 4.21% RMSE reduction relative to the full-ring DNN at substantially higher computational cost.

We do not report a hardware-independent FLOP figure for these two implementations because sparse GraphSAGE aggregation, dense layers, framework kernels, and batching are executed by different software operators and a nominal operation count would not constitute an end-to-end latency measurement. More importantly, the reported wall-clock values begin after input-feature availability and exclude online ring construction, deterministic GNN-descriptor construction, graph preparation, memory-access or transfer overheads, model loading, fixed-point or quantized implementation, and production scheduling. No specific High-Level Trigger latency budget is therefore claimed to be satisfied by either configuration. The measurements are used only as within-environment relative throughput benchmarks; deployment conclusions require end-to-end measurement on the target software and hardware stack.

2.7.3. Bin-Wise Calibration-Factor Evaluation

After the global ranking, the top-performing configurations are evaluated locally within the $(|\eta|, E_T)$ regions defined in Table 2. This bin-wise analysis is performed to verify whether the globally trained models remain stable across different detector and energy regions.

The predicted calibration factor α_{calib} is compared with the target calibration factor α_{true}, defined in Eq. 9. For each bin, three metrics are computed: IQR, Mean Absolute Error (MAE), and RMSE. The IQR quantifies the dispersion of the predicted calibration factors, whereas MAE and RMSE measure the deviation from the ideal multiplicative correction.

This localized evaluation is important because global metrics may hide instabilities or systematic deviations in specific regions of the detector acceptance. Therefore, the bin-wise analysis complements the global ranking by assessing the regional consistency of the selected calibration strategies.

2.7.4. Energy-Resolution Evaluation

Energy resolution is used as an additional performance indicator to characterize the precision of the calibrated energy estimates over the full energy range [18, 3]. For each event, the calibrated response is defined as

$$r = \frac{E_{\text{calib}}}{E_{\text{true}}}, \qquad (40)$$

with the corresponding relative residual

$$\delta = \frac{E_{\text{calib}} - E_{\text{true}}}{E_{\text{true}}} = r - 1. \qquad (41)$$

The energy resolution is estimated from the dispersion of r, or equivalently δ, within each energy interval. When

the response distribution is approximately Gaussian, this dispersion can be represented by the standard deviation, such that

$$\frac{\sigma_E}{E} \approx \sigma_r \approx \sigma_\delta. \tag{42}$$

Because calorimetric residuals may exhibit non-Gaussian tails or asymmetries, a robust quantile-based estimator is also considered. The interquartile range is defined as

$$\mathrm{IQR}(X) = Q_{75}(X) - Q_{25}(X), \tag{43}$$

and may be converted into a Gaussian-equivalent width through

$$\sigma_{\mathrm{eq}} \approx \frac{\mathrm{IQR}}{1.349}. \tag{44}$$

In this work, the energy resolution is estimated empirically from the relative-response distribution on the independent test set, following the same convention adopted in the preceding Lorenzetti calibration study [14]. For a method m, define

$$r_m^{(i)} = \frac{E_m^{(i)}}{E_{\mathrm{true}}^{(i)}}, \tag{45}$$

where E_m denotes Standard FastCalo or a calibrated energy estimate. Let $\mathcal{B}_k$ denote the set of test events belonging to the k-th E_{true} interval used to construct Fig. 15. The relative energy resolution in that interval is

$$\left(\frac{\sigma_E}{E}\right)_k = \mathrm{Std}\left(\left\{ r_m^{(i)} \;\middle|\; i \in \mathcal{B}_k \right\}\right). \tag{46}$$

Thus, Fig. 15 reports the energy dependence of the empirical relative-response width rather than a single resolution value over the complete energy range.

For the compact whole-sample summary reported in Table 5, the same definition is applied directly to all $N_{\mathrm{test}} = 90{,}786$ events of the independent test partition,

$$\left(\frac{\sigma_E}{E}\right)_{\mathrm{global}} = \mathrm{Std}\left(\left\{r_m^{(i)}\right\}_{i=1}^{N_{\mathrm{test}}}\right). \tag{47}$$

This dimensionless quantity is not obtained by averaging the energy-bin values. The uncertainty reported with the global width follows the standard-error expression used in the original analysis,

$$\mathrm{SE}\left[\left(\frac{\sigma_E}{E}\right)_{\mathrm{global}}\right] \approx \frac{\left(\sigma_E/E\right)_{\mathrm{global}}}{\sqrt{2\left(N_{\mathrm{test}}-1\right)}}. \tag{48}$$

The resulting quantity is a truth-referenced empirical response-width measure for the Lorenzetti benchmark. It is not interpreted here as a fitted experimental detector-resolution parameter, and no stochastic–constant–noise decomposition $a/\sqrt{E} \oplus b \oplus c/E$ is assigned to the whole-sample summary. The energy-dependent curve in Fig. 15 is retained as the primary representation of the resolution as a function of energy.

2.7.5. Acceptance-Window Evaluation

The curves referred to as *turn-on curves* in the preceding Lorenzetti-based studies are used here as a signal-only, truth-referenced acceptance-window diagnostic of the reconstructed transverse-energy response. They should not be interpreted as the efficiency curve of a production trigger with a single fixed threshold.

For method $m \in \{\mathrm{calib}, \mathrm{fast}\}$, define the event-wise transverse-energy residual

$$\varepsilon_m^{(i)} = E_{\mathrm{true}}^{(i)} - E_m^{(i)}. \tag{49}$$

For a positive tolerance $\Delta\lambda$, event i is counted as accepted when

$$\left|\varepsilon_m^{(i)}\right| < \Delta\lambda, \tag{50}$$

which is equivalent to requiring

$$E_{\mathrm{true}}^{(i)} - \Delta\lambda < E_m^{(i)} < E_{\mathrm{true}}^{(i)} + \Delta\lambda. \tag{51}$$

The denominator is the complete independent test sample, and exactly the same held-out events and the same particle-gun E_{true} spectrum are used for every compared method. The event acceptance rate is

$$\mathrm{EAR}_m(\Delta\lambda_s) = \frac{1}{N_{\mathrm{test}}} \sum_{i=1}^{N_{\mathrm{test}}} \mathbb{I}\left(|\varepsilon_m^{(i)}| < \Delta\lambda_s\right), \tag{52}$$

where $\mathbb{I}(\cdot)$ is the indicator function.

For each model–baseline comparison, the scan range is

$$\Delta\lambda_{\max} = 6 \max\left(\sigma_{\varepsilon_{\mathrm{calib}}}, \sigma_{\varepsilon_{\mathrm{fast}}}\right), \tag{53}$$

and $[0, \Delta\lambda_{\max}]$ is discretized into 1000 equally spaced points. For the curves reported in Fig. **??**, the resulting scan spacing is approximately 17.05 MeV. For an acceptance level q,

$$\Delta\lambda_{q,m} = \min\left\{\Delta\lambda_s : \mathrm{EAR}_m(\Delta\lambda_s) \geq q\right\}, \tag{54}$$

and this work reports $q = 0.90$. Thus, a smaller $\Delta\lambda_{90}$ means that 90% of the signal events are contained inside a narrower symmetric residual window around Monte-Carlo truth. Because the construction is signal-only, truth-centered, and bilateral, it is a residual-containment/threshold-response proxy rather than a complete trigger-efficiency or signal-versus-background performance measurement.

2.7.6. Statistical Validation

A statistical validation is performed to determine whether the observed improvements over the Standard FastCalo baseline are significant and not attributable to random fluctuations. Since the calibrated and uncalibrated estimates are obtained from the same physical events, all comparisons are treated as paired tests.

The paired error difference is defined as

$$d = \varepsilon_{\mathrm{calib}} - \varepsilon_{\mathrm{fast}}. \tag{55}$$

For the mean error, the directional hypothesis is

$$H_0 : \mu_d \geq 0 \quad \text{vs.} \quad H_1 : \mu_d < 0, \tag{56}$$

where μ_d is the mean of the paired differences. A large-sample paired z-test is used as the primary test, supported by a paired Student's t-test and a Wilcoxon signed-rank test as robustness checks [38, 39, 40].

For the error variance, the directional hypothesis is

$$H_0 : \frac{\sigma^2_{\text{calib}}}{\sigma^2_{\text{fast}}} \geq 1 \quad \text{vs.} \quad H_1 : \frac{\sigma^2_{\text{calib}}}{\sigma^2_{\text{fast}}} < 1. \tag{57}$$

The Pitman–Morgan test is used as the primary paired variance test, complemented by paired bootstrap confidence intervals for the variance ratio and by the classical F-test as a reference comparison [41, 42, 43, 38].

Normality is assessed using the D'Agostino–Pearson K^2 test and the Anderson–Darling test, complemented by Q–Q plots [44, 45, 46]. The Shapiro–Wilk test is not used for samples larger than $n = 5000$, due to its known limitations in very large datasets [46]. All tests are performed at a significance level of $\alpha = 0.05$. For each comparison, the test statistic, p-value, and one-sided 95% confidence interval are reported.

The predefined inferential comparisons in this work are restricted to each selected calibration model versus the Standard FastCalo baseline. No direct hypothesis test is used to establish superiority of the GNN over the full-ring DNN. In particular, the RMSE standard deviations reported in Table 4 are obtained from five disjoint subsets of the fixed independent test predictions; they quantify descriptive variability across test subsets and are not independent-training-run uncertainties. They should therefore not be used as the sampling distribution for a GNN–DNN architectural superiority test. The observed GNN–DNN differences in RMSE, IQR, and d_0 are reported as descriptive hold-out differences between the two trained pipelines. A statistically controlled architectural comparison would require a pre-specified direct paired analysis and, for training-procedure uncertainty, repeated independent model fitting or repeated nested selection. This distinction is maintained throughout the Results and Conclusions.

3. Results

This section presents the experimental evaluation of the proposed neural calibration strategies and the reference baseline models. The analysis first characterizes the simulated fast-energy response and the ring-based input representation. It then evaluates feature-selection and feature-extraction strategies, global calibration-error metrics, accuracy–efficiency trade-offs, bin-wise stability, energy resolution, acceptance-window behavior, statistical significance, and the impact of the pile-up-inclusive benchmark conditions.

Fig. 5a presents the distributions of the true transverse energy E_{true} and the Standard FastCalo transverse-energy estimate E_{fast}. Although the two histograms cover similar energy ranges, the scatter plot in Fig. 5b reveals systematic deviations from the ideal response $E_{\text{fast}} = E_{\text{true}}$. These deviations include changes in the mean response, asymmetric dispersion, and a visible step-like structure around 250 GeV.

The step-like structure around 250 GeV is part of the controlled fast-response benchmark used in this work. It emulates a localized non-smooth response distortion and tests whether the calibration models can correct nonlinear effects beyond smooth global trends. It is therefore used as a robustness stress test, rather than as a claim of reproducing a specific detector artifact.

The same controlled response structure and dataset were previously adopted in a Lorenzetti-based calibration study [13]. They are retained here to preserve a consistent and reproducible benchmark for assessing methodological developments under identical simulated response conditions. In the present work, this benchmark is extended by introducing Deep Neural Network and Graph Neural Network calibration models, fixed-cardinality compact representations, and a broader measurement-oriented evaluation protocol. The observed response structure should not be interpreted as a detector-specific artifact or as a complete model of realistic calorimeter-response imperfections. Instead, it defines a controlled nonlinear and non-smooth stress test under which all calibration models and the Standard FastCalo baseline are evaluated. Consequently, the reported improvements should be interpreted as relative performance gains within this controlled benchmark. A dedicated comparison using otherwise identical simulations with and without this response structure would be required to isolate its effect explicitly and is left for future work.

Further analysis in Fig. 6a compares three event-level quantities derived from E_{true} and E_{fast}: the signed residual $E_{\text{true}} - E_{\text{fast}}$, the relative residual $(E_{\text{true}} - E_{\text{fast}})/E_{\text{true}}$, and the calibration factor $\alpha_{\text{true}} = E_{\text{true}}/E_{\text{fast}}$. For visualization, the distributions are standardized using the z-score, defined as the number of standard deviations by which a value differs from its mean [38].

Fig. 6b shows that the signed residual $E_{\text{true}} - E_{\text{fast}}$ provides the most stable error variable among the evaluated formulations. The relative residual and the energy-ratio variables exhibit stronger skewness and larger deviations, especially in regions where the denominator is small. The signed residual is therefore adopted as the primary event-level error variable for the calibration-error analysis. However, the residual distributions still exhibit non-Gaussian behavior, as later confirmed by the normality tests and Q–Q plots. For this reason, the statistical validation does not rely exclusively on Gaussian assumptions, but combines parametric reference tests with paired non-parametric and bootstrap-based procedures.

The calibration factor α_{true}, crucial for ML methods estimating α_{calib}, is examined in Fig. 7. The boxplots across different E_T and $|\eta|$ bins reveal that estimated calibration factors should ideally match α_{true} to minimize calibration errors ($\varepsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}}$).

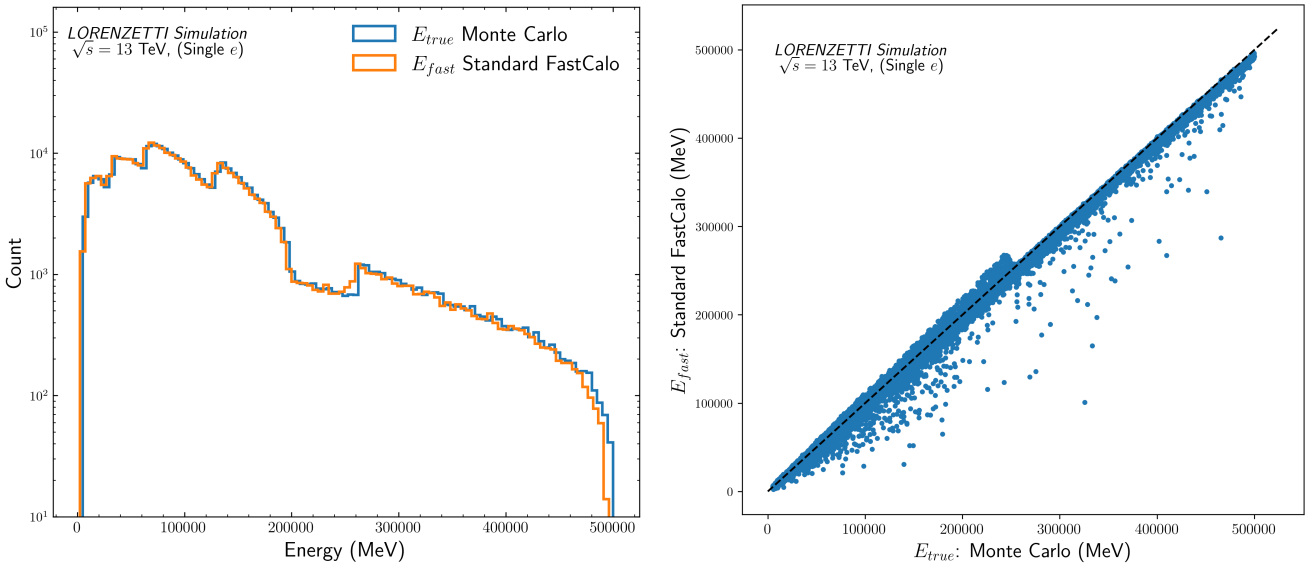


Figure 5: Comparison of true and fast reconstructed energies: (a) energy histogram comparing the true transverse energy E_{true} and the Standard FastCalo energy E_{fast} generated with the Lorenzetti simulator over the 0–500 GeV range; (b) scatter plot of E_{true} versus E_{fast}. The scatter plot shows an overall correlation between the two quantities, but also exhibits a visible step-like nonlinear distortion around 250 GeV, together with changes in the mean response and variance at higher energies. This step-like structure is part of the controlled benchmark response used to evaluate the robustness of the proposed calibration methods against nonlinear and non-smooth fast-response effects.

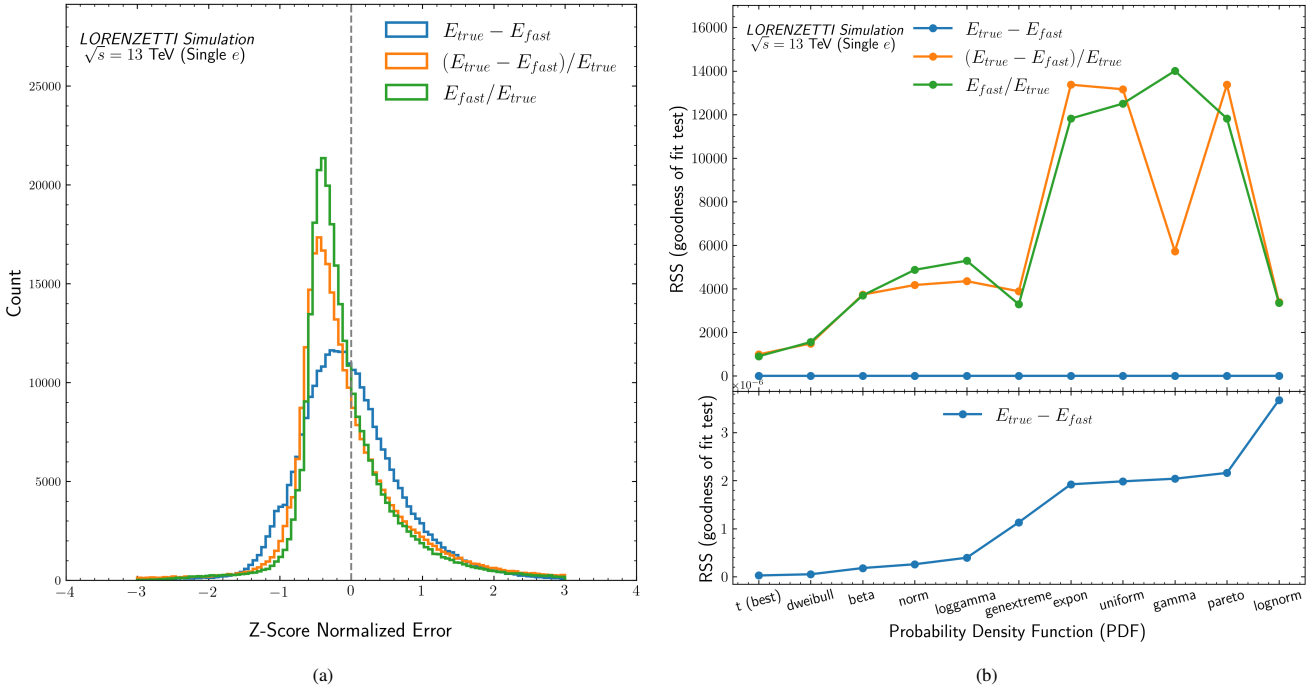


Figure 6: Comparison of event-level error formulations derived from E_{true} and E_{fast}: (a) z-score-standardized distributions of the signed residual $E_{\text{true}} - E_{\text{fast}}$, the relative residual $(E_{\text{true}} - E_{\text{fast}})/E_{\text{true}}$, and the energy ratio $E_{\text{fast}}/E_{\text{true}}$; (b) residual-sum-of-squares goodness-of-fit comparison for the same error formulations. Among the evaluated quantities, the signed residual $E_{\text{true}} - E_{\text{fast}}$ provides the most stable event-level error variable for the subsequent calibration-error analysis. Nevertheless, the residual distributions remain non-Gaussian, motivating the use of complementary paired non-parametric and bootstrap-based statistical procedures in addition to parametric reference tests.

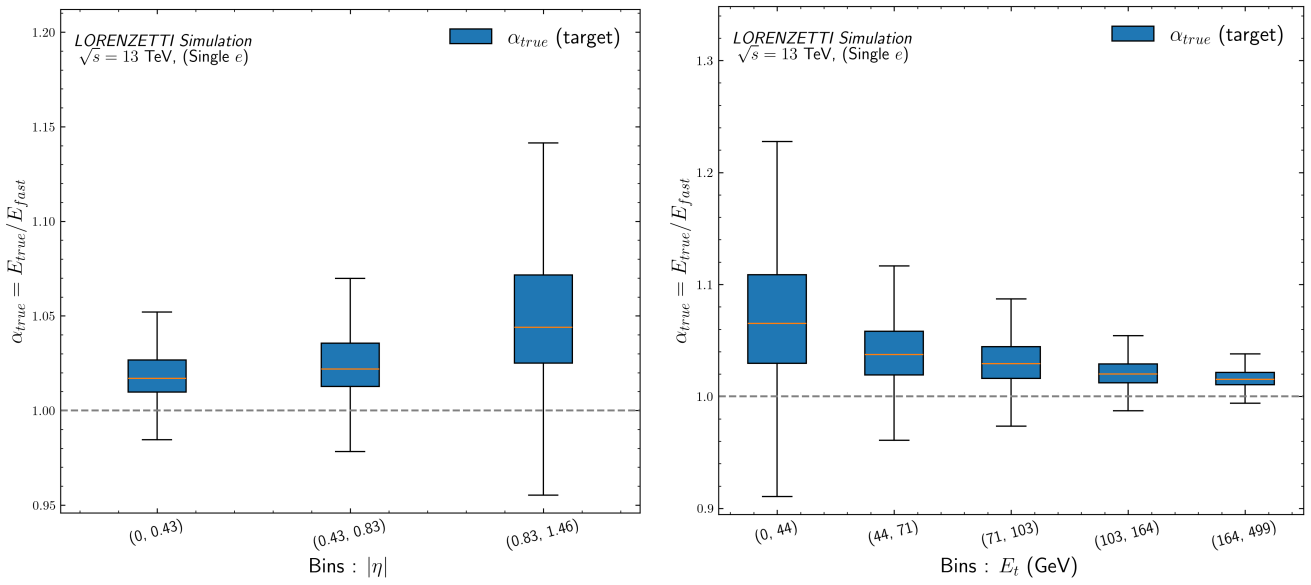


Figure 7: Boxplots of the true calibration factor α_{true} across (a) the pseudorapidity $|\eta|$, and (b) transverse energy E_T bins. The boxplots illustrate the distribution of calibration factors within each bin, where the central line represents the median, the box denotes the interquartile range (IQR), and the whiskers extend to 1.5 times the IQR from the lower and upper quartiles. This visualization highlights the uncertainty and variability of the calibration factors per bin, which serve as the target for the ML calibration methods.

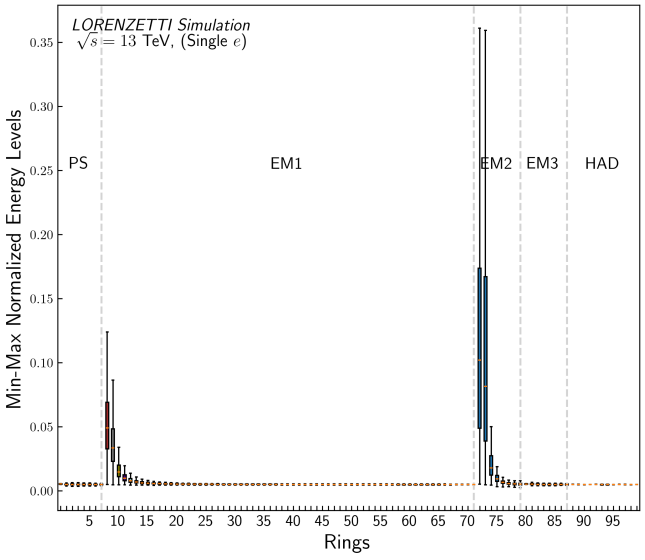


Figure 8: Boxplot of visualization-scaled ring-energy levels across the 100 concentric rings defined by the *Ringer* algorithm for 300,000 simulated single-electron events. Each box represents the distribution of one ring variable across all events after applying a min–max scaling for visualization purposes only. Vertical dashed lines delineate the approximate positions of the calorimeter layers: presampler (PS), Electromagnetic Layers (EM1–EM3), and Hadronic Layers (HAD). The concentration of energy in EM2 is consistent with typical electromagnetic shower development.

Fig. 8 presents the distribution of ring-energy depositions across the 100 concentric rings generated by the *Ringer* algorithm for the full set of simulated events. For visualization only, the ring values shown in this figure are min–max scaled before plotting. This graphical scaling is distinct from the ring normalization defined in Eq. 5, which is the representation used by the calibration models.

The vertical dashed lines indicate the approximate boundaries corresponding to the different longitudinal layers of the calorimeter: the Pre-Sampler (PS), the Electromagnetic layers (EM1, EM2, EM3), and the Hadronic layers (HAD). As expected for electromagnetic showers initiated by electrons, the largest energy concentrations occur in the EM2 region, followed by smaller but non-negligible contributions in the PS and EM1 regions. Minimal energy is deposited in the EM3 and hadronic layers, consistent with the containment of the electromagnetic shower within the electromagnetic calorimeter.

This visualization highlights the characteristic energy deposition pattern across calorimeter layers, and illustrates how the *Ringer* algorithm captures radial information effectively in a compact 100-dimensional feature space. These transformed features serve as the foundation for the subsequent analysis, where we evaluate various methods of feature selection and extraction to identify the most informative representations for energy calibration.

3.1. Feature Selection and Extraction Evaluation

Feature-selection and feature-extraction methods are analyzed through the heatmaps shown in Fig. 9. These maps

summarize the target relevance of the variables or transformed components used by CORR, MUTI, PCA, and ICA. For CORR and MUTI, the rows correspond to original ring variables, preserving their direct association with calorimeter regions. For PCA and ICA, the rows correspond to transformed latent components, which do not have a one-to-one correspondence with physical ring positions. In all cases, the relevance values are used only to define the ranked compact representations described in Section 2.4.

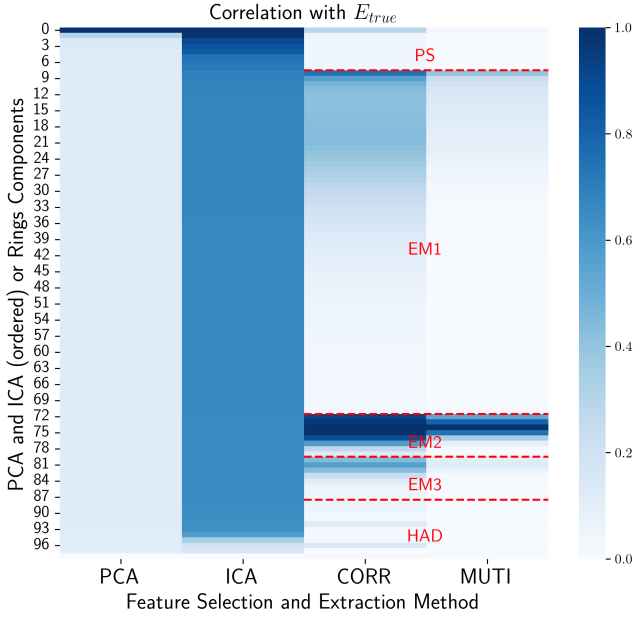


Figure 9: Heatmap of the relationship between the 100 input variables or transformed components and the true energy E_{true} for four feature-selection or feature-extraction strategies: PCA, ICA, CORR, and MUTI. Each column corresponds to one method, and each row to a variable or component index. Darker blue shades indicate stronger target relevance. The vertical red dashed lines indicate the approximate calorimeter-region boundaries, which are physically meaningful only for CORR and MUTI because they operate directly on the original ring energies. PCA and ICA components are transformed latent variables and therefore do not map directly to specific calorimeter regions.

Table 3 summarizes the sets of features selected or extracted by each dimensionality reduction method evaluated in this study. The first row (`Rings[1:100]`) represents the original input space, composed of all 100 normalized ring energies generated by the Lorenzetti Showers framework. The second row, `Rings/2`, denotes an expert-guided 50-ring subset selected from the original 100-ring representation. This subset was defined by inspecting the visualization-scaled ring-energy profiles and retaining rings associated with calorimeter regions exhibiting larger energy depositions. It is included as a physically motivated compact reference, distinct from the data-driven CORR, MUTI, PCA, and ICA strategies.

Table 3
Selected/extracted features and number of features (F) by correlation, mutual information, PCA and ICA approaches.

Selection/Extraction	Features	F
Rings[1:100]	Rings[1:100]	100
Rings/2	Rings[15, 9:40, 73:76, 81:84, 89:94]	50
CORR-50	Rings[1, 9:42, 73:85, 93, 97]	50
CORR-10	Rings[9:11, 73:78, 82]	10
CORR-5	Rings[73:77]	5
CORR-2	Rings[75, 76]	2
MUTI-50	Rings[1, 9:39, 71:85, 89, 93, 97, 98]	50
MUTI-10	Rings[9:13, 73:77]	10
MUTI-5	Rings[9, 73:76]	5
MUTI-2	Rings[74, 75]	2
PCA-50	PCA[1:50]	50
PCA-10	PCA[1:10]	10
PCA-5	PCA[1:5]	5
PCA-2	PCA[1:2]	2
ICA-50	ICA[1:50]	50
ICA-10	ICA[1:10]	10
ICA-5	ICA[1:5]	5
ICA-2	ICA[1:2]	2

The subsequent rows summarize the feature subsets selected by the correlation-based (CORR) and mutual information based (MUTI) strategies at different selection levels. For instance, `CORR-50` retains the 50 rings exhibiting the largest absolute Pearson correlation coefficients with the true energy. As the number of retained features is reduced (e.g., `CORR-2`), the input representation becomes increasingly compact, retaining only the two most strongly correlated ring components (e.g., `Rings[75, 76]`).

Similarly, the `MUTI`-based method selects features according to their mutual information with the true energy. As the retained cardinality m decreases, the representation becomes increasingly compact, with `MUTI-2` retaining only the two most informative rings, `Rings[74,75]`. It is noteworthy that the strong correlation and mutual information observed for Rings 72–78 can be attributed to their spatial arrangement within the calorimeter layers. These rings are positioned close to each other, leading to similar responses to incident particles due to shared energy deposition characteristics. This spatial relationship increases the likelihood of correlated measurements, as they capture similar energy distributions from particle showers.

For the PCA and ICA methods, which perform feature extraction rather than direct variable selection, the table lists the top $m \in \{50, 10, 5, 2\}$ transformed components after E_{true}-relevance ordering. For example, `PCA-5` retains the five principal components with the largest absolute correlation with E_{true}, rather than necessarily the five components with the largest explained variance. The same ranking principle is applied to ICA components. This target-relevance ordering is fitted only within the training partition, as described in Section 2.4.

Table 3 highlights how different dimensionality-reduction methods reduce the feature space from 100 original inputs to as few as two features or components. This compactness is relevant for inference-oriented studies because it can

reduce input dimensionality, memory footprint, and feature-transfer requirements, although the final prediction time also depends on the regression model and implementation.

It is also important to interpret the compact PCA and ICA-based representations in the context of the simulated detector conditions. These representations are not used here as explicit pile-up-suppression mechanisms, but as dimensionality-reduction strategies for the ring-based input space. This reduction may be useful in high-throughput calorimetric systems because storing or processing a small number of latent components can reduce storage requirements and inference cost when compared with the full 100-ring representation. Accordingly, the compact representations should be interpreted as preserving part of the calibration-relevant shower information under the simulated conditions considered here, rather than as dedicated pile-up-mitigation methods.

3.2. Global Energy-Error Evaluation

This section evaluates the event-wise calibration error obtained with full-ring and compact ring-based representations. Deep Neural Network models are applied to both complete and reduced input spaces, whereas the Graph Neural Network model is evaluated on the complete 100-ring representation. Tree-based and hybrid tree-based regressors are retained as reference baselines to contextualize the neural results under the same benchmark and evaluation protocol. The analysis is based on the calibration error

$$\varepsilon_{\mathrm{calib}} = E_{\mathrm{true}} - E_{\mathrm{calib}}, \tag{58}$$

and compares how each representation–model combination reduces both the average error magnitude and the central dispersion of the residual distribution.

Fig. 10 shows the calibration-error distributions and the corresponding boxplots for the evaluated feature-selection and feature-extraction strategies. The histograms provide a global view of the residual distributions, while the boxplots summarize their central tendency and dispersion. In each boxplot, the central line represents the median, the box width represents the interquartile range, and the black marker indicates the RMSE.

Across all representation families, the Standard FastCalo baseline exhibits the broadest residual distributions and the largest deviations from the zero-error region. The ML-based calibrators reduce these residuals, but the magnitude of the improvement depends jointly on the input representation and on the regression model. This behavior highlights the role of the representation stage as part of the overall information-fusion pipeline.

For correlation-based feature selection, Fig. 10a shows that both hybrid tree-based and neural models benefit from retaining calibration-relevant ring variables. The strongest result within the `CORR` group is obtained by `RFBDT no-Bin: CORR-50`, which reduces the IQR by 36.18% and the RMSE by 53.40% relative to Standard FastCalo. However, the compact neural configuration `DNN no-Bin: CORR-10` provides a more relevant accuracy–compactness compromise for the present neural-fusion study, reducing the IQR by 28.83% and the RMSE by 51.96% while using only 10 selected ring variables.

For mutual-information-based selection, Fig. 10b shows that the highest accuracy in this family is achieved by `DNN no-Bin: MUTI-50`, with reductions of 33.33% in IQR and 59.95% in RMSE. The compact `DNN no-Bin: MUTI-10` configuration also remains effective, reducing the IQR by 26.42% and the RMSE by 48.11%. These results indicate that nonlinear relevance criteria can identify informative subsets of ring variables, although the best performance in this family still benefits from retaining a larger selected feature set.

For PCA-based feature extraction, Fig. 10c shows that `DNN no-Bin: PCA-50` provides the best result within the PCA group, reducing the IQR by 28.46% and the RMSE by 54.41%. The ten-component configuration, `DNN no-Bin: PCA-10`, remains close in performance, with reductions of 27.64% in IQR and 51.89% in RMSE. This suggests that a large fraction of the calibration-relevant information captured by PCA is concentrated in a compact transformed representation.

For ICA-based feature extraction, Fig. 10d shows the strongest reduced-representation performance among the evaluated feature-extraction strategies. The `DNN no-Bin: ICA-50` configuration reduces the IQR by 42.28% and the RMSE by 58.94%, while the compact `DNN no-Bin: ICA-10` configuration still reduces the IQR by 28.61% and the RMSE by 55.76%. The performance retained by `ICA-10` indicates that compact independent-component representations can preserve substantial calibration information for nonlinear neural fusion.

Overall, the reduced-representation results show that Deep Neural Network calibrators can exploit both selected ring variables and transformed latent components. The strongest reduced configurations are obtained with `DNN no-Bin: MUTI-50`, `DNN no-Bin: ICA-50`, and the compact `DNN no-Bin: ICA-10` model. Tree-based and hybrid tree-based regressors remain competitive reference baselines, particularly in the correlation-based scenario, but the main trend relevant to this work is that neural models preserve substantial calibration performance even under strong input compression. This motivates the subsequent ranking based on d_0 and D_0, where predictive accuracy, residual dispersion, input dimensionality, and prediction time are considered jointly.

3.2.1. Global Ranking and Accuracy–Efficiency Trade-Off

To compare the calibration strategies under both predictive and efficiency-oriented criteria, all evaluated configurations were ranked using the two composite metrics defined in Section 2.7. The d_0 metric combines RMSE and IQR, and therefore emphasizes calibration accuracy and residual dispersion. The D_0 metric additionally includes the number of input features F and the amortized trained-model prediction time per event T, providing an equal-weight summary of accuracy, compactness, and CPU-based inference cost.

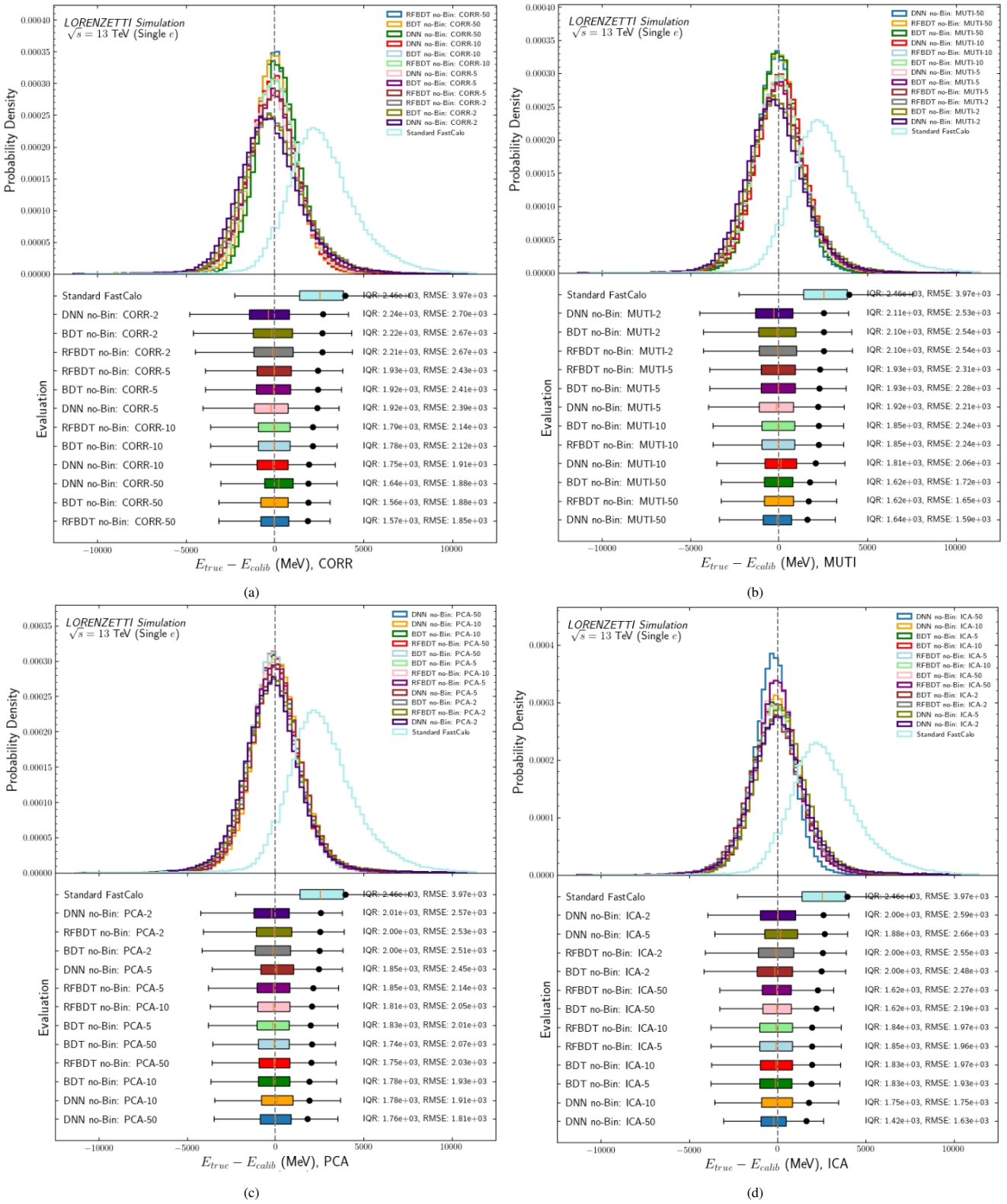


Figure 10: Calibration-error distributions for $E_{\text{true}} - E_{\text{calib}}$ obtained with different feature-selection and feature-extraction strategies: (a) correlation-based selection (CORR), (b) mutual-information-based selection (MUTI), (c) principal component analysis (PCA), and (d) independent component analysis (ICA). The upper panels show the error histograms, while the lower panels show the corresponding boxplots. In each boxplot, the box represents the interquartile range, the central line indicates the median, and the black marker denotes the RMSE.

Table 4
Performance summary of the four top-ranked calibration configurations in terms of interquartile range (IQR), root mean squared error (RMSE), amortized prediction time per event T, number of external ring-based input features F, and the composite metrics d_0 and D_0. Results were obtained from the independent test set, corresponding to 30% of the full dataset. Standard FastCalo (Std) is used as the reference baseline for all relative comparisons. IQR and RMSE values are reported as mean and standard deviation obtained from a five-fold resampling procedure on the test set. The prediction time T is computed as the total wall-clock prediction time for the test batch divided by the number of events and is reported in milliseconds per event.

Method	IQR (MeV)	RMSE (MeV)	IQR (% Std)	RMSE (% Std)	T (ms/event)	F	d_0 (MeV)	D_0
Standard FastCalo	2457.11 $\pm$ 9.89	3967.12 $\pm$ 6.74	-	-	0.00e+00	1	4666.42	1.0
DNN no-Bin: CORR-10	1748.78 $\pm$ 6.95	1905.63 $\pm$ 6.31	28.83% $\pm$ 0.40%	51.96% $\pm$ 0.18%	1.38e-03	10	2586.43	0.25
DNN no-Bin: ICA-10	1754.19 $\pm$ 6.81	1754.95 $\pm$ 5.82	28.61% $\pm$ 0.40%	55.76% $\pm$ 0.16%	1.52e-03	10	2481.34	0.21
DNN no-Bin: Rings[1:100]	1419.48 $\pm$ 5.73	1439.35 $\pm$ 4.75	42.23% $\pm$ 0.33%	63.72% $\pm$ 0.13%	4.05e-03	100	2021.55	1.0
GNN no-Bin: Rings[1:100]	1404.90 $\pm$ 5.28	1378.77 $\pm$ 4.57	42.82% $\pm$ 0.32%	65.25% $\pm$ 0.13%	1.53e-01	100	1968.44	1.17

The comparison includes the full-ring representation, `Rings[1:100]`, and compact representations obtained through `CORR`, `MUTI`, `PCA`, and `ICA`. This design distinguishes configurations that maximize absolute calibration performance from those that provide more favorable accuracy–efficiency compromises under the benchmark conditions considered here.

Fig. 11 summarizes the top-ranked configurations under both criteria. Panels (a) and (b) show the 10 best configurations according to d_0 and D_0, respectively, expressed as percentage improvements relative to Standard FastCalo. Panels (c) and (d) show the corresponding calibration-error distributions and boxplots. In these boxplots, the IQR represents the central residual dispersion, while the black markers indicate the RMSE.

Table 4 reports four representative configurations that summarize the two main regimes identified by the ranking analysis. The full-ring models, `GNN no-Bin: Rings[1:100]` and `DNN no-Bin: Rings[1:100]`, achieve the best absolute calibration accuracy and therefore dominate the d_0-based ranking. In contrast, the compact configurations, `DNN no-Bin: ICA-10` and `DNN no-Bin: CORR-10`, achieve the most favorable D_0 values because they preserve competitive calibration performance while using substantially fewer input features.

Relative to Standard FastCalo, all four ML-based configurations substantially reduce both IQR and RMSE. The numerically best d_0 value is obtained by `GNN no-Bin: Rings[1:100]`, which reduces d_0 from 4666.42 MeV to 1968.44 MeV, corresponding to a 57.82% reduction. This configuration also reduces the IQR by 42.82% and the RMSE by 65.25%. The full-ring DNN provides closely comparable accuracy, with reductions of 42.23% in IQR and 63.72% in RMSE.

The difference between the two full-ring neural pipelines is modest: the fixed hold-out predictions yield an RMSE difference of 60.58 MeV, an IQR difference of 14.58 MeV, and a d_0 difference of 53.11 MeV in favor of the GNN configuration. These are descriptive differences for the selected trained models, not inferential evidence that the GNN architecture is statistically superior to the DNN. The predefined statistical protocol compares each selected calibrator with Standard FastCalo and does not include a direct GNN–DNN superiority test or repeated independent neural-network training runs.

The compact configurations exhibit a different but practically relevant behavior. Although their absolute error reductions are smaller than those of the full-ring models, `DNN no-Bin: ICA-10` and `DNN no-Bin: CORR-10` still reduce the RMSE by more than 51% relative to Standard FastCalo while using only 10 input components. This indicates that a substantial fraction of the calibration-relevant information can be retained in a low-dimensional representation. As a result, these configurations are favored by the D_0 criterion, which includes input dimensionality and trained-model prediction time.

The reported prediction times correspond to the trained-model prediction stage under the CPU-based benchmark described in Section 2.7.2. They were computed as amortized per-event times by dividing the total wall-clock prediction time of the test batch by the number of events. Thus, they provide a consistent relative throughput comparison among the evaluated trained models within the same computational environment. They should not be interpreted as direct estimates of single-event production-trigger latency, which would also depend on feature construction, memory access, numerical representation, batching policy, parallelization strategy, and target hardware.

The systematic deviations observed in the Standard FastCalo baseline are consistent with the nonlinear response patterns shown in Fig. 5b and with the broader residual distributions reported in Figs. 10a–10d. These deviations arise from the combined effects of detector response, electronic noise, pile-up, and limitations of the fast reconstruction chain. The ML-based calibrators reduce these effects by learning event-level correction factors from the ring-based shower topology, thereby bringing E_{calib} closer to E_{true} and narrowing the residual distributions.

The ranking results also support the use of compact dimensionality-reduced representations. In particular, `ICA-10` and `CORR-10` remain competitive despite their reduced dimensionality, suggesting that the corresponding transformations concentrate substantial calibration-relevant information into a small number of inputs. This behavior is relevant for future trigger-oriented studies, where memory footprint, data movement, and prediction cost may become limiting factors.

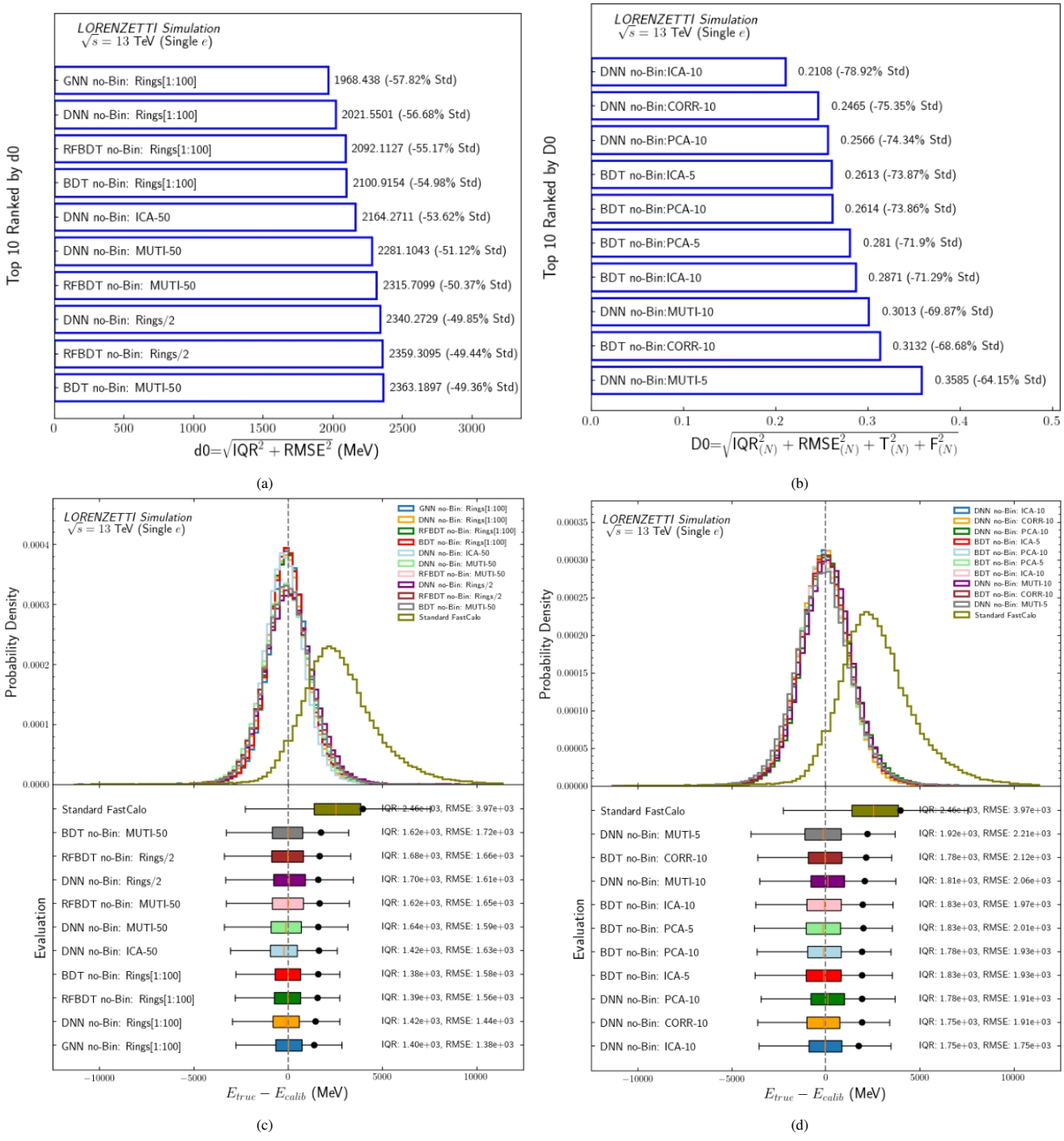
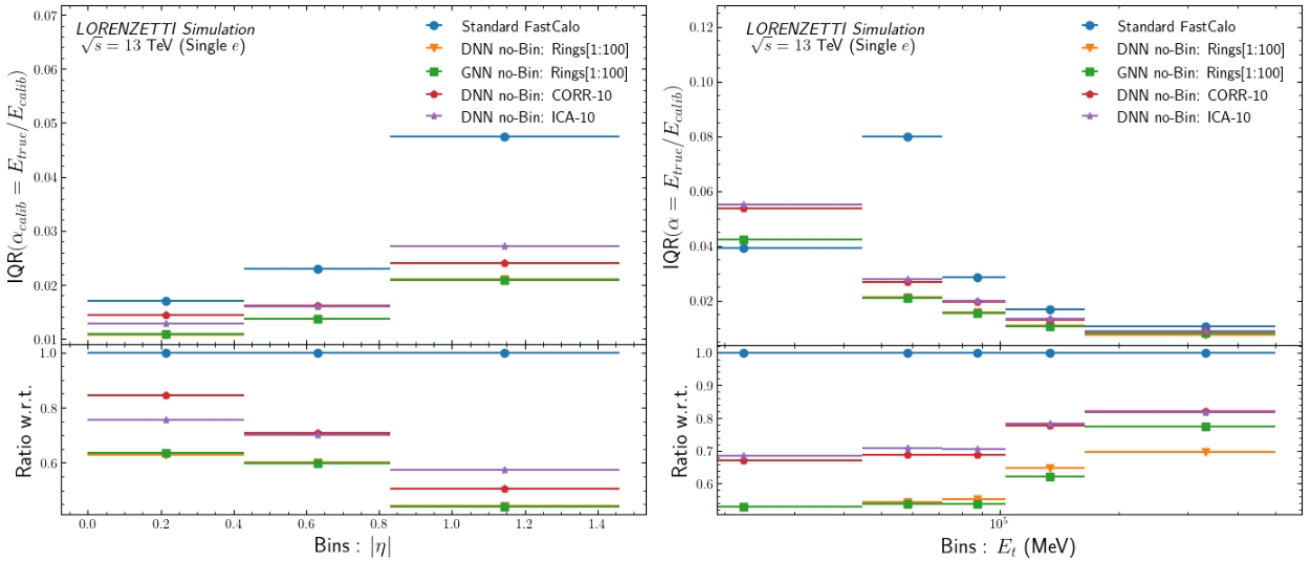


Figure 11: Ranked-method comparison according to the d_0 and D_0 metrics. Panels (a) and (b) show the top-10 configurations ranked by d_0 and D_0, respectively, with percentage improvements relative to Standard FastCalo. Panels (c) and (d) show the corresponding distributions of the calibration error $E_{\text{true}} - E_{\text{calib}}$ and the associated boxplots. The IQR quantifies the central dispersion of the residuals, while the black markers indicate the RMSE.

All four selected neural configurations exhibit amortized trained-model prediction times below 0.2 ms per event on the CPU-based benchmark, but their computational profiles differ substantially. The GNN requires 0.153 ms/event compared with 0.00405 ms/event for the full-ring DNN, while its RMSE is 4.21% lower than that DNN. Thus, the GNN favors the lowest observed absolute error in this benchmark, whereas the full-ring DNN and especially the ten-component DNN configurations offer substantially more favorable throughput. Because the GNN and DNN differ in internal feature engineering and capacity, this comparison does not isolate the benefit of graph modeling itself.



(a) (b)

Figure 12: Bin-wise comparison of the interquartile range of the calibration factor, IQR(α), for the top-ranked calibration methods across (a) $|\eta|$ and (b) transverse-energy E_T bins. The upper panels show the absolute IQR values, while the lower panels show the values relative to the Standard FastCalo reference. In the relative representation, the Standard FastCalo reference is fixed at 1.0, and values below this baseline indicate reduced dispersion of the calibration-factor distribution.

3.2.2. Bin-Wise Calibration-Factor Evaluation

The bin-wise analysis provides a diagnostic evaluation of the regional stability of the globally trained calibration models. Unlike the global RMSE, IQR, and energy-resolution analyses, which are computed from the calibrated energy, this study focuses on the predicted calibration factor itself. For this purpose, the calibration factors predicted by the top-ranked configurations are compared with the reference factor

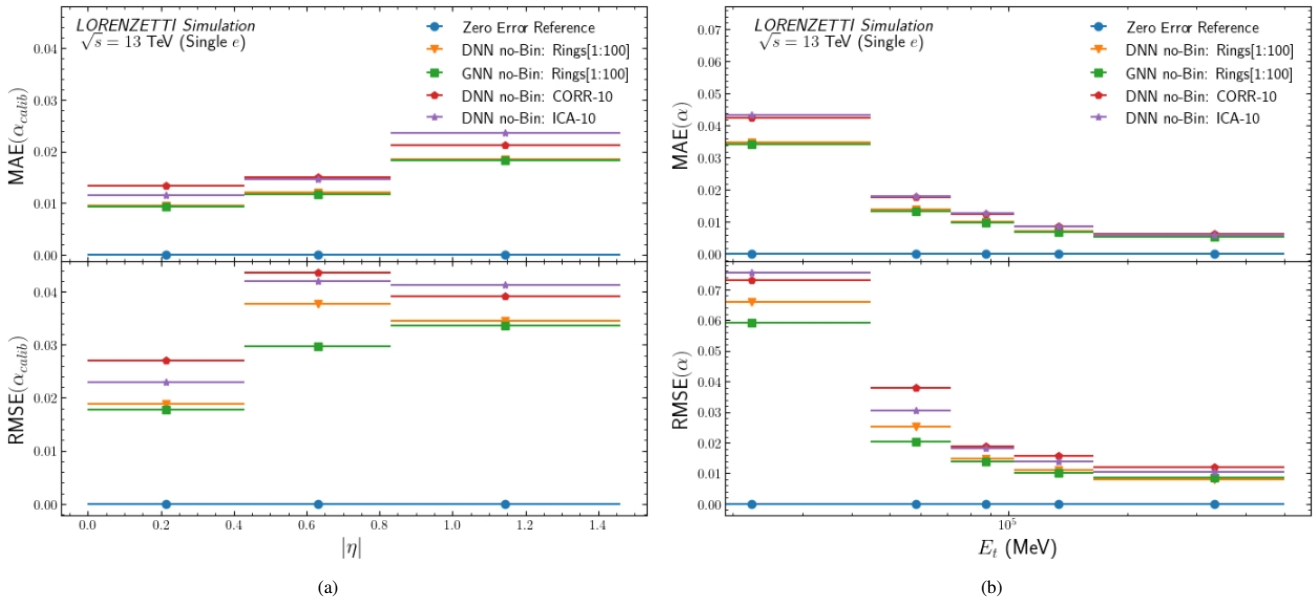
$$\alpha_{\text{true}} = \frac{E_{\text{true}}}{E_{\text{fast}}}, \tag{59}$$

which represents the event-wise multiplicative correction that would exactly transform the Standard FastCalo estimate into the true transverse energy. The comparison is performed in the $(|\eta|, E_T)$ regions defined previously, using the IQR of the predicted calibration-factor distributions and the MAE and RMSE between α_{calib} and α_{true}. This factor-level analysis is intended to assess regional consistency, while the practical calibration performance is evaluated primarily through the energy-level metrics reported in the previous sections.

Fig. 12 shows the IQR of the calibration-factor distributions as a function of $|\eta|$ and E_T. The dispersion generally increases with $|\eta|$, indicating larger variability of the required correction away from the central detector region. Conversely, the dispersion decreases as E_T increases, reflecting the reduced relative impact of fluctuations at higher transverse energies. Across the evaluated regions, the ML-based methods follow the same broad trend as the reference calibration factor, indicating that the globally trained models preserve the main regional dependence of the correction.

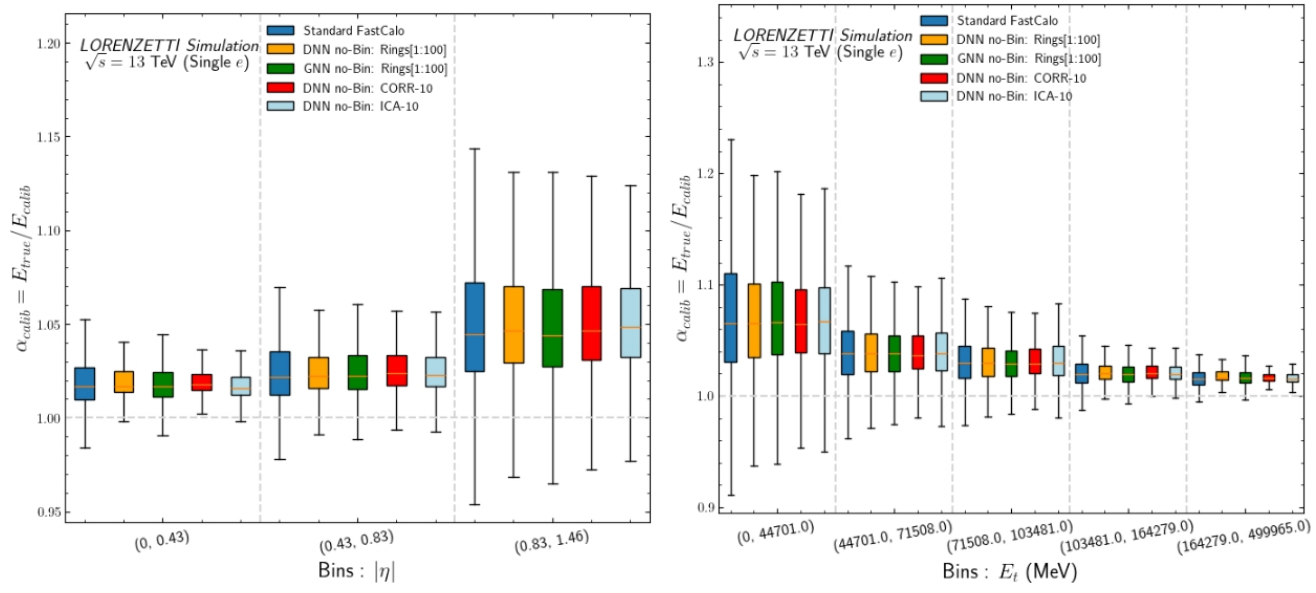
In the lowest transverse-energy interval, Standard FastCalo may appear locally more stable than some ML-based configurations in terms of IQR(α). This does not contradict the global calibration results, because Fig. 12 evaluates the dispersion of the multiplicative calibration factor, not the direct calibrated-energy error. At low transverse energy, the ratio $\alpha_{\text{true}} = E_{\text{true}}/E_{\text{fast}}$ becomes more sensitive to small absolute fluctuations in E_{fast}, since E_{fast} appears in the denominator. According to standard uncertainty propagation for quotient-based measurements, the relative uncertainty of a ratio is directly affected by the relative uncertainty of its denominator [47]. This effect is amplified in calorimetric measurements at low energy, where stochastic and noise terms contribute more strongly to the relative resolution [2, 3]. Therefore, estimating multiplicative correction factors is intrinsically more challenging in the lowest E_T bin, and a limited local advantage of the baseline in IQR(α) remains compatible with the overall energy-level improvement achieved by the ML-based calibrators.

Fig. 13 complements this analysis by reporting the MAE and RMSE of the predicted calibration factors with respect to α_{true}. The same broad regional behavior is observed: the factor-level errors tend to increase with $|\eta|$ and decrease with increasing E_T. This indicates that the calibration-factor estimation problem is more challenging in regions of larger pseudorapidity and lower transverse energy, where detector-response variations and relative fluctuations are more pronounced. Nevertheless, the selected ML configurations remain stable across the evaluated bins, suggesting that the globally trained models do not introduce strong localized instabilities in the correction factor.



(a) (b)

Figure 13: Bin-wise comparison of the MAE and RMSE of the predicted calibration factors with respect to the target correction factor $\alpha_{\text{true}} = E_{\text{true}}/E_{\text{fast}}$, evaluated across (a) $|\eta|$ and (b) E_T bins. The Zero Error Reference corresponds to the ideal case $\alpha_{\text{calib}} = \alpha_{\text{true}}$ and is included only as a visual reference. These metrics quantify factor-level prediction errors and should be interpreted as regional stability diagnostics complementary to the energy-level performance metrics.



(a) (b)

Figure 14: Boxplots of the predicted calibration factors α_{calib} for the top-ranked methods across (a) $|\eta|$ and (b) E_T bins. The box width represents the interquartile range, and the dashed vertical lines indicate the bin boundaries.

The distributions of the predicted calibration factors are further summarized in Fig. 14. The boxplots confirm the trends observed in the IQR, MAE, and RMSE analyses. The dispersion of α_{calib} generally increases with $|\eta|$ and decreases with increasing E_T. Importantly, the medians of the ML-predicted calibration factors remain aligned with the expected regional behavior, indicating that the globally trained models preserve the main dependence of the correction factor across the evaluated detector and energy regions.

These results support the use of global, non-binned training in the present benchmark. Although bin-specific training may improve local specialization, it can also introduce boundary discontinuities and reduce statistical robustness in sparsely populated regions. By contrast, the global models learn a continuous calibration function over the full feature

space, while the bin-wise evaluation provides an a posteriori diagnostic of regional stability. The bins are therefore not used to train independent models, but to verify whether the globally trained calibrators preserve consistent correction behavior across detector and energy regions.

A more complete evaluation of segmentation-aware calibration is left for future work. Such an analysis would require larger simulated samples with more homogeneous populations across the $(|\eta|, E_T)$ regions, allowing local or region-adaptive models to be trained without excessive statistical fragmentation. This would clarify whether segmentation-aware calibration can provide additional gains beyond the globally trained strategies evaluated here.

3.3. Energy Resolution

Energy resolution provides an energy-dependent assessment of the precision of the calibrated response and complements the global RMSE and IQR analyses. While the global metrics summarize the overall agreement between E_{calib} and E_{true}, the resolution curve shows how the dispersion of the reconstructed response evolves across the transverse-energy range. This is particularly relevant for calorimetric measurements, where the relative impact of noise, shower fluctuations, and reconstruction nonlinearities may vary with energy.

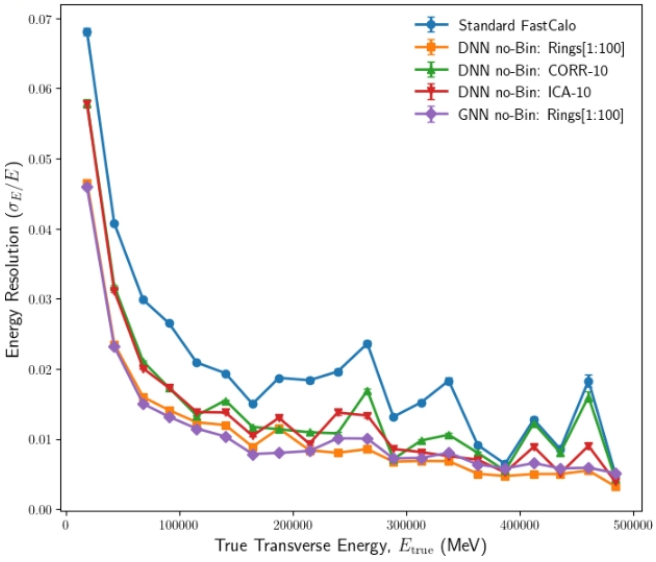


Figure 15: Relative energy resolution, expressed as σ_E/E, as a function of the true transverse energy E_{true} for the Standard FastCalo baseline and the four top-ranked ML-based calibration methods. The curves show the energy-dependent dispersion of the calibrated response over the range considered in this study. The ML-based methods generally provide improved resolution with respect to Standard FastCalo, with the largest gains observed in the low- and intermediate-energy regions.

Fig. 15 shows the relative energy resolution as a function of E_{true} for the Standard FastCalo baseline and the four selected ML-based calibration methods. The ML-based configurations generally reduce the resolution values relative to Standard FastCalo over the analyzed spectrum, indicating that the global reductions in RMSE and IQR also translate into improved energy-dependent precision. The largest gains are observed in the low- and intermediate-energy regions, where relative fluctuations and reconstruction nonlinearities have a stronger impact on the measured response. At higher energies, the differences among the calibrated methods become less pronounced, and local fluctuations are more visible.

Among the evaluated methods, the two full-ring neural configurations, `GNN no-Bin: Rings[1:100]` and `DNN no-Bin: Rings[1:100]`, provide the lowest resolution values over most of the energy range. The GNN configuration shows the most favorable behavior in much of the low- and intermediate-energy region, whereas the full-ring DNN becomes comparable, and in some intervals slightly lower, at higher energies. This indicates that preserving the complete ring representation is beneficial for energy-resolution performance, while the relative ordering between the two full-ring neural models may depend on the energy interval.

The compact configurations, `DNN no-Bin: ICA-10` and `DNN no-Bin: CORR-10`, also improve the resolution relative to Standard FastCalo while using only 10 input components. Their performance indicates that low-dimensional representations retain a substantial fraction of the calibration-relevant information contained in the full ring profile. Larger local fluctuations are observed in some higher-energy intervals, particularly for `DNN no-Bin: CORR-10`, which is compatible with the reduced input dimensionality and the smaller event populations typically available in the upper energy range.

Overall, the energy-resolution curves show that the ML-based calibrators reduce the relative-response width over most of the simulated energy range. The whole-sample summary reported in Table 5 is the dimensionless quantity $\left(\sigma_E/E\right)_{\text{global}}$ of Eq. 47, computed directly from the complete independent test sample rather than by averaging the energy-bin values. Its 51.46% reduction for the GNN should not be expected to equal the 42.82% reduction in global IQR. The energy-resolution quantity is the standard deviation of the dimensionless relative response E_m/E_{true}, whereas the global IQR used in the ranking is computed from the absolute energy residual in MeV and characterizes only the central 50% of that distribution. The two metrics therefore quantify different and complementary properties of the calibrated response.

3.4. Truth-Referenced Acceptance-Window Curves

The truth-referenced acceptance-window curves evaluate how tightly the reconstructed transverse-energy estimates are contained around E_{true} as the residual-window half-width $\Delta\lambda$ is varied. While RMSE, IQR, and energy resolution quantify complementary aspects of reconstruction accuracy and dispersion, the acceptance-window analysis measures the fraction of the common independent test events satisfying the bilateral residual condition defined in Section 2.7.5. It is therefore a signal-only residual-containment

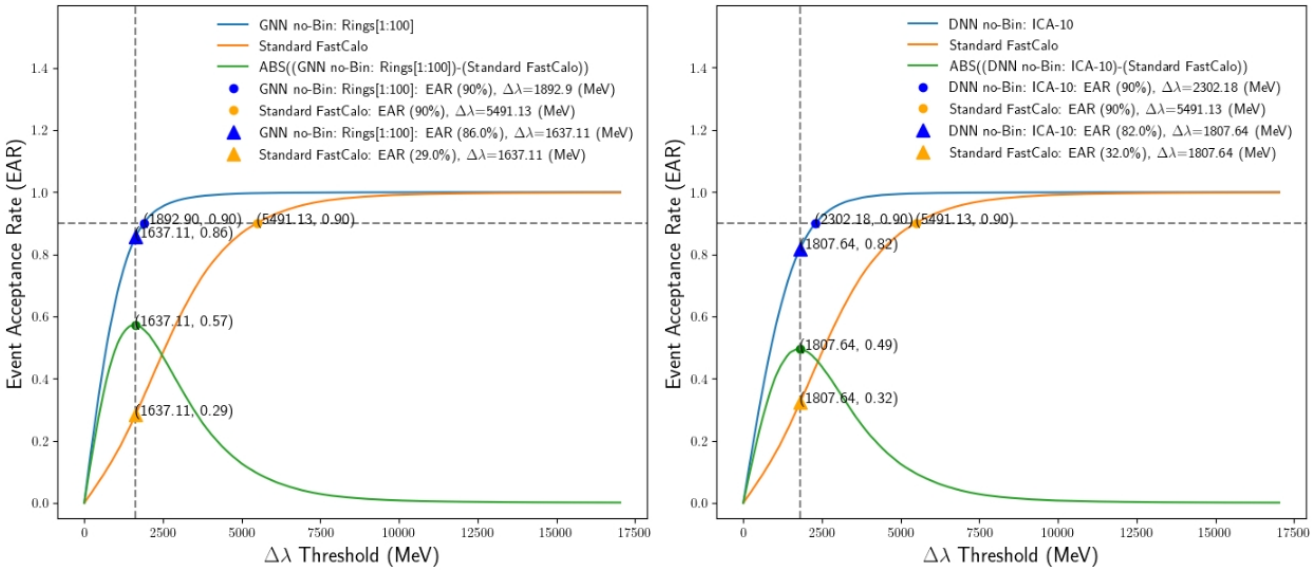


Figure 16: Truth-referenced acceptance-window curves showing the event acceptance rate (EAR) as a function of the residual tolerance $\Delta\lambda$ for Standard FastCalo and two representative ML-based calibration methods. Panel (a) compares `GNN no-Bin: Rings[1:100]` with Standard FastCalo, while panel (b) compares `DNN no-Bin: ICA-10` with Standard FastCalo. At each $\Delta\lambda$, EAR is the fraction of the common independent test events satisfying $|E_{\text{true}} - E_m| < \Delta\lambda$. The green curve is the absolute EAR difference between the ML method and the baseline. Dashed lines and markers identify representative points on the finite 1000-point scan.

diagnostic, not the efficiency curve of a production trigger with a fixed online threshold.

Fig. **??** shows the event acceptance rate (EAR) as a function of the residual-window half-width $\Delta\lambda$ for the Standard FastCalo baseline and two representative ML-based calibrators. In this representation, a smaller value of $\Delta\lambda$ required to reach a given EAR indicates tighter truth-referenced residual containment. The curves are evaluated on the same independent signal events for all methods and are not interpreted as a signal-versus-background trigger-efficiency measurement.

A clear improvement over the baseline is observed for both ML-based configurations. The full-ring GNN model, `GNN no-Bin: Rings[1:100]`, reaches 90% event acceptance at $\Delta\lambda = 1892.90$ MeV, whereas Standard FastCalo requires $\Delta\lambda = 5491.13$ MeV to reach the same operating point. This corresponds to a 65.53% reduction in $\Delta\lambda_{90}$. At the representative fixed offset $\Delta\lambda = 1637.11$ MeV, the same ML-based method achieves an EAR of 86%, compared with 29% for the baseline.

The compact configuration `DNN no-Bin: ICA-10` also yields tighter residual containment. It reaches 90% acceptance at $\Delta\lambda = 2302.18$ MeV, corresponding to a 58.07% reduction relative to Standard FastCalo. At the representative fixed half-width $\Delta\lambda = 1807.64$ MeV, this compact model achieves an EAR of 82%, whereas the baseline reaches 32%. These results show that a ten-component compact representation retains sufficient calibration information to narrow the truth-referenced acceptance window substantially within the evaluated benchmark.

The energy-resolution and acceptance-window metrics are summarized in Table 5. The full-ring GNN and full-ring DNN both yield $\Delta\lambda_{90} = 1892.90$ MeV on the finite scan grid. We verified the underlying curves and stored dataframe: this equality is not a transcription error. The two acceptance curves are different at neighboring scan points, but the first grid value for which each curve reaches or exceeds 90% acceptance is the same. The result should therefore be interpreted as a grid-resolution tie in the $\Delta\lambda_{90}$ summary, not as identical response functions. The compact ICA-10 and CORR-10 configurations yield 2302.18 and 2251.02 MeV, respectively.

These curves characterize signal-only residual containment relative to the raw Standard FastCalo reference. They do not measure background rejection, jet-fake rate, ROC performance, or the efficiency of a production trigger menu with a fixed online threshold. Moreover, because no constant-scale recalibration control is included in the present experimental design, the reduction in $\Delta\lambda_{90}$ should not be interpreted as the incremental contribution uniquely attributable to event-wise machine learning beyond all possible bias-only recalibration.

3.5. Integrated Performance and Accuracy–Efficiency Analysis

This section provides an integrated visual summary of the calibration results. The previous sections evaluated global error metrics, bin-wise stability, energy resolution, acceptance-window response, and CPU-based prediction time separately. Here, scatter plots, compact performance summaries, and accuracy–efficiency diagrams are used to connect these complementary findings and to highlight

Table 5
Energy-resolution and acceptance-window summary for Standard FastCalo (Std) and the four selected ML-based calibration configurations. The quantity $(\sigma_E/E)_{\text{global}}$ is the standard deviation of the relative response E_m/E_{true} computed over the complete independent test set; it is dimensionless and is not an average of the energy-bin values shown in Fig. 15. The reported ± term is the standard error of this estimated width. This whole-sample value is retained only as a compact truth-referenced summary and is not interpreted as a fitted experimental stochastic–constant–noise resolution parameter. The table also reports $\Delta\lambda_{90}$, the minimum scanned residual-window half-width containing at least 90% of the same independent test events. Standard FastCalo is the reference for the relative percentage comparisons.

Method	$(\sigma_E/E)_{\text{global}}$	$(\sigma_E/E)_{\text{global}}$ (% Std)	$\Delta\lambda_{90}$ (MeV)	$\Delta\lambda_{90}$ (% Std)
Standard FastCalo	2.46e-02 ± 5.80e-05	-	5491.13	0.0%
DNN no-Bin: CORR-10	1.65e-02 ± 3.90e-05	32.98% ± 0.22%	2251.02	59.01%
DNN no-Bin: ICA-10	1.52e-02 ± 3.60e-05	38.20% ± 0.21%	2302.18	58.07%
DNN no-Bin: Rings[1:100]	1.24e-02 ± 2.90e-05	49.46% ± 0.17%	1892.90	65.53%
GNN no-Bin: Rings[1:100]	1.19e-02 ± 2.80e-05	51.46% ± 0.16%	1892.90	65.53%

the benchmark-level trade-off between calibration accuracy, input dimensionality, and trained-model prediction cost.

Fig. 17 compares Standard FastCalo with two representative ML-based calibration methods. In both panels, the dashed diagonal line represents the ideal response, $E_{\text{calib}} = E_{\text{true}}$. The Standard FastCalo points exhibit a broader spread around this line, whereas the calibrated predictions are more tightly concentrated around the ideal response. This indicates reduced event-by-event dispersion after calibration.

The full-ring configuration, `GNN no-Bin: Rings[1:100]`, shows the strongest visual concentration around the diagonal, consistent with its leading d_0, RMSE, IQR, energy-resolution, and acceptance-window results. The compact configuration, `DNN no-Bin: CORR-10`, also improves the response relative to Standard FastCalo despite using only 10 input features. This confirms that compact selected-ring representations can retain substantial calibration-relevant information, although the full-ring graph-based model provides the strongest absolute correction.

Fig. 18 summarizes the main calibration gains across RMSE, IQR, relative energy resolution σ_E/E, and the acceptance-window half-width $\Delta\lambda_{90}$. The values are expressed as percentage improvements relative to Standard FastCalo, so larger bars indicate stronger improvement. The full-ring neural configurations provide the largest overall gains, while the compact `DNN no-Bin: ICA-10` and `DNN no-Bin: CORR-10` models retain competitive performance with substantially reduced input dimensionality.

A complementary view of the accuracy–efficiency trade-off is shown in Fig. 19. The vertical axis represents the joint RMSE–IQR metric d_0, for which lower values indicate better calibration accuracy and residual dispersion. The horizontal axis shows the amortized CPU-based prediction time per event in logarithmic scale, and the marker size encodes the number of input features. The dashed horizontal line indicates the Standard FastCalo d_0 reference.

This representation highlights the benchmark-level compromise between predictive performance, trained-model prediction cost, and input dimensionality. Compact configurations such as `DNN no-Bin: CORR-10` and `DNN no-Bin: ICA-10` use only 10 input features and achieve very low amortized prediction times, but they retain higher d_0 values than the full-ring models. The full-ring configurations use 100 input features and achieve the lowest d_0 values, with `GNN no-Bin: Rings[1:100]` providing the best absolute accuracy and `DNN no-Bin: Rings[1:100]` offering a similar accuracy level with lower CPU-based prediction time.

The integrated performance analysis confirms the two main regimes identified in this study. Full-ring neural models, especially `GNN no-Bin: Rings[1:100]`, provide the strongest absolute calibration performance. Compact DNN configurations, particularly `DNN no-Bin: ICA-10` and `DNN no-Bin: CORR-10`, preserve competitive gains with substantially lower input dimensionality and very low amortized CPU-based prediction time. These results support the benchmark-level conclusion that neural ring-based calibration can improve fast calorimetric energy estimates while offering flexible accuracy–compactness trade-offs for future low-latency reconstruction studies.

3.6. Statistical Validation

The normality assessment was performed for the uncalibrated residuals, $\varepsilon_{\text{fast}} = E_{\text{true}} - E_{\text{fast}}$, the calibrated residuals, $\varepsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}}$, and their paired differences, $d = \varepsilon_{\text{calib}} - \varepsilon_{\text{fast}}$. For all top-ranked configurations, the D'Agostino–Pearson and Anderson–Darling tests rejected the null hypothesis of normality, with $p < 10^{-16}$. Representative Q–Q plots for Standard FastCalo, `GNN no-Bin: Rings[1:100]`, and `DNN no-Bin: CORR-10` are shown in Fig. 20. The deviations from the theoretical normal reference line, especially in the tails, confirm the non-Gaussian character of the residual distributions.

Because calibrated and uncalibrated residuals are computed from the same events, all statistical comparisons are treated as paired comparisons. Parametric paired tests are therefore complemented by non-parametric and bootstrap-based procedures to provide robustness against the observed departures from normality.

3.6.1. Mean Error Comparison

Table 6 summarizes the results of three paired procedures applied to the signed differences $d = \varepsilon_{\text{calib}} - \varepsilon_{\text{fast}}$: the large-sample paired z-test, the paired Student's t-test, and

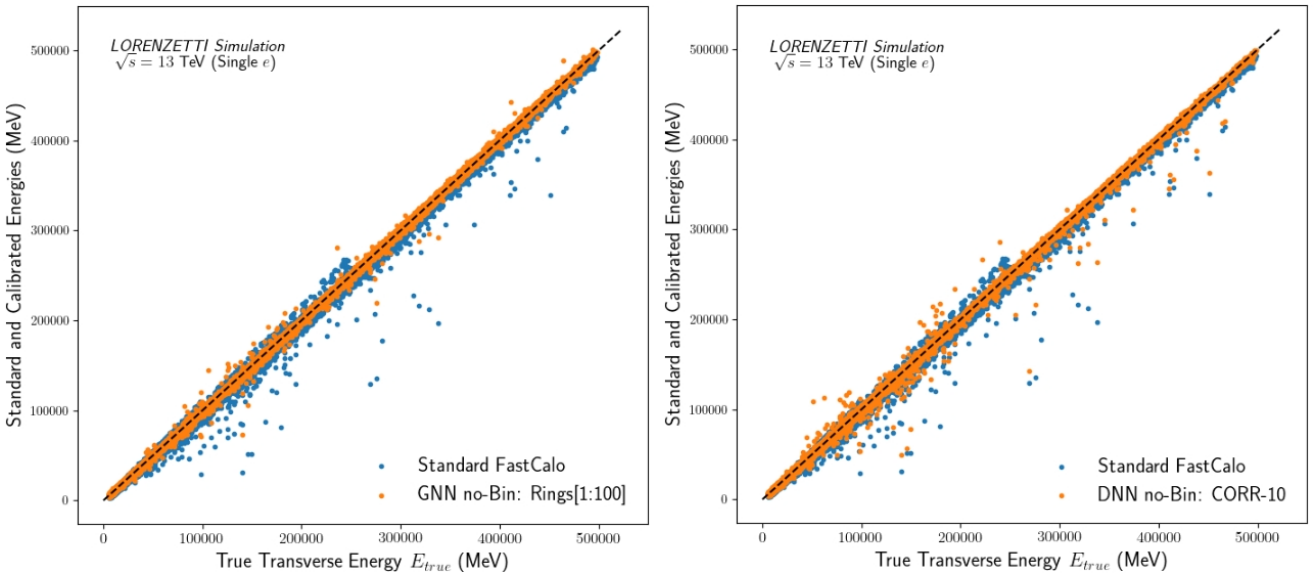


Figure 17: Scatter plots of reconstructed or calibrated transverse energy as a function of true transverse energy for Standard FastCalo and two representative ML-based calibration methods. Panel (a) compares Standard FastCalo with `GNN no-Bin: Rings[1:100]`, while panel (b) compares Standard FastCalo with `DNN no-Bin: CORR-10`. The dashed diagonal line indicates the ideal response, $E_{\text{calib}} = E_{\text{true}}$. In both panels, the ML-based calibrators show a tighter concentration around the diagonal and a reduced spread relative to Standard FastCalo.

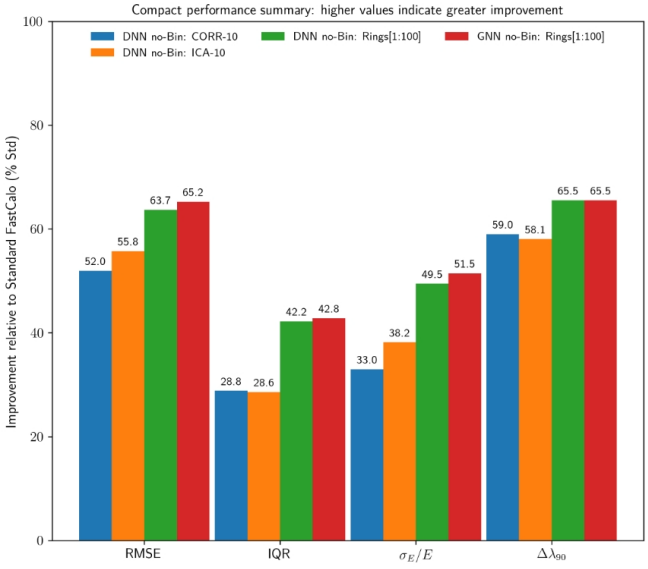


Figure 18: Compact performance summary of representative top-performing calibration configurations. Bars show the percentage improvement relative to Standard FastCalo for RMSE, IQR, relative energy resolution σ_E/E, and the acceptance-window half-width $\Delta\lambda_{90}$. Higher values indicate greater improvement. Full-ring configurations achieve the largest overall gains, while compact low-dimensional models retain competitive performance with reduced input dimensionality.

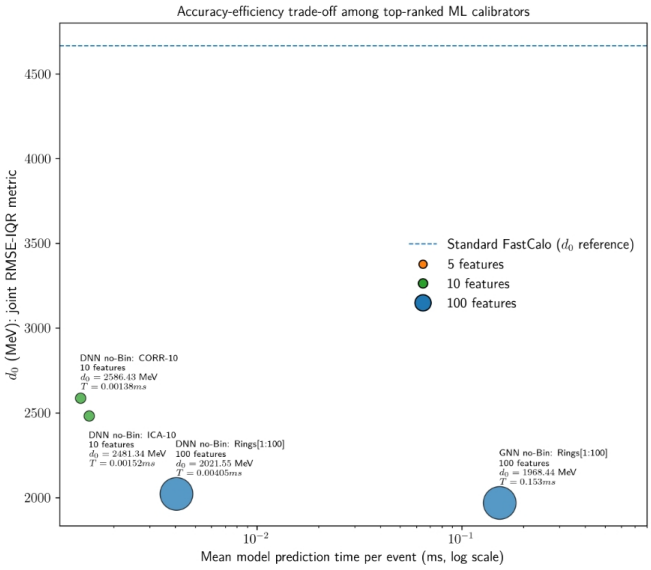


Figure 19: Accuracy–efficiency trade-off among representative top-ranked machine-learning calibrators. The vertical axis shows the joint RMSE–IQR metric d_0, where lower values indicate better calibration accuracy and residual dispersion. The horizontal axis shows the amortized CPU-based prediction time per event in logarithmic scale. Marker size represents the number of input features used by each configuration, and the dashed horizontal line indicates the Standard FastCalo d_0 reference. Compact low-dimensional models provide the lowest CPU-based prediction times, whereas full-ring models achieve the lowest d_0 values.

the Wilcoxon signed-rank test. These tests assess whether calibration induces a statistically significant signed-error shift relative to Standard FastCalo.

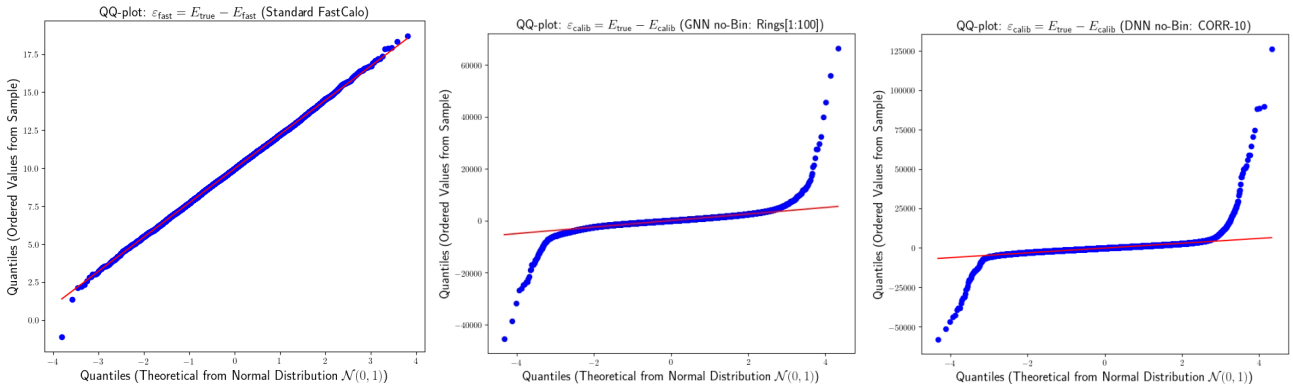


Figure 20: Representative Q–Q plots for $\varepsilon_{\text{fast}}$ and $\varepsilon_{\text{calib}}$ with respect to a standard normal distribution. Panel (a) shows the Standard FastCalo residuals, while panels (b) and (c) show the calibrated residuals for `GNN no-Bin: Rings[1:100]` and `DNN no-Bin: CORR-10`, respectively. Deviations from the theoretical reference line, especially in the tails, indicate departures from Gaussian behavior. This motivates the use of complementary non-parametric and bootstrap-based paired statistical procedures in addition to parametric reference tests.

For all selected configurations, the null hypothesis $H_0 : \mu_d \geq 0$ was rejected at $\alpha = 0.05$. The parametric tests yielded $p < 10^{-300}$, while the Wilcoxon signed-rank test gave consistent conclusions without assuming normality. In all cases, the one-sided 95% confidence intervals for μ_d, or for the median of d in the Wilcoxon test, lay entirely below zero.

The paired mean-error tests should be interpreted as tests of signed residual displacement, not as standalone measures of practical calibration quality. For `GNN no-Bin: Rings[1:100]`, the one-sided confidence interval lies below -2.72 GeV, indicating a statistically significant event-wise shift of the signed residual relative to the uncalibrated baseline. Similar behavior is observed for `DNN no-Bin: Rings[1:100]`, `DNN no-Bin: ICA-10`, and `DNN no-Bin: CORR-10`.

The practical relevance of the calibration is therefore assessed jointly through RMSE, IQR, energy resolution, acceptance-window behavior, and residual-variance reduction, rather than from the signed mean-error test alone.

3.6.2. Variance Ratio Analysis

Table 7 reports the variance-ratio analysis for $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$. The Pitman–Morgan test is used as the primary paired variance test, while the one-sided F-test is reported only as a classical reference. A paired bootstrap BCa procedure is also included as a robustness check because it preserves the event-wise pairing between calibrated and uncalibrated residuals and does not rely on a Gaussian residual model.

Across all selected configurations, the null hypothesis $H_0 : \sigma^2_{\text{calib}}/\sigma^2_{\text{fast}} \geq 1$ was rejected at $\alpha = 0.05$. The Pitman–Morgan and F-test reference comparisons yielded $p < 10^{-300}$. For the paired bootstrap BCa procedure, the smallest attainable p-value is limited by the number of bootstrap resamples; with $B = 5000$, this lower bound is $1/B = 0.0002$. The bootstrap intervals nevertheless remained entirely below one, supporting the conclusion that the calibrated residuals have lower variance than the Standard FastCalo residuals.

The variance-ratio statistic is numerically the same in the F-test and bootstrap rows because both are computed from the observed ratio $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$. The difference lies in the inferential framework: the F-test uses a parametric reference distribution, whereas the bootstrap procedure estimates the sampling behavior empirically from paired resamples. The agreement between these approaches reinforces the robustness of the variance-reduction conclusion.

For `GNN no-Bin: Rings[1:100]`, the calibrated residual variance corresponds to 23.56% of the Standard FastCalo residual variance, representing a 76.44% variance reduction. Similar reductions are observed for `DNN no-Bin: Rings[1:100]` (74.45%), `DNN no-Bin: ICA-10` (61.80%), and `DNN no-Bin: CORR-10` (55.08%). These results show that all selected neural configurations substantially reduce residual variance, with the full-ring GNN achieving the strongest variance suppression among the evaluated models.

3.6.3. Statistical Interpretation

Taken together, the statistical tests show that the selected calibration models produce a significant paired shift in the signed residuals and, more importantly for practical performance, a substantial reduction in residual variance relative to Standard FastCalo.

From a statistical standpoint, for a residual variable ε, the mean squared error admits the exact decomposition [38, 22]

$$\text{MSE}(\varepsilon) = \text{Var}(\varepsilon) + [\mathbb{E}(\varepsilon)]^2 . \tag{60}$$

This identity also clarifies the interpretation of Table 6. The reported quantity $\mu_d = \mathbb{E}(\varepsilon_{\text{calib}} - \varepsilon_{\text{fast}})$ is the difference between the mean calibrated and uncalibrated residuals; it is not the mean bias of Standard FastCalo itself. Therefore, μ_d^2 cannot be interpreted as the bias-squared contribution of the Standard FastCalo MSE. The RMSE reductions reported in Table 4 establish a reduction of total mean squared error, while the variance-ratio results in Table 7 independently

Table 6
Statistical evaluation of mean calibration error using paired one-sided t, z, and Wilcoxon signed-rank tests. Hypotheses: $H_0 : \mu_d \geq 0$ vs. $H_1 : \mu_d < 0$, where μ_d is the mean (or median for Wilcoxon) of the paired differences $d = \varepsilon_{\text{calib}} - \varepsilon_{\text{fast}}$, with $\varepsilon_{\text{calib}} = E_{\text{true}} - E_{\text{calib}}$ and $\varepsilon_{\text{fast}} = E_{\text{true}} - E_{\text{fast}}$. All confidence intervals are one-sided at 95%. Units for μ_d and CIs are in MeV.

Method (z-test: primary)	z-statistic	p-value	95% one-sided CI for μ_d [MeV]	Decision
DNN no-Bin: Rings[1:100]	-324.5005	$< 10^{-300}$	$(-\infty, -2866.32)$	Reject H0
DNN no-Bin: CORR-10	-392.2214	$< 10^{-300}$	$(-\infty, -2859.67)$	Reject H0
DNN no-Bin: ICA-10	-325.9147	$< 10^{-300}$	$(-\infty, -2779.70)$	Reject H0
GNN no-Bin: Rings[1:100]	-317.7842	$< 10^{-300}$	$(-\infty, -2721.46)$	Reject H0
Method (t-test: reference)	**t-statistic**	**p-value**	**95% one-sided CI for μ_d [MeV]**	**Decision**
DNN no-Bin: Rings[1:100]	-324.5005	$< 10^{-300}$	$(-\infty, -2866.32)$	Reject H0
DNN no-Bin: CORR-10	-392.2214	$< 10^{-300}$	$(-\infty, -2859.67)$	Reject H0
DNN no-Bin: ICA-10	-325.9147	$< 10^{-300}$	$(-\infty, -2779.69)$	Reject H0
GNN no-Bin: Rings[1:100]	-317.7842	$< 10^{-300}$	$(-\infty, -2721.47)$	Reject H0
Method (Wilcoxon signed-rank test: robustness)	**W-statistic**	**p-value**	**95% one-sided CI for median(d) [MeV]**	**Decision**
DNN no-Bin: Rings[1:100]	75816027.0000	$< 10^{-300}$	$(-\infty, -2604.96)$	Reject H0
DNN no-Bin: CORR-10	68487095.0000	$< 10^{-300}$	$(-\infty, -2603.16)$	Reject H0
DNN no-Bin: ICA-10	74775877.0000	$< 10^{-300}$	$(-\infty, -2588.42)$	Reject H0
GNN no-Bin: Rings[1:100]	83841186.0000	$< 10^{-300}$	$(-\infty, -2476.35)$	Reject H0

show that the calibrated residual distributions are substantially narrower than the Standard FastCalo residual distribution.

The present experimental design, however, does not include a dedicated constant-scale recalibration or a low-capacity regressor using only (E_{fast}, η). Consequently, the reported gains quantify the improvement of the ring-based calibration pipelines relative to the raw Standard FastCalo reference, but do not isolate the fraction of that gain that would remain after an independently optimized bias-only or low-capacity recalibration. Such controls require a separate, pre-defined comparison in which the constant-scale, (E_{fast}, η)-only, and ring-based models are trained and evaluated under the same development/test protocol. This controlled decomposition is identified as a priority for future ablation work rather than inferred retrospectively from the paired mean-residual shifts.

At the candidate level, the combined reductions in RMSE, robust dispersion, and residual variance indicate a more concentrated calibrated response within the simulated benchmark. The broader limits of this interpretation, including physics-process validation and simulation-to-data transfer, are consolidated in Section 3.8.

3.7. Impact of Pile-Up on Calibration Performance

Pile-up is a major source of degradation in calorimetric energy measurements. Additional proton–proton interactions can introduce energy deposits unrelated to the primary particle of interest, distort the reconstructed shower topology, broaden the calorimetric response, and increase event migration across selection thresholds [2, 3, 48, 49]. These effects are particularly relevant in high-luminosity conditions, where several tens of interactions may occur within the same or nearby readout time windows.

As described in Section 2.1, the Lorenzetti configuration used in this work includes pile-up by merging minimum-bias events with the primary single-electron event [15, 17]. The simulated pile-up profile shown in Fig. 1 has mean $\langle\mu\rangle = 38.94$ and standard deviation 9.999. Therefore, the calibration models are trained and evaluated under a pile-up-inclusive benchmark with substantial additional collision activity. This level is of the same order as pile-up conditions reported in recent LHC operation periods, including Run 2 and Run 3 scenarios [48, 49].

The reported RMSE, IQR, energy-resolution, and acceptance-window results therefore already include the pile-up treatment implemented in the adopted Lorenzetti simulation.

Nevertheless, the present study does not perform a controlled scan of calibration performance as a function of $\langle\mu\rangle$ or of detector-response perturbations. Consequently, the reported values quantify performance under the pile-up-inclusive Lorenzetti distribution actually simulated, not invariance with respect to changing pile-up or detector conditions. A controlled robustness study would require dedicated samples generated or stratified at pre-defined low-, medium-, and high-pile-up regimes together with systematic response/noise variations, followed by the same fixed evaluation protocol. Such a study addresses a distinct domain-robustness question and is identified in Section 3.8 as required future validation.

3.8. Scope, Limitations, and Transfer to Experimental Data

The present study estimates a point calibration factor for each electromagnetic candidate. The reported statistical uncertainties, confidence intervals, bootstrap analyses, and regional summaries characterize population-level performance of the calibration procedure; they are not per-event predictive uncertainty intervals. A deployment-oriented uncertainty model would require an additional training and

Table 7
Statistical evaluation of the error variance ratio using Pitman–Morgan, one-sided F, and paired bootstrap tests. Hypotheses: $H_0 : \sigma^2_{\text{calib}}/\sigma^2_{\text{fast}} \geq 1$ vs. $H_1 : \sigma^2_{\text{calib}}/\sigma^2_{\text{fast}} < 1$, with σ^2_{calib} denoting the variance of $\varepsilon_{\text{calib}}$ and σ^2_{fast} the variance of $\varepsilon_{\text{fast}}$. All confidence intervals are one-sided at 95% and expressed in dimensionless units (variance ratios).

Method (Pitman-Morgan: primary)	PM-statistic	p-value	95% one-sided CI for $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$	Decision
DNN no-Bin: Rings[1:100]	-238.2553	$< 10^{-300}$	(0.0, 0.2850)	Reject H0
DNN no-Bin: CORR-10	-159.5687	$< 10^{-300}$	(0.0, 0.4861)	Reject H0
DNN no-Bin: ICA-10	-168.5690	$< 10^{-300}$	(0.0, 0.4174)	Reject H0
GNN no-Bin: Rings[1:100]	-260.4883	$< 10^{-300}$	(0.0, 0.2555)	Reject H0
Method (F-test: reference)	F-statistic	p-value	95% one-sided CI for $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$	Decision
DNN no-Bin: Rings[1:100]	0.2555	$< 10^{-300}$	(0.0, 0.2582)	Reject H0
DNN no-Bin: CORR-10	0.4492	$< 10^{-300}$	(0.0, 0.4541)	Reject H0
DNN no-Bin: ICA-10	0.3820	$< 10^{-300}$	(0.0, 0.3861)	Reject H0
GNN no-Bin: Rings[1:100]	0.2356	$< 10^{-300}$	(0.0, 0.2382)	Reject H0
Method (Paired bootstrap BCa test: robustness)	$(\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}})$-statistic	p-value	95% one-sided CI for $\sigma^2_{\text{calib}}/\sigma^2_{\text{fast}}$	Decision
DNN no-Bin: Rings[1:100]	0.2555	0.0002	(0.0, 0.2847)	Reject H0
DNN no-Bin: CORR-10	0.4492	0.0002	(0.0, 0.4851)	Reject H0
DNN no-Bin: ICA-10	0.3820	0.0002	(0.0, 0.4174)	Reject H0
GNN no-Bin: Rings[1:100]	0.2356	0.0002	(0.0, 0.2551)	Reject H0

validation layer, for example through quantile regression, heteroscedastic likelihoods, ensembles, Bayesian approximations, or conformal prediction, together with explicit coverage and calibration tests. Such per-event uncertainty quantification is outside the present experimental protocol and is identified as a separate extension of the calibration framework.

No external deep-learning calorimeter-regression result is numerically reproduced in the Lorenzetti benchmark, and no claim of detector-independent numerical superiority is made. The controlled comparisons are restricted to methods evaluated with the same Lorenzetti data and protocol. Likewise, the GNN–DNN comparison is a comparison of complete trained pipelines and not a causal ablation of graph topology, deterministic descriptors, or model capacity.

The study is also simulation based. Application to experimental data would require a dedicated simulation-to-data validation strategy. A practical sequence would include comparison of low-level and ring-level observables between simulation and collision data, validation of the response versus E_T, η, pile-up, and detector conditions, closure tests in control samples such as $Z \to e^+e^-$, and uncertainty or correction procedures for residual domain mismatch. Domain adaptation, reweighting, detector-response nuisance variations, and data-driven recalibration could then be evaluated if statistically significant discrepancies remained. None of these steps is replaced by the present Lorenzetti benchmark.

Finally, the CPU timing results cover trained-model prediction after feature availability. They do not include online ring construction, deterministic descriptor construction, graph preparation, memory movement, or production scheduling. Therefore, the present work does not establish end-to-end trigger latency or compliance with a specific online latency budget.

3.9. Implications for Physics-Process Reconstruction and Invariant-Mass Propagation

The results reported so far quantify the candidate-level calibration performance under the single-electron Lorenzetti benchmark. This scope is intentional. The objective of the present study is to evaluate the event-wise calibration mapping from ring-based calorimetric information to a multiplicative transverse-energy correction factor. This mapping is defined at the electromagnetic-candidate level and does not require the candidate to originate from a specific hard-scattering process.

This property is relevant for physics processes such as $Z \to e^+e^-$, where calibrated electron and positron energies are propagated to resonance-level observables such as the di-electron invariant mass [3, 18]. In such a process, each electromagnetic candidate would be calibrated independently using the same event-wise mapping:

$$E_{\text{calib},a} = \alpha_{\text{calib},a} E_{\text{fast},a}, \qquad E_{\text{calib},b} = \alpha_{\text{calib},b} E_{\text{fast},b}, \quad (61)$$

where a and b denote the two electromagnetic candidates in the pair. In the notation used throughout this work, E_{fast} and E_{calib} denote transverse-energy estimates. The electric charge affects the pairing and event-selection logic, but it is not an input to the calorimetric energy-calibration mapping evaluated in this work.

For two approximately massless electromagnetic candidates, the invariant mass can be written in terms of the calibrated transverse energies as

$$m^2_{ee} = 2E_{\text{calib},a} E_{\text{calib},b} \left[\cosh(\Delta\eta) - \cos(\Delta\phi)\right], \quad (62)$$

where $\Delta\eta$ and $\Delta\phi$ are the pseudorapidity and azimuthal-angle differences between the two candidates. Thus, improvements in the event-wise calibrated transverse-energy response are expected to affect the energy-driven component of the reconstructed invariant-mass distribution.

A first-order uncertainty-propagation argument can be used to illustrate this connection. Defining

$$A = \cosh(\Delta\eta) - \cos(\Delta\phi), \tag{63}$$

Eq. 62 can be written as

$$m_{ee} = \left(2E_{\mathrm{calib},a}E_{\mathrm{calib},b}A\right)^{1/2}. \tag{64}$$

For small perturbations and fixed angular terms, standard first-order logarithmic uncertainty propagation gives [47]

$$\frac{\delta m_{ee}}{m_{ee}} \approx \frac{1}{2}\left(\frac{\delta E_a}{E_a} + \frac{\delta E_b}{E_b}\right), \tag{65}$$

where $\delta E_a/E_a$ and $\delta E_b/E_b$ are the relative calibrated-energy perturbations of the two electromagnetic candidates. Under the additional simplifying assumption that the two candidate-energy residuals have similar variance and weak correlation, this gives

$$\mathrm{Var}\left(\frac{\delta m_{ee}}{m_{ee}}\right) \approx \frac{1}{2}\mathrm{Var}\left(\frac{\delta E}{E}\right). \tag{66}$$

This relation should be interpreted only as an analytical propagation of the candidate-level energy term, not as a prediction of the full reconstructed Z-boson mass resolution. Within this approximation, the reduction of the relative energy-resolution term reported in Table 5 would reduce the energy-driven contribution to σ_m/m. For example, the full-ring GNN reduces the candidate-level relative energy resolution from 2.46×10^{-2} to 1.19×10^{-2}. In the simplified equal-resolution and weak-correlation limit, this changes the approximate energy-driven contribution to σ_m/m from $2.46\times10^{-2}/\sqrt{2}$ to $1.19\times10^{-2}/\sqrt{2}$. This calculation is illustrative and does not include angular uncertainties, candidate correlations, event selection, background contamination, or invariant-mass fitting effects.

A complete $Z \to e^+e^-$ validation therefore remains a distinct resonance-level study. Such an analysis would need to include resonance production, two-candidate pairing, angular correlations, charge requirements, final-state radiation, selection efficiencies, background treatment, mass-window definitions, and invariant-mass fitting procedures [3, 1]. These ingredients are essential for physics-performance validation, but they introduce process-dependent effects that are deliberately factorized out of the present candidate-level calibration benchmark.

The single-electron particle-gun benchmark is therefore appropriate for isolating the calibration mapping itself. It provides broad coverage of the transverse-energy range and allows the model to learn how ring-based shower information corrects the fast reconstructed transverse energy independently of a particular resonance topology. The present work establishes the candidate-level calibration mechanism required for subsequent resonance-level validation; a dedicated $Z \to e^+e^-$ study is left for future work.

4. Conclusions

This work formulated fast transverse-energy calibration in segmented particle-detector calorimeters as a multi-stage information-fusion problem. The methodological contribution is the organization of the calibration task as a sequence of complementary fusion levels operating on the same detector information: segmented calorimeter-cell measurements are first aggregated into structured ring-based shower descriptors; the resulting representation is either preserved in full or transformed into compact selected or extracted feature spaces; neural models subsequently fuse the available shower information into an event-level calibration factor; and, in the graph-based configuration, regional relational embeddings are further combined with deterministic tabular shower summaries. The estimated factor is finally integrated with the Standard FastCalo energy to obtain the calibrated transverse-energy estimate. This formulation provides a unified information-fusion view connecting detector-level aggregation, representation construction, nonlinear feature integration, relational modeling, and energy calibration within a common pipeline.

Within this formulation, the DNN and GNN instantiate two complementary neural fusion strategies. The DNN performs nonlinear fusion directly over complete or compact ring-derived representations, allowing the effect of representation compactness on calibration performance and computational cost to be evaluated. The relational–tabular GNN preserves the complete 100-ring representation and combines two internal information paths: regional GraphSAGE message passing over the ordered PS, EM, and HAD ring structures, and tabular processing of the full ring profile together with deterministic regional, longitudinal, and positional shower descriptors. Their fusion produces a common event-level calibration estimate. The contribution demonstrated here is therefore the formulation and benchmark evaluation of these complete neural information-fusion pipelines; the present experiments do not causally isolate the individual contribution of the graph, tabular, descriptor, or fusion branches.

Among the evaluated configurations, the lowest observed RMSE, IQR, and d_0 values were obtained by `GNN no-Bin: Rings[1:100]`. Relative to the raw Standard FastCalo reference under the simulated benchmark conditions considered here, this configuration reduced d_0 from 4666.42 MeV to 1968.44 MeV, RMSE by 65.25%, and IQR by 42.82%. These percentages quantify improvement relative to the unmodified Standard FastCalo estimate. Because the present protocol does not include a separately optimized constant-scale or $(E_{\mathrm{fast}}, \eta)$-only recalibration control, they should not be interpreted as the incremental gain uniquely attributable to event-wise ring information beyond all possible low-capacity recalibration. The full-ring DNN achieved closely comparable performance, and the GNN–DNN difference is treated as a descriptive numerical difference between complete pipelines rather than evidence of intrinsic architectural superiority.

The compact neural configurations also showed strong accuracy–efficiency behavior. In particular, `DNN no-Bin: ICA-10` and `DNN no-Bin: CORR-10` retained substantial calibration gains while using only 10 input components. These configurations achieved RMSE reductions above 51% relative to Standard FastCalo and were favored by the D_0 criterion because of their reduced input dimensionality and low amortized CPU-based prediction time. This result indicates that compact ring-derived representations can preserve a large fraction of the calibration-relevant information, which is relevant for future studies involving memory, data-movement, and throughput constraints.

The results reveal a clear trade-off between absolute accuracy and compactness. Full-ring neural models provide the strongest calibration performance and dominate the d_0-based ranking, while compact ten-component DNN configurations provide the most favorable D_0-based accuracy–efficiency compromise. All four selected neural configurations exhibited amortized trained-model prediction times below 0.2 ms per event on the CPU-based hardware used in this study. These measurements should be interpreted as relative batched-throughput benchmarks after feature availability, not as direct estimates of production trigger latency.

The statistical validation supports the robustness of the observed improvements within the evaluated benchmark. The residual distributions were found to be non-Gaussian, motivating the use of complementary paired statistical procedures, including large-sample z-tests, paired t-tests, Wilcoxon signed-rank tests, Pitman–Morgan tests, F-test references, and paired bootstrap BCa variance-ratio analyses. Across the selected neural configurations, the calibrated residuals exhibited statistically significant paired signed-error shifts and substantially reduced variance relative to the Standard FastCalo residuals. For `GNN no-Bin: Rings[1:100]`, the calibrated-error variance corresponded to 23.56% of the uncalibrated variance, representing a 76.44% variance reduction.

From an information-fusion perspective, the simultaneous improvements in RMSE, IQR, residual variance, relative energy resolution, and acceptance-window half-width show that the successive representation and neural-processing stages preserve calibration-relevant information while producing a more concentrated final energy response under the simulated detector-response and pile-up conditions considered here. The results also expose an important fusion trade-off: retaining the complete structured representation favors maximum predictive accuracy, whereas compact selected or extracted representations preserve substantial calibration information with lower input dimensionality and prediction cost. These observations concern the complete evaluated pipelines and should not be interpreted as causal attribution of the gain to any individual fusion component.

The results remain benchmark-level evidence obtained within the controlled Lorenzetti simulation environment. The principal limitations and the required extensions—including simulation-to-data validation, controlled pile-up studies, per-event uncertainty quantification, architectural ablations, external numerical reproduction, and end-to-end target-hardware timing—are consolidated in Section 3.8. These items define the validation program required before detector-independent or deployment-level conclusions can be drawn.

Overall, this study shows how highly segmented detector measurements can be organized as a hierarchical information-fusion problem in which low-level observations are progressively aggregated, represented, transformed, and combined before producing a final calibrated estimate. The results support ring-based deep neural fusion and relational–tabular graph fusion as viable strategies within this formulation, while also showing that different fusion representations lead to distinct accuracy–compactness–cost trade-offs. Beyond the specific calorimetric benchmark, the principal methodological contribution is therefore the demonstration of a structured multi-stage fusion framework for combining spatially distributed detector information, compact feature representations, and relational and tabular neural processing within a common regression pipeline.

A. Repository and Benchmark Availability

The Lorenzetti Showers framework used to generate the simulation benchmark is publicly available, including source code, installation instructions, and simulation examples [16]. The Lorenzetti-generated dataset used in this work is also publicly available in the corresponding data repository [19]. These repositories provide the open benchmark interface used in the present study, including the calorimetric observables required to construct the ring-based representations evaluated in the calibration pipeline.

The complete implementation scripts are not distributed as part of this manuscript because they are part of an ongoing research code base. However, the public benchmark data, the Lorenzetti simulation framework, and the implementation-independent specification provided here allow the neural calibration workflow to be independently reimplemented. The manuscript specifies the calibration target, input representations, feature-selection and feature-extraction procedures, graph construction, DNN and GNN architectures, selected hyperparameters, training protocol, evaluation metrics, and statistical-validation procedures. Therefore, the results are reproducible at the methodological level, even though the authors' internal implementation scripts are not released.

B. Derived Shower-Shape Descriptors Used by the GNN

The relational–tabular GNN does not rely exclusively on the graph representation of the calorimeter rings. In parallel with the graph branch, the architecture constructs a tabular representation that preserves the complete 100-ring profile and augments it with 20 deterministic shower-shape descriptors derived from the same calorimetric information. These descriptors are intended to summarize complementary properties of the shower that are otherwise distributed across the ordered ring representation.

The resulting descriptor vector combines four classes of information: regional energy content; relative energy concentration among calorimeter regions; the energy content of the EM1, EM2, and EM3 electromagnetic samplings and their relative contributions to the complete electromagnetic region; and weighted positional moments that characterize the distribution of the ring response within the PS, EM, and HAD regions. Therefore, the derived quantities do not introduce additional detector measurements; rather, they provide deterministic summaries of the information already contained in the 100-ring representation.

Let the three coarse calorimeter regions used by the relational–tabular architecture be defined by the index sets $\mathcal{I}_{\mathrm{PS}} = \{1, \ldots, 8\}, \mathcal{I}_{\mathrm{EM}} = \{9, \ldots, 88\}, \mathcal{I}_{\mathrm{HAD}} = \{89, \ldots, 100\}$.

For each region $q \in \{\mathrm{PS}, \mathrm{EM}, \mathrm{HAD}\}$, the regional sum is defined as

$$S_q^{(i)} = \sum_{j \in \mathcal{I}_q} \tilde{E}_{r,j}^{(i)}, \tag{67}$$

where $\tilde{E}_{r,j}^{(i)}$ denotes the normalized value of the j-th calorimeter ring for event i.

The total contribution of the three regions is then

$$S_{\mathrm{tot}}^{(i)} = S_{\mathrm{PS}}^{(i)} + S_{\mathrm{EM}}^{(i)} + S_{\mathrm{HAD}}^{(i)}. \tag{68}$$

From these quantities, three regional fractions are constructed,

$$\mathrm{Frac}_q^{(i)} = \frac{S_q^{(i)}}{S_{\mathrm{tot}}^{(i)}}, \qquad q \in \{\mathrm{PS}, \mathrm{EM}, \mathrm{HAD}\}, \tag{69}$$

providing a compact measure of how the total ring response is distributed among the presampler, electromagnetic, and hadronic regions.

Two additional inter-region ratios characterize the relative contribution of the PS and HAD responses with respect to the electromagnetic region:

$$R_{\mathrm{HAD/EM}}^{(i)} = \frac{S_{\mathrm{HAD}}^{(i)}}{S_{\mathrm{EM}}^{(i)}}, \qquad R_{\mathrm{PS/EM}}^{(i)} = \frac{S_{\mathrm{PS}}^{(i)}}{S_{\mathrm{EM}}^{(i)}}. \tag{70}$$

In addition to these coarse regional quantities, the electromagnetic block is resolved according to the three physical EM samplings represented in the Ringer organization. The corresponding partial sums are compactly defined as

$$S_{\upsilon}^{(i)} = \sum_{j \in \mathcal{J}_{\upsilon}} \tilde{E}_{r,j}^{(i)}. \tag{71}$$

where $\upsilon \in \{\mathrm{EM1}, \mathrm{EM2}, \mathrm{EM3}\}$, $\mathcal{J}_{\mathrm{EM1}} = \{9, \ldots, 72\}$, $\mathcal{J}_{\mathrm{EM2}} = \{73, \ldots, 80\}$, and $\mathcal{J}_{\mathrm{EM3}} = \{81, \ldots, 88\}$.

Thus, $S_{\mathrm{EM1}}^{(i)}$, $S_{\mathrm{EM2}}^{(i)}$, and $S_{\mathrm{EM3}}^{(i)}$ represent the accumulated ring response associated with the first, second, and third electromagnetic samplings, respectively. Unlike the coarser PS–EM–HAD partition used by the graph branch, these three quantities preserve the finer longitudinal organization of the electromagnetic calorimeter represented by the Ringer descriptor.

The relative contributions of the three electromagnetic samplings to the complete EM region are then given by

$$R_{\upsilon/\mathrm{EM}}^{(i)} = \frac{S_{\upsilon}^{(i)}}{S_{\mathrm{EM}}^{(i)}}, \qquad \upsilon \in \{\mathrm{EM1}, \mathrm{EM2}, \mathrm{EM3}\}. \tag{72}$$

These quantities provide complementary longitudinal information: while $S_{\mathrm{EM}}^{(i)}$ summarizes the complete electromagnetic response, the three partial sums and their corresponding ratios describe how that response is distributed among EM1, EM2, and EM3.

The remaining six descriptors characterize the positional distribution of the ring response within the PS, EM, and HAD regions. These descriptors are based on the first and second weighted moments of the ordered ring sequence. Because the Ringer representation preserves an ordered spatial organization within each calorimeter region, these moments provide compact measures of the location and spread of the deposited response along that ordered representation.

For a region q containing n_q rings, let

$$p_{q,j} = \frac{j-1}{n_q - 1}, \qquad j = 1, \ldots, n_q, \tag{73}$$

denote the normalized ordinal coordinate of the j-th ring within that region. Here, j is a local index defined separately inside PS, EM, and HAD, rather than the global ring index in the complete 100-ring sequence. Consequently, $p_{q,j} \in [0, 1]$ describes the relative ordinal position of a ring within its corresponding calorimeter region and should not be interpreted as a physical geometrical distance in the detector.

The weights used to compute the positional moments are obtained by applying a min–max transformation to the corresponding ring response:

$$w_{q,j}^{(i)} = \frac{\tilde{E}_{r,j}^{(i)} - \min\left(\tilde{E}_{r,j}\right)}{\max\left(\tilde{E}_{r,j}\right) - \min\left(\tilde{E}_{r,j}\right)}. \tag{74}$$

The minimum and maximum values correspond to ring j in the sample used to estimate the transformation. This min–max mapping is employed exclusively to construct the positional weights and maps the corresponding ring response to the interval $[0, 1]$.

Using these weights, the weighted positional mean of region q is

$$\bar{p}_q^{(i)} = \frac{\sum_{j=1}^{n_q} w_{q,j}^{(i)} p_{q,j}}{\sum_{j=1}^{n_q} w_{q,j}^{(i)}}, \tag{75}$$

and the corresponding weighted positional width is

$$\sigma_{p,q}^{(i)} = \sqrt{\frac{\sum_{j=1}^{n_q} w_{q,j}^{(i)} \left(p_{q,j} - \bar{p}_q^{(i)}\right)^2}{\sum_{j=1}^{n_q} w_{q,j}^{(i)}}}. \tag{76}$$

The three values $\bar{p}^{(i)}_{\mathrm{PS}}$, $\bar{p}^{(i)}_{\mathrm{EM}}$, and $\bar{p}^{(i)}_{\mathrm{HAD}}$ therefore summarize the weighted position of the response within each coarse calorimeter region, whereas $\sigma^{(i)}_{p,\mathrm{PS}}$, $\sigma^{(i)}_{p,\mathrm{EM}}$, and $\sigma^{(i)}_{p,\mathrm{HAD}}$ describe the corresponding positional spread.

With these definitions, the complete deterministic descriptor vector used by the relational–tabular architecture is

$$\begin{aligned}\mathbf{d}^{(i)} = \big[&S^{(i)}_{\mathrm{PS}}, S^{(i)}_{\mathrm{EM}}, S^{(i)}_{\mathrm{HAD}},\\ &\mathrm{Frac}^{(i)}_{\mathrm{PS}}, \mathrm{Frac}^{(i)}_{\mathrm{EM}}, \mathrm{Frac}^{(i)}_{\mathrm{HAD}},\\ &R^{(i)}_{\mathrm{HAD/EM}}, R^{(i)}_{\mathrm{PS/EM}},\\ &S^{(i)}_{\mathrm{EM1}}, S^{(i)}_{\mathrm{EM2}}, S^{(i)}_{\mathrm{EM3}},\\ &R^{(i)}_{\mathrm{EM1/EM}}, R^{(i)}_{\mathrm{EM2/EM}}, R^{(i)}_{\mathrm{EM3/EM}},\\ &\bar{p}^{(i)}_{\mathrm{PS}}, \bar{p}^{(i)}_{\mathrm{EM}}, \bar{p}^{(i)}_{\mathrm{HAD}},\\ &\sigma^{(i)}_{p,\mathrm{PS}}, \sigma^{(i)}_{p,\mathrm{EM}}, \sigma^{(i)}_{p,\mathrm{HAD}}\big]^{T} \in \mathbb{R}^{20}.\end{aligned} \tag{77}$$

The descriptor vector therefore contains 3 regional sums, 3 regional fractions, 2 inter-region ratios, 3 electromagnetic-sampling sums, 3 corresponding EM-sampling ratios, and 6 positional moments, for a total of 20 deterministic quantities.

Because all of these descriptors are deterministic functions of the original 100 calorimeter rings, they do not constitute additional independent measurements and do not modify the external input dimensionality adopted for comparison among models, which remains $F = 100$. Their role is to make selected regional, longitudinal, and positional relationships explicitly available to the tabular branch instead of requiring the network to reconstruct all such quantities exclusively through its internal nonlinear transformations.

The tabular representation is finally constructed by concatenating the complete ring representation with the 20 derived descriptors:

$$\mathbf{x}^{(i)}_{\mathrm{tab}} = \left[\mathbf{x}^{(i)}_{\mathrm{full}} \| \mathbf{d}^{(i)}\right] \in \mathbb{R}^{120}. \tag{78}$$

This 120-dimensional internal representation is subsequently processed by the tabular branch and fused with the relational information extracted by the graph branch.

C. Selected Neural-Model Hyperparameters

Tables 8 and 9 summarize the final hyperparameter settings used for the representative neural configurations discussed in the global ranking. The appendix focuses on the Deep Neural Network and Graph Neural Network models because these architectures constitute the main methodological extension of the present work.

The tree-based and hybrid tree-based regressors are retained as reference baselines. Their hyperparameter-search spaces and optimized configurations were described in detail in the previous Lorenzetti-based calibration study in which those baselines were introduced and evaluated [14]. Therefore, the present appendix reports only the neural-model hyperparameters needed to make the newly evaluated DNN and GNN configurations explicit.

References

[1] ATLAS Collaboration, Evidence for Higgs boson decays to a low-mass dilepton system and a photon in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector, Physics Letters B 819 (2021) 136412. doi:https://doi.org/10.1016/j.physletb.2021.136412.

[2] C. W. Fabjan, F. Gianotti, Calorimetry for particle physics, Reviews of Modern Physics 75 (2003) 1243–1286. doi:10.1103/RevModPhys.75.1243.

[3] ATLAS Collaboration, Electron and photon energy calibration with the ATLAS detector using LHC Run 2 data, Journal of Instrumentation 19 (2024) P02009. doi:10.1088/1748-0221/19/02/P02009.

[4] J. V. da Fonseca Pinto, on behalf of the ATLAS Collaboration, Ring-shaped Calorimetry Information for a Neural EGamma Identification with ATLAS Detector, Technical Report, CERN, Geneva, 2016. URL: https://cds.cern.ch/record/2142761. doi:10.1088/1742-6596/762/1/012049.

[5] G. E. Orellana, ATLAS Collaboration (ATLAS), Projected ATLAS Electron and Photon Trigger Performance in Run 3, Technical Report, CERN, Geneva, 2021. URL: https://cds.cern.ch/record/2730417. doi:10.22323/1.382.0244.

[6] F. Castanedo, A review of data fusion techniques, The Scientific World Journal 2013 (2013) 704504. doi:10.1155/2013/704504.

[7] T. Meng, X. Jing, Z. Yan, W. Pedrycz, A survey on machine learning for data fusion, Information Fusion 57 (2020) 115–129. URL: https://www.sciencedirect.com/science/article/pii/S1566253519303902. doi:https://doi.org/10.1016/j.inffus.2019.12.001.

[8] Z. Chen, L. Fu, J. Yao, et al., Learnable graph convolutional network and feature fusion for multi-view learning, Information Fusion 95 (2023) 109–119. URL: https://www.sciencedirect.com/science/article/pii/S1566253523000532. doi:https://doi.org/10.1016/j.inffus.2023.02.013.

[9] Q. Zhang, Y. Yang, Y. Cheng, et al., Information fusion for multi-scale data: Survey and challenges, Information Fusion 100 (2023) 101954. URL: https://www.sciencedirect.com/science/article/pii/S1566253523002701. doi:https://doi.org/10.1016/j.inffus.2023.101954.

[10] D. Belayneh, F. Carminati, A. Farbin, et al., Calorimetry with deep learning: particle simulation and reconstruction for collider physics, European Physical Journal C 80 (2020) 688. doi:10.1140/epjc/s10052-020-8251-9.

[11] N. Akchurin, C. Cowden, J. Damgov, et al., On the use of neural networks for energy reconstruction in high-granularity calorimeters, Journal of Instrumentation 16 (2021) P12036. doi:10.1088/1748-0221/16/12/P12036.

[12] P. Simkina, F. Couderc, J. Malclès, et al., Reconstruction of electromagnetic showers in calorimeters using deep learning, European Physical Journal C 84 (2024) 639. doi:10.1140/epjc/s10052-024-12978-1.

[13] P. R. A. Silva, E. F. Simas Filho, P. C. M. A. Farias, E. E. P. d. Souza, J. L. Marin, J. M. Seixas, Machine learning for energy calibration in high energy physics experiments using the lorenzetti showers framework, IEEE Transactions on Nuclear Science (2026). doi:10.1109/TNS.2026.3686269.

[14] P. R. A. Silva, E. F. Simas Filho, P. C. Farias, E. E. Souza, J. L. Marin, J. M. de Seixas, Hybrid tree-based machine learning for calorimetric energy calibration using the lorenzetti showers framework, Measurement 290 (2026) 122771. URL: https://www.sciencedirect.com/science/article/pii/S0263224126024802. doi:https://doi.org/10.1016/j.measurement.2026.122771.

[15] M. Araújo, M. Begalli, W. Freund, G. Gonçalves, M. Khandoga, B. Laforge, A. Leopold, J. Marin, B.-M. Peralva, J. Pinto, M. S. J. Seixas, E. S. Filho, E. Souza, Lorenzetti showers - a general-purpose framework for supporting signal reconstruction and triggering with calorimeters, Computer Physics Communications 286 (2023) 108671. URL: https://www.sciencedirect.com/science/article/abs/pii/S0010465523000164. doi:https://doi.org/10.1016/j.cpc.2023.108671.

Table 8
Selected hyperparameters used by the `GNN no-Bin: Rings[1:100]` configuration discussed in the global ranking. Structural details of the graph construction and relational–tabular architecture are described in Section 2.5.2.

Hyperparameter	GNN no-Bin: Rings[1:100]	Purpose
`node_hidden`	64	Hidden dimensionality of the node encoder applied to each calorimeter-ring node before graph message passing.
`graph_hidden`	128	Hidden dimensionality of the GraphSAGE layers used to extract regional graph embeddings from the ring topology.
`tab_hidden`	256	Hidden dimensionality of the tabular branch used to encode the complete 100-ring representation and the 20 deterministic shower-shape descriptors before fusion with the graph embedding.
`num_graph_layers`	2	Number of GraphSAGE message-passing layers applied independently to the presampler, electromagnetic, and hadronic ring subgraphs.
`ring_hops`	1	Maximum sequential ring-index distance used to define graph edges between adjacent positions within each calorimeter region.
`graph_scale_init`	0.05	Initial scaling factor applied to the graph embedding before fusion with the tabular representation, controlling the initial contribution of the graph branch.
`dropout`	0.05	Dropout probability used in the tabular and fusion branches to regularize the model and reduce overfitting.
`epochs`	65	Maximum number of training epochs allowed for the GNN model.
`batch_size`	256	Number of samples processed per optimization step during mini-batch training.
`lr`	0.001	Learning rate used by the AdamW optimizer. It controls the step size during gradient-based optimization.
`weight_decay`	1×10^{-6}	L2 weight regularization applied by the AdamW optimizer to improve generalization.
`lambda_mse`	0.70	Weight assigned to the mean squared error component in the combined MSE–Smooth-L1 objective.
`huber_beta`	1.10	Transition parameter of the Smooth-L1 component. It controls the point at which the loss changes from quadratic to approximately linear behavior.
`scale_y`	True	Enables standardization of the target variable during training, improving numerical stability.
`max_train_samples`	300000	Upper cap on the number of training samples accepted by each GNN training call. The cap of 300000 exceeds the number of development events available to any training fold and therefore does not induce subsampling in the experiments reported here.
`validation_fraction`	0.10	Fraction of the training data internally reserved for validation during GNN training and early stopping.
`early_stopping_patience`	12	Stops training if the validation loss does not improve after the specified number of epochs.
`min_delta`	1×10^{-8}	Minimum validation-loss improvement required to reset the early-stopping patience counter.
`random_state`	0	Seed used to ensure reproducibility of stochastic procedures, including initialization and train-validation splitting.
`device`	cpu	Computational device used for model training and inference.
`verbose`	False	Disables epoch-level training logs during nested cross-validation execution.

Table 9
Selected hyperparameters used by the DNN no-Bin configurations discussed in the global ranking. The table summarizes the representative full-ring and compact DNN settings retained after hyperparameter selection.

Hyperparameter	DNN no-Bin configurations	Purpose
`activation`	relu	Activation function used in the hidden layers. The ReLU function introduces nonlinearity while preserving computational efficiency.
`solver`	adam	Optimization algorithm used to update the network weights through stochastic gradient-based learning.
`learning_rate_init`	0.001	Initial learning rate used by the Adam optimizer. It controls the step size during weight updates.
`alpha`	0.001	L2 regularization parameter applied to the network weights. It penalizes large weights and helps reduce overfitting.
`batch_size`	512	Number of samples processed per mini-batch during training. It controls the stochasticity and computational efficiency of the optimization process.
`max_iter`	200	Maximum number of training iterations allowed for the DNN model.
`early_stopping`	True	Enables internal validation-based early stopping to avoid unnecessary training after validation performance stops improving.
`validation_fraction`	0.1	Fraction of the training data internally reserved for validation during early stopping.
`n_iter_no_change`	20	Number of consecutive iterations without validation improvement allowed before stopping the training process.
`n_layers`	16	Number of hidden layers in the neural network. It controls the depth and representational capacity of the DNN.
`min_width`	16	Minimum number of neurons allowed in the hidden layers when using architectures with decreasing layer widths.
`shape`	constant for Rings[1:100]; halving for CORR-10 and ICA-10	Defines the hidden-layer width pattern. The constant shape keeps the same number of neurons across layers, whereas the halving shape progressively reduces the layer width.
`width`	100 for Rings[1:100]; 200 for CORR-10 and ICA-10	Initial hidden-layer width. It defines the number of neurons in each layer for the constant architecture or the starting width for the halving architecture.
`random_state`	0	Seed used to ensure reproducibility of stochastic training procedures, including weight initialization and validation splitting.

[16] J. V. da Fonseca Pinto, jlieberm, W. Freund, G. G. Milan, M. V. de Araújo, G. Gonçalves, lorenzetti-hep/lorenzetti: Version 2.0.0, 2022. URL: https://github.com/lorenzetti-hep/lorenzetti. doi:10.5281/zenodo.7494463, accessed: 2025-04-30.

[17] ATLAS Collaboration, The ATLAS trigger system for LHC Run 3 and trigger performance in 2022, Journal of Instrumentation 19 (2024) P06029. URL: http://dx.doi.org/10.1088/1748-0221/19/06/P06029. doi:10.1088/1748-0221/19/06/p06029.

[18] ATLAS Collaboration, Electron and photon performance measurements with the ATLAS detector using the 2015–2017 LHC proton-proton collision data, Journal of Instrumentation 14 (2019) P12006–P12006. URL: https://iopscience.iop.org/article/10.1088/1748-0221/14/12/P12006. doi:10.1088/1748-0221/14/12/p12006.

[19] P. R. A. da Silva, E. F. de Simas Filho, Lorenzetti shower simulation dataset, https://github.com/PauloRASpro/Lorenzetti-Shower-Simulation, 2025.

[20] J. Hoya, on behalf of the ATLAS Collaboration, The ATLAS Electron and Photon Trigger in Run-2, in: PoS(LHCP2018), 2018, p. 027. URL: https://cds.cern.ch/record/2637126/files/ATL-DAQ-PROC-2018-020.pdf. doi:10.22323/1.321.0027.

[21] J. Seixas, L. Caloba, M. Souza, A. Braga, A. Rodrigues, Neural second-level trigger system based on calorimetry, Computer Physics Communications 95 (1996) 143–157. URL: https://www.sciencedirect.com/science/article/pii/0010465596000124. doi:https://doi.org/10.1016/0010-4655(96)00012-4.

[22] T. Hastie, R. Tibshirani, J. Friedman, The Elements of Statistical Learning, second ed., Springer, 2009.

[23] A. Hyvärinen, J. Karhunen, E. Oja, Independent Component Analysis, John Wiley & Sons, 2001. URL: https://onlinelibrary.wiley.com/doi/book/10.1002/0471221317. doi:10.1002/0471221317.

[24] T. M. Cover, J. A. Thomas, Elements of Information Theory, John Wiley & Sons, 1991.

[25] I. Goodfellow, Y. Bengio, A. Courville, Deep Learning, first ed., The MIT Press, 2016.

[26] K. Albertsson, P. Altoe, D. Anderson, et al., Machine learning in high energy physics community white paper, Journal of Physics: Conference Series 1085 (2018) 022008. URL: https://iopscience.iop.org/article/10.1088/1742-6596/1085/2/022008. doi:10.1088/1742-6596/1085/2/022008.

[27] D. Belayneh, F. Carminati, A. Farbin, et al., Calorimetry with deep learning: Particle simulation and reconstruction for collider physics, Eur. Phys. J. C 80 (2020) 688. URL: https://cds.cern.ch/record/2706000. doi:10.1140/epjc/s10052-020-8251-9. arXiv:1912.06794.

[28] P. W. Battaglia, J. B. Hamrick, V. Bapst, et al., Relational inductive biases, deep learning, and graph networks, arXiv preprint arXiv:1806.01261 (2018). doi:https://doi.org/10.48550/arXiv.1806.01261.

[29] J. Shlomi, P. Battaglia, J.-R. Vlimant, Graph neural networks in particle physics, Machine Learning: Science and Technology 2 (2020) 021001. URL: https://doi.org/10.1088/2632-2153/abbf9a. doi:10.1088/2632-2153/abbf9a.

[30] S. R. Qasim, N. Chernyavskaya, J. Kieseler, et al., End-to-end multi-particle reconstruction in high occupancy imaging calorimeters with graph neural networks, The European Physical Journal C 82 (2022) 753. doi:https://doi.org/10.1140/epjc/s10052-022-10665-7.

[31] W. L. Hamilton, R. Ying, J. Leskovec, Inductive representation learning on large graphs, in: Proceedings of the 31st International Conference on Neural Information Processing Systems, volume 30 of *NIPS'17*, 2017, p. 1025–1035. doi:https://doi.org/10.48550/arXiv.1706.02216.

[32] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778. doi:10.1109/CVPR.2016.90.

[33] J. H. Friedman, Greedy function approximation: A gradient boosting machine, The Annals of Statistics 29 (2001) 1189–1232. URL: https://projecteuclid.org/journals/annals-of-statistics/volume-29/issue-5/Greedy-function-approximation-A-gradient-boosting-machine/10.1214/aos/1013203451.full. doi:10.1214/aos/1013203451.

[34] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, T.-Y. Liu, Lightgbm: A highly efficient gradient boosting decision tree, Advances in neural information processing systems 30 (2017) 3146–3154.

[35] L. Breiman, Random forests, Machine Learning 45 (2001) 5–32. URL: https://link.springer.com/article/10.1023/A:1010933404324. doi:10.1023/A:1010933404324.

[36] P. Florek, M. Korytkowski, R. Scherer, Benchmarking state-of-the-art gradient boosting algorithms for classification, arXiv preprint arXiv:2305.17094 (2023). doi:10.48550/arXiv.2305.17094.

[37] H. Alshari, A. Y. Saleh, A. Odabaş, Comparison of gradient boosting decision tree algorithms for cpu performance, Journal of the Institute of Science and Technology 37 (2021) 157–168. URL: https://www.researchgate.net/publication/351133481_Comparison_of_Gradient_Boosting_Decision_Tree_Algorithms_for_CPU_Performance.

[38] G. Casella, R. L. Berger, Statistical Inference, 2 ed., Duxbury, Pacific Grove, CA, 2002.

[39] E. L. Lehmann, J. P. Romano, Testing Statistical Hypotheses, 3 ed., Springer, New York, 2005.

[40] F. Wilcoxon, Individual comparisons by ranking methods, Biometrics Bulletin 1 (1945) 80–83. URL: https://www.jstor.org/stable/3001968. doi:https://doi.org/10.2307/3001968.

[41] E. J. G. Pitman, A note on normal correlation, Biometrika 31 (1939) 9–12. URL: https://www.jstor.org/stable/2334971. doi:https://doi.org/10.2307/2334971.

[42] W. A. Morgan, A test for the significance of the difference between the two variances in a sample from a normal bivariate population, Biometrika 31 (1939) 13–19. URL: https://www.jstor.org/stable/2334972. doi:https://doi.org/10.2307/2334972.

[43] A. C. Davison, D. V. Hinkley, Bootstrap Methods and Their Application, Cambridge University Press, Cambridge, 1997. doi:10.1017/CBO9780511802843.

[44] R. B. D'Agostino, E. S. Pearson, Tests for departure from normality. empirical results for the distributions of b_2 and $\sqrt{b_1}$, Biometrika 60 (1973) 613–622. URL: https://www.jstor.org/stable/2335012. doi:https://doi.org/10.2307/2335012.

[45] T. W. Anderson, D. A. Darling, A test of goodness of fit, Journal of the American Statistical Association 49 (1954) 765–769. URL: https://www.jstor.org/stable/2281537. doi:https://doi.org/10.2307/2281537.

[46] N. M. Razali, Y. B. Wah, Power comparisons of shapiro–wilk, kolmogorov–smirnov, lilliefors and anderson–darling tests, Journal of Statistical Modeling and Analytics 2 (2011) 21–33. URL: https://www.researchgate.net/publication/267205556.

[47] Joint Committee for Guides in Metrology, Evaluation of Measurement Data—Guide to the Expression of Uncertainty in Measurement, Technical Report JCGM 100:2008, JCGM, 2008. GUM 1995 with minor corrections.

[48] ATLAS Collaboration, Using pile-up collisions as an abundant source of low-energy hadronic physics processes in atlas and an extraction of the jet energy resolution, JHEP 2412 (2024) 032. URL: https://cds.cern.ch/record/2904445. doi:10.1007/JHEP12(2024)032. arXiv:2407.10819.

[49] ATLAS Collaboration, The atlas experiment at the cern large hadron collider: A description of the detector configuration for run 3, Journal of Instrumentation 19 (2024) P05063. URL: https://iopscience.iop.org/article/10.1088/1748-0221/19/05/P05063. doi:10.1088/1748-0221/19/05/P05063.